# ULTRAVIOLET BOUNDS AND THE POSITIVITY OBSTRUCTION IN BRANE-CLUSTER GRAVITY

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Abstract. A Poisson brane gas yields an ultraviolet-suppressed propagator with a single graviton pole. We realize its transmission multiplier as a unitary matrix coefficient and prove that the resulting gauge-invariant curvature covariance violates reflection positivity.

## 1. Introduction

Einstein gravity quantized around flat space fails by power counting. The propagator falls like $p^{-2}$, the vertices grow like $p^2$, and the superficial degree of divergence of an $L$-loop graph in four dimensions is $2 + 2L$. Pure gravity is finite at one loop on shell, but not once matter is added [1], and the two-loop counterterm of pure gravity does not vanish [2]. Curvature-squared terms make the theory renormalizable [3, 60, 61] at the price of a massive spin-two state with negative residue, the state that every higher-derivative action has carried since Pais and Uhlenbeck [58, 59]. The state can be kept out of the asymptotic spectrum by the prescription of Lee and Wick [20, 62] or by Anselmi's fakeon projection [63, 64], and local theories with complex-conjugate pairs of such poles have been built to be finite [65, 66]; what one then gives up is microcausality at the scale of the massive state [67]. More derivatives do not cure the ghost. A propagator $1/(p^2 a(p^2))$ with $a$ a polynomial has as many extra poles as $a$ has zeros, their residues add up to $-1$, and at least one of them is a ghost [4].

A way around the problem goes back to Efimov [5] and was developed for gauge theories and gravity by Krasnikov, Kuz'min, Tomboulis, Biswas, Mazumdar, Siegel, Modesto and others [6–12]. One replaces the polynomial by an entire function without zeros, $a = e^H$ with $H$ entire. The propagator then keeps a single pole, at $p^2 = 0$, while $a$ can still grow like a power

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at large Euclidean momenta, and with enough growth the theory is super-renormalizable [68–70]. Modesto and Rachwał showed further that the one-loop divergences which remain can be removed with a few operators of higher order in curvature, so that all beta functions vanish [12]. The same class of form factors makes the field of a point mass finite at the origin and removes the singularities of the classical solutions [10, 71, 72].

In all of this $H$ is an input. Tomboulis proposed a family built from the incomplete gamma function because it has the right asymptotics [8], and later work took that family or variants of it. Nothing in the construction says where such a function should come from. A polynomial, which is what any finite number of higher-derivative corrections produces, is precisely what has to be avoided. Entire form factors had been used before as a regulator of gauge theories by Moffat, Evens, Kleppe and Woodard [73–75]; there the scale in the form factor is a regulator mass, sent to infinity at the end. Here $M$ is a physical scale, and it stays.

We revisit the intersecting-brane proposal of Ref. [85]. The original framework uses incidence, operator algebra, and curvature-coupled modes to organize brane intersections. These data give a geometric effective theory. They leave the sign and high-momentum size of the spin-two response to be determined. We compute the exact sign of Gaussian cluster exchange and then study a different, multiplicative response in which brane scattering is represented by a Poisson gas. Section 3 derives a four-dimensional curvature test after null projection and compactification. Section 4.9 constructs the one-brane response as a unitary matrix coefficient. In one realization of the gas, each brane contributes the exponential of its response, and the logarithms add. For the cell model the resulting form factor is

$$a(z) = \exp\Big[\frac{n}{2}\,\mathrm{Ein}\Big(\frac{z^2}{M^4}\Big) + R(z)\Big], \tag{1}$$

where $\mathrm{Ein}(w) = \int_0^w (1 - e^{-t})\,\mathrm{d}t/t$ is entire, $z = p^2$ on the Euclidean section, and $R$ is the centered Poisson fluctuation. On closed cones about the real axes the form factor grows as $c(\pm z/M^2)^n$, with the sign chosen separately on each cone. The mean logarithm is the same incomplete-gamma form used in nonlocal gravity [8, 11, 12]. Here it follows from the size intensity and single-brane response.

Three conclusions have different hypotheses. First, the Poisson construction gives an entire, zero-free form factor for almost every realization. For power-law fine graining it also gives an entire, zero-free averaged inverse and an exact asymptotic remainder for every positive inverse moment (Sec. 4.12). Second, the Euclidean hard-subspace bounds isolate the possible one-loop subdivergences at strength $n \geq 4$. Dimensional regularization matches the one-loop pole to that of a local higher-derivative comparison theory. Three quartic curvature terms give an invertible tuning in the background curvature sector. Section 8 distinguishes this cancellation from the forest and BRST hypotheses needed for a claim about every physical amplitude. Third, the

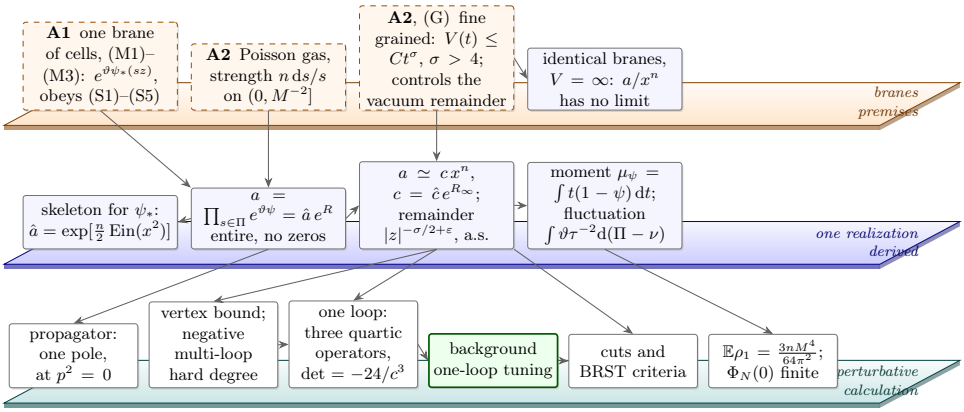



FIGURE 1. Logical structure of the paper, in three layers. The top layer holds the premises about branes (dashed boxes): the response of one brane, A1, derived from a model of cells, and the gas, A2, whose fine-graining condition (G) controls the remainder for a single realization and the subtracted vacuum integral; without it, for identical branes, the form factor has no power-law asymptotics (box on the right). The middle layer holds the properties of the form factor of one realization, derived for every admissible response and almost every realization (Secs. 4.5–4.8), with the closed-form skeleton for $\psi_*$ as one instance. The bottom layer holds the field theory, which uses the form factor only through the middle layer with power counting at $n \ge 4$ and a separate one-loop tuning. Arrows point from what is used to what is derived.

spin-two covariance fails reflection positivity on a compactly supported, linearized curvature test (Sec. 9.6). The failure is stable in any limit preserving that free curvature two-point function. A positive-metric nonperturbative Euclidean completion with that free limit is thereby excluded. Section 9.8 constructs an interacting curvature law in a bounded spacetime window, proves convergence of its cutoff approximations, and shows that the negative reflected norm persists at small coupling.

The cluster geometry is developed first, together with the effective stress tensor and an exact heavy-sector integration. The brane response and Poisson size law are model inputs. The graviton action is constructed from the resulting quadratic factor and a covariant curvature ansatz. The following sections treat power counting, the background-field pole, perturbative cutting, the vacuum subtraction and the Newtonian limit. The appendices derive the angular and divided-difference formulas and examine the geometric claims about Regge curvature, stationary entropy and information.

## 2. Intersection complexes and their effective fields

The starting point is the intersecting-brane construction of Ref. [85]. The finite intersection complex, its induced algebra, and the curvature-coupled cluster fields are three distinct structures. We first specify what follows from the intersection data, then compute what happens when a positive cluster sector is integrated out. This gives a test of the proposed link between topology and ultraviolet suppression.

2.1. **Incidence, harmonic charges, and products.** For branes $B_1, \dots, B_N$ in an ambient space, let $K$ have a simplex $[i_0, \dots, i_q]$ whenever $B_{i_0} \cap \cdots \cap B_{i_q} \neq \varnothing$. All its faces also have nonempty intersection. Orient simplices by a fixed order of the indices and set

$$\partial_q[i_0, \dots, i_q] = \sum_{j=0}^{q} (-1)^j [i_0, \dots, \widehat{i_j}, \dots, i_q], \qquad \partial_0 = 0.$$

Deleting two distinct vertices in opposite orders gives coefficients of opposite sign, so $\partial_{q-1}\partial_q = 0$. A cycle is a compatible collection of oriented intersections with vanishing net boundary. Its class is unchanged when a boundary is added. This is a combinatorial charge. Its identification with a physical conserved charge requires a coupling to a conserved current.[1]

Give the oriented basis any positive diagonal inner product and write

$$L_q = \partial_q^* \partial_q + \partial_{q+1} \partial_{q+1}^*.$$

Then $\langle c, L_q c \rangle = \|\partial_q c\|^2 + \|\partial_{q+1}^* c\|^2$. As $\operatorname{im} \partial_{q+1} \subseteq \ker \partial_q$, orthogonal projection of a cycle off $\operatorname{im} \partial_{q+1}$ gives the unique harmonic representative of its class. In particular,

$$C_q(K; \mathbb{R}) = \operatorname{im} \partial_{q+1} \oplus \ker L_q \oplus \operatorname{im} \partial_q^*.$$

For a triangular pattern without a triple overlap, $z = [12] + [23] - [13]$ represents a nonzero $H_1$ class. Filling the triple overlap makes $z = \partial[123]$. The incidence geometry changes discontinuously while all three pairwise overlaps persist.

For cochains on the ordered nerve, the cup product has the explicit formula

$$(\alpha \smile \beta)[i_0, \dots, i_{p+q}] = \alpha[i_0, \dots, i_p]\, \beta[i_p, \dots, i_{p+q}].$$

The boundary formula gives $\delta(\alpha \smile \beta) = \delta\alpha \smile \beta + (-1)^p \alpha \smile \delta\beta$. In the two sums on the right the terms corresponding to the shared vertex cancel. This makes the product independent of the cochain representatives in cohomology. An operator algebra is a separate choice. For an associative algebra $A$, the insertion of a $q$-cochain $g$ into a $p$-cochain $f$ at slot $i+1$ has sign $(-1)^{i(q-1)}$, and the difference of the two signed insertion sums is

$$[f, g] = f \circ g - (-1)^{(p-1)(q-1)} g \circ f.$$

[1] The complex is the nerve of the intersection pattern. Its homology is the homology of the nerve itself. A comparison with the union of branes requires additional hypotheses on the intersections and the cover.

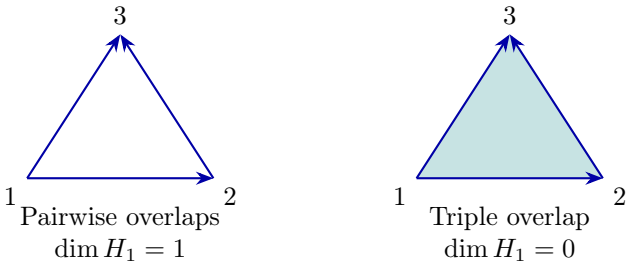



FIGURE 2. A nontrivial nerve cycle becomes a boundary when the triple overlap is filled. Metric or kinetic information is additional data.

These are operations on Hochschild cochains $C^p(A,A) = \mathrm{Hom}(A^{\otimes p}, A)$. The Hochschild differential is induced by the multiplication cochain, whose self-bracket vanishes by associativity. Gerstenhaber's theorem gives the degree $-1$ bracket on Hochschild cohomology [88]. Neither this bracket nor the cup product gives a bound on spacetime momentum.

2.2. **Curvature coupling and stress tensor.** Write $\Phi$ for one real cluster mode in Lorentzian signature,

$$S_\Phi = -\int \sqrt{-g}\,\left[\tfrac{1}{2}\nabla_\mu\Phi\nabla^\mu\Phi + U(\Phi) + \tfrac{1}{2}\xi R\Phi^2\right]\mathrm{d}^4x.$$

Here $U$ includes any quadratic mass term. With compactly supported variations, $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}\,g_{\mu\nu}\delta g^{\mu\nu}$ and $\delta R = R_{\mu\nu}\delta g^{\mu\nu} + (g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu)\delta g^{\mu\nu}$. Integration by parts gives

$$E_\Phi = \Box\Phi - U'(\Phi) - \xi R\Phi, \tag{2}$$

$$T^\Phi_{\mu\nu} = \nabla_\mu\Phi\nabla_\nu\Phi - g_{\mu\nu}\left[\tfrac{1}{2}(\nabla\Phi)^2 + U(\Phi)\right] + \xi\big[G_{\mu\nu}\Phi^2 + (g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu)\Phi^2\big]. \tag{3}$$

Diffeomorphism invariance gives $\nabla^\mu T^\Phi_{\mu\nu} = E_\Phi\nabla_\nu\Phi$. For instance the divergence of the nonminimal term is $-\frac{1}{2}\xi R\nabla_\nu\Phi^2$, by the contracted Bianchi identity and the commutator on a gradient. Thus the stress tensor is conserved on solutions of the cluster equation. Varying the Einstein action with the same metric convention gives $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T^\Phi_{\mu\nu}$. For several modes one sums the kinetic and nonminimal terms and replaces $U$ by their joint potential.

2.3. **Exact positive-sector elimination.** In a Euclidean polarization write $z = |p|^2 > 0$. For a real light amplitude $h$ and a positive heavy sector with quadratic action

$$\tfrac{1}{2}\langle h, A(z)h\rangle + \mathrm{Re}\langle\chi, B(z)h\rangle + \tfrac{1}{2}\langle\chi, D(z)\chi\rangle, \qquad D(z) > 0,$$

the substitution $\chi \mapsto \chi + D^{-1}Bh$ gives the effective inverse covariance

$$A_{\mathrm{eff}}(z) = A(z) - B(z)^* D(z)^{-1}B(z) \;\le A(z).$$

For finite matrices the determinant from Gaussian integration is independent of $h$. For a positive operator with a bounded inverse the quadratic-form conclusion follows from the same identity. Whenever $A_{\mathrm{eff}} > 0$, the covariance in the light sector is greater than or equal to $A^{-1}$ in quadratic-form order.

Take $A = z$, $B = zb$, and $D = z+T$, where $T = m_0^2 I + \lambda L_q$, $m_0 > 0, \lambda \geq 0$. The spectral measure $\mu_b(U) = \langle b, \mathbf{1}_U(T) b\rangle$ gives

$$\frac{A_{\mathrm{eff}}(z)}{z} = 1 - \int_{m_0^2}^{\infty} \frac{z}{z+t}\, \mathrm{d}\mu_b(t). \tag{4}$$

The right side is nonincreasing and lies between $1-\|b\|^2$ and 1. It is uniformly positive for $\|b\|^2 < 1$. Formula (4) also holds for a countable heavy spectrum with a self-adjoint $T \geq m_0^2$ and $b$ of finite norm. For $N \geq 0$, division of $1/(z+t)$ gives

$$\langle b, (T+z)^{-1} b\rangle = \sum_{j=0}^{N} (-z)^j \langle b, T^{-j-1} b\rangle + (-z)^{N+1} \int \frac{\mathrm{d}\mu_b(t)}{t^{N+1}(z+t)}.$$

The absolute remainder is bounded by $z^{N+1}\|b\|^2/m_0^{2N+4}$. Thus the same exchange has a controlled low-momentum derivative expansion and a sign obstruction to hardening the light kinetic term. With a scalar curvature source $J$ the covariant Gaussian exchange term is $-\frac{1}{2}\langle J, (T-\nabla^2)^{-1} J\rangle$. A coupling to $R^{(1)}$ alone vanishes on a transverse traceless polarization. Tensor curvature sources are needed to test the spin-two channel. The Poisson construction below prescribes the response of that channel through multiplicative scattering.

## 3. Null projection and a four-dimensional curvature test

The proposed $11+3$ worldvolume inside a $27+3$ construction and a $27+3$ field theory are different candidates for the constituent of the gas. The former has a precedent in Sezgin's null-projected equations [97, 98]. We first test the residual geometry. We then calculate the quadratic metric response of an explicit four-dimensional reduction.

3.1. **The residual signature.** Write $\mathbb{R}^{s,t} = \mathbb{R}^s \oplus \mathbb{R}^t$ with metric $\langle (x,y),(x',y')\rangle = x\cdot x' - y\cdot y'$. On a totally isotropic real subspace $L$, projection to either factor is injective. For instance, a vector with $y = 0$ and vanishing norm has $x = 0$. Hence $k = \dim L \leq \min(s,t)$. For a basis $u_i$ of $L$, choose $v_i$ with $\langle u_i, v_j\rangle = \delta_{ij}$ by nondegeneracy. Replacing $v_i$ by $v_i - \frac{1}{2}\sum_j \langle v_i, v_j\rangle u_j$ makes the $v_i$ mutually null and leaves their pairings with the $u_i$ unchanged. The $k$ resulting hyperbolic planes remove one positive and one negative direction each. Their orthogonal complement represents $L^{\perp}/L$, of signature $(s-k, t-k)$.

The classification in Ref. [96, Theorem 3] uses a full projected Majorana–Weyl gaugino, the specified transformation law, a nonzero residual gauge variation and translation bracket, and absorption of the cubic fermion term.

We use its reduction to the residual Clifford identity under those hypotheses. Here is the dimension calculation that enters that test. In a complex orthonormal residual basis take symmetric chiral blocks $\gamma_\mu$ and dual blocks $\widetilde{\gamma}_\mu$ with $\gamma_\mu\widetilde{\gamma}_\nu + \gamma_\nu\widetilde{\gamma}_\mu = 2\delta_{\mu\nu}I$ and the relation with the blocks interchanged. The dual blocks are symmetric, since $\widetilde{\gamma}_\mu = \gamma_\mu^{-1}$ for each basis index. Let $\Gamma_\mu$ be the corresponding off-diagonal Clifford matrix on the sum of both chiral spaces. For distinct $\mu, \nu$ choose $\rho$ different from both. The product $\Gamma_\rho\Gamma_\mu$ preserves each chiral space and conjugates $\Gamma_\mu\Gamma_\nu$ to its negative. Its restriction therefore has trace zero. For equal indices the Clifford square is the identity. These two cases give $\operatorname{tr}(\gamma_\mu\widetilde{\gamma}_\nu) = N\delta_{\mu\nu}$. Summing the Clifford relation gives

$$\sum_\mu \gamma_\mu\widetilde{\gamma}_\nu\gamma_\mu = 2\gamma_\nu - \gamma_\nu\sum_\mu \widetilde{\gamma}_\mu\gamma_\mu = (2-d)\gamma_\nu.$$

If the symmetrized cubic tensor is proportional to the source tensor, write the relation in the form

$$\tfrac{1}{2}\sum_\mu \big((\gamma_\mu)_{ab}(\gamma_\mu)_{cd} + (\gamma_\mu)_{ac}(\gamma_\mu)_{bd}\big) = j\sum_\mu (\gamma_\mu)_{ad}(\gamma_\mu)_{bc}.$$

Contraction with $(\widetilde{\gamma}_\nu)^{cd}$ gives $N-(d-2) = -2j(d-2)$, and contraction with $(\widetilde{\gamma}_\nu)^{bc}$ gives $jN = -(d-2)$. Eliminating $j$ gives $(N-2(d-2))(N+d-2) = 0$. For $d > 2$ and a nonzero module this forces $N = 2(d-2)$ and $j = -1/2$. For a full even-dimensional complex Weyl module $N = 2^{d/2-1}$, obtained by taking the even or odd part of the exterior algebra on $d/2$ creation operators. For $d = 2r \geq 4$ the equality is $2^{r-1} = 4(r-1)$. Direct substitution for $r = 2, 3, 4, 5$ leaves $r = 5$. The ratio $2^{r-1}/(4(r-1))$ strictly increases for $r \geq 3$, so no larger $r$ works.

For $27+3$, even the maximal $k = 3$ leaves $d = 24$. For $19+3$ the corresponding minimum is 16. Thus neither ambient signature passes the residual test for the specified null-projected field theory. In $11+3$, by contrast, choose orthonormal spacelike $e_i$ and timelike $f_i$ and put

$$u_i = \frac{e_i + f_i}{\sqrt{2}}, \qquad v_i = \frac{e_i - f_i}{\sqrt{2}}, \qquad i = 1, 2.$$

The remaining vectors $e_3, \ldots, e_{11}, f_3$ have signature $(9, 1)$. This is the rank-two residual signature in Sezgin's construction. A further ordinary reduction can use six spacelike directions. It requires a separate choice of background and field truncation.[2]

3.2. **A curvature coupling after torus reduction.** For an explicit test, take the residual ten-dimensional space to be a product of four-dimensional spacetime and a flat six-torus of volume $V_6$. Retain gauge fields independent of the torus coordinates and work in a commuting gauge sector. Write the

[2]The residual-dimension test concerns a particular field theory. An abstract superalgebra or a $27+3$ bulk theory containing a smaller worldvolume need not satisfy its hypotheses.

ten-dimensional potential as $(A_\mu, \phi_i)$. In Euclidean signature with a product metric $(g_{\mu\nu}, \delta_{ij})$, the field strengths are $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, $F_{\mu i} = \partial_\mu \phi_i$, and $F_{ij} = 0$. The reduced quadratic action is therefore

$$\frac{V_6}{4g_{10}^2} \int \sqrt{g} \left( g^{\mu\rho} g^{\nu\sigma} F_{\mu\nu} F_{\rho\sigma} + 2 g^{\mu\nu} \sum_i \partial_\mu \phi_i \, \partial_\nu \phi_i \right) \mathrm{d}^4 x.$$

Each term contains two gauge or scalar fields. Around the zero-field vacuum and $g = \delta + h$, its Hessian has no mixed entry between $h$ and a gauge or scalar fluctuation. In particular, minimizing or integrating the quadratic zero-mode action cannot change the graviton propagator at tree level. The minimally coupled matter determinant can contribute at loop order through stress-tensor correlations.

A direct bilinear curvature coupling can be tested as well. In a flat Euclidean Fourier mode,

$$R^{(1)}_{\mu\nu} = \tfrac{1}{2} \left( z h_{\mu\nu} - p_\mu p^\rho h_{\rho\nu} - p_\nu p^\rho h_{\rho\mu} + p_\mu p_\nu h^\rho{}_\rho \right).$$

For transverse traceless $h$ this is $z h_{\mu\nu}/2$ and $R^{(1)} = 0$. The scalar source $R^{(1)}\phi$ therefore vanishes in the physical polarization. For a positive Euclidean tensor sector $\chi_{\mu\nu}$ with quadratic operator $z + T$, $T \geq m_0^2 > 0$, take a bilinear source $2\langle \chi_{\mu\nu}, b \rangle R^{(1)\mu\nu}$. In a unit transverse traceless polarization the mixed block is $zb$. With bare metric block $z$, completing the square gives

$$z_{\text{eff}} = z - z^2 \langle b, (z + T)^{-1} b \rangle.$$

Thus $z_{\text{eff}} \leq z$ for every $z > 0$. Where this quadratic form is stable, its propagator is at least $1/z$. For comparison, the gas response with a nonzero cell strength requires $z_{\text{eff}} > z$ at every positive $z$. The sign bound is the same for an infinite positive heavy sector with a bounded source, by the spectral theorem.

There is a separate obstruction for a finite number of finite-derivative local fields. In a fixed scalar polarization, if the momentum blocks are matrices polynomial in $z$, then $z_{\text{eff}}$ is rational, by the adjugate formula. The same holds for $z_{\text{eff}}/z$. If this ratio were equal to the single-cell-brane factor $\exp(\vartheta(1 - e^{-s^2 z^2}))$ on a real interval, the identity theorem would make all its finite poles removable. An entire rational function is a polynomial, and a polynomial with a finite limit as $z \to +\infty$ is constant. The required factor equals 1 at 0 and tends to $e^\vartheta > 1$. This excludes such a finite tree-level representation even when the positive heavy-sector hypothesis is dropped. It does not address nonlocal bare blocks or loops.

## 4. A gas of branes with no preferred size

We work in Euclidean signature, with $z = p^2 \geq 0$ for real momenta, $\Box \to -p^2$, $x = z/M^2$ on the real axis and $w = z/M^2$ in the complex plane. Lorentzian statements follow by continuing $z$ to negative values; the response

used for the closed forms depends on $z^2$, so for it the two half-axes behave alike.

4.1. **The response of one brane.** The word brane denotes here an extended object that a graviton crosses and that acts on it multiplicatively. It is not the brane gas of string cosmology [76, 77], whose constituents are the D-branes of string theory [56, 78] and whose role is thermodynamic, although both pictures fill space with a gas of extended objects and both use the fact that a gas has no preferred member. Consider a graviton mode crossing a stack of thin branes. Let $t_i$ and $r_i$ be the forward transmission and reflection amplitudes of brane $i$, and $\delta$ the phase accumulated between neighbors. Summing the multiple reflections between two branes gives $t_{12} = t_1 t_2/(1 - r_1' r_2 e^{2i\delta})$, where $r_1'$ is the reflection amplitude of the first brane seen from the right. In the corresponding expansion for $N$ branes every term other than $\prod_i t_i$ contains at least one round trip, and hence two reflection amplitudes, so that

$$t_{1\cdots N} = \Big(\prod_{i=1}^{N} t_i\Big)\Big[1 + O\Big(\sum_{i<j} |r_i||r_j|\Big)\Big]. \tag{5}$$

This is a multiplicative transmission model motivated by multiple scattering [17, 18]. For weakly reflecting branes the kinetic operator of the graviton is dressed multiplicatively, $p^2 \to p^2 \prod_i t_i^{-1}$, and the problem reduces to the single factor $t_i^{-1}$.[3] We first state what the field theory requires of that factor, and call the requirement assumption A1. Sections 4.7 and 4.13 show that the field theory needs nothing more than these conditions, and Sec. 4.2 derives them from a model of the brane.

A brane has a size $s > 0$, with dimensions of length squared, and a strength $\vartheta > 0$, and it multiplies the spin-two and spin-zero parts of the graviton kinetic operator by

$$t^{-1} = e^{\vartheta\, \psi(sz)}, \tag{6}$$

where

(S1) $\psi$ is entire and $\psi(0) = 0$;

(S2) $\psi$ is real on the real axis, with $0 \le \psi(t) \le 1$ there;

(S3) $1 - \psi$ and all its derivatives decay along the real axis faster than any power of $1/|t|$;

(S4) the decay in (S3) holds uniformly in closed sectors around the real axis: there is an opening $\theta > 0$ such that for every $\delta \in (0, \theta)$ and every $N$, $|1 - \psi(w)| \le C_{\delta,N} |w|^{-N}$ for $|w| \ge 1$ and $|\arg(\pm w)| \le \theta - \delta$.

[3] The gases below contain infinitely many branes, most of them much smaller than the wavelength of a given mode. The correction in Eq. (5) is bounded by $(\sum_i |r_i|)^2$, which stays small if the reflection of a brane falls off for modes much longer than the brane at least as fast as its effect on the kinetic operator does. In the model of Sec. 4.2 there is no reflection, and the product is exact. The sum of the effects themselves converges almost surely (Sec. 4.5).

On the real axis the factor (6) rises from 1, for modes much longer than the brane, to $e^{\vartheta}$, for modes much shorter, so $\vartheta$ is the saturated logarithmic strength of the brane. Condition (S1) says that a brane has a single length scale and leaves long modes untouched. Condition (S2) is passivity: along the real axis, spacelike and timelike alike, a brane can suppress a mode but never amplify it. Condition (S3) says that the suppression saturates once the mode is much shorter than the brane. Condition (S4) says that saturation is not special to real momenta: a mode whose squared momentum is slightly complex, as it is for the loop energies of Sec. 9 near the ends of their contours, is saturated as well.[4] Dependence on the momentum through $p^2$ alone requires the branes to respect four-dimensional Poincaré invariance, as branes that fill the observed four dimensions and are scattered in the transverse ones do.

The exponential in Eq. (6) is not a separate choice. It is equivalent to a fifth condition,

(S5) the factor $t^{-1}$ has no zeros in the complex plane of $z$,

which says that a brane has no resonances: a zero of $t^{-1}$ is a pole of the transmission amplitude at complex momentum, a quasi-bound state of the brane. Suppose a brane multiplies the kinetic operator by $1+\varphi(sz)$, with $\varphi$ entire, real on the real axis, $\varphi(0)=0$, and $0\le\varphi\le\varphi_\infty$ on the real axis, saturating at $\varphi_\infty$ as in (S3) and (S4). If $1+\varphi$ has no zeros it is the exponential of an entire function, $1+\varphi=e^h$ with $h(0)=0$ [19], and $h=\log(1+\varphi)$ on the real axis, where $1+\varphi>0$. Put $\vartheta=\log(1+\varphi_\infty)$ and $\psi=h/\vartheta$. Then $\psi$ satisfies (S1) and (S2), and $1-\psi=-\vartheta^{-1}\log[1-(\varphi_\infty-\varphi)/(1+\varphi_\infty)]$ is bounded by $2|\varphi_\infty-\varphi|/\vartheta$ wherever $|\varphi_\infty-\varphi|\le\frac12(1+\varphi_\infty)$, which gives (S3) and (S4).[5] Conversely $e^{\vartheta\psi}$ is entire and has no zeros. Conditions (S1)–(S5) on a factor $1+\varphi$ are therefore the same as (S1)–(S4) on the exponent $\psi$ of Eq. (6).

Condition (S5) is what lets a single realization of the gas define a theory, and it fails for the most obvious saturating factor. The function $1+\varphi=2-e^{-s^2z^2}$ obeys every other condition, but it vanishes when

$$s^2z^2=-\log 2-2\pi ik,\qquad k\in\mathbb{Z}. \tag{7}$$

These resonances lie in $\pi/4<|\arg z|<3\pi/4$, away from both real half-axes, but there are infinitely many of them, and a product over a gas of such branes vanishes at the resonances of every brane in it. A gas with $n$ such branes per $e$-fold of size below $1/M^2$ has on average $nR^2/(\pi M^4)$ zeros in the disk

[4] Condition (S4) does not follow from (S3). The entire function $\psi(w)=1-e^{-w^2}\cos^2(e^w-1)$ satisfies (S1)–(S3), but off the real axis $|\cos^2(e^w-1)|$ grows like $\exp(2e^{\mathrm{Re}\,w}|\sin\mathrm{Im}\,w|)$, which defeats the Gaussian in every sector. For the response $\psi_*$ used below the opening is $\theta=\pi/4$.

[5] In a sector, at large $|w|$, $h-\vartheta$ equals the principal logarithm of $(1+\varphi)/(1+\varphi_\infty)$ plus $2\pi ik$ with an integer $k$ that is constant on the connected set where the ratio is close to 1; that set touches the real axis, where $k=0$.

$|z| < R$ for $R \gg M^2$,[6] each a complex pole of the propagator. Complex pairs of this kind are what Lee and Wick [20] and, in a modern form, Anselmi and Piva [21, 22] proposed to treat with a modified prescription. Branes obeying (S5) have none, and no such prescription is needed.

The simplest response with all five properties, and the one we use from Sec. 4.6 on, has the exponent

$$\psi_*(w) = 1 - e^{-w^2}, \qquad t^{-1} = \exp\bigl[\vartheta\bigl(1 - e^{-s^2 z^2}\bigr)\bigr]. \tag{8}$$

Its derivative expansion is $t^{-1} = 1 + \vartheta s^2 z^2 + O(z^4)$. The more obvious exponent $1 - e^{-w}$, which would give the incomplete gamma function directly, violates (S2): on the negative axis it is unbounded below, so the factor falls to zero faster than any power along the timelike axis and no polynomial bound holds there.[7]

Conditions (S1) and (S2) already have a consequence. On the real axis $\psi \geq 0 = \psi(0)$, so the origin is a minimum of a smooth function and $\psi'(0) = 0$. The leading correction that a passive brane makes to the kinetic operator is therefore of order $s^2 z^2$ and not $sz$: a passive brane does not generate a curvature-squared term at leading order.

4.2. **A brane made of cells.** Conditions (S1)–(S5) say what a brane must do. A simple model of what a brane is made of produces all five, and it produces the response $\psi_*$ of Eq. (8) instead of leaving it as a choice. Take a brane of size $s$ to be a thin layer of microscopic cells, and assume three things about them.

(M1) The cells are placed independently of one another, so that a mode crossing the brane at a given point of its worldvolume meets a Poisson number $N$ of cells, with a mean $\vartheta$ that is the same at every point.

(M2) A cell neither absorbs nor reflects. It shifts the proper time of the mode that crosses it by an amount $Y$, a delay or an advance.

(M3) The shifts of different cells are independent, with a common law that is Gaussian with mean zero and variance $2s^2$.

The first assumption is the independence that made the gas a Poisson process, applied now inside one brane. The second fixes how a cell acts. In the proper-time representation of the propagator a mode of squared momentum $z$ propagates for a proper time $T$ with the amplitude $e^{-iTz}$, so a shift of $T$ by $Y$ multiplies it by the phase $e^{-iYz}$, of modulus one for real $z$ of either

[6] A brane of size $s$ contributes the zeros with $|\log 2 + 2\pi i k| \leq s^2 R^2$ inside the disk $|z| \leq R$, about $2s^2R^2/\pi$ of them counting both signs of $z$. Averaging with $n\,\mathrm{d}s/s$ over $s \leq M^{-2}$ gives $nR^2/(\pi M^4)$. The numerical check in Appendix D reproduces this count to 0.03% at $R = 60M^2$.

[7] The exponent of the gas, computed as in Sec. 4.5, is then $n\,\mathrm{Ein}(z/M^2)$, and along the negative real axis $\mathrm{Ein}(-t) = -\int_0^t (e^u - 1)\,\mathrm{d}u/u$ tends to $-\infty$ faster than any power. The Gaussian response is even, so both half-axes have the same saturation.

sign.[8] On shell, $z = 0$, the phase is 1: a cell does not act on a free graviton, as it must not if the brane is invariant under the Poincaré group of its worldvolume. Off shell the phases of the cells that a mode crosses add up. The third assumption is what the central limit theorem gives when the shift of a cell is the sum of many small independent shifts of its constituents, and the symmetry $Y \to -Y$ says that a cell has no preferred direction of proper time. The size of the brane is the spread of the shift, $\mathbb{E}Y^2 = 2s^2$, which is why $s$ has the dimensions of a proper time, a length squared.

A mode crossing the brane is a plane wave across its worldvolume, and different points of the worldvolume meet independent sets of cells. What continues as the same mode is the coherent part of the transmitted wave, the average of the phase over the worldvolume; the remainder is scattered into other modes. This is the coherent wave of the theory of multiple scattering [17, 18], and Fig. 3 shows the construction. For a brane whose extent is large compared with the spacing of its cells, the average over the worldvolume is the expectation over the cells, by the law of large numbers. Conditioning on $N$,

$$\begin{aligned} t &= \mathbb{E}\exp\Big(-iz\sum_{j=1}^{N} Y_j\Big) = \sum_{N\geq 0} e^{-\vartheta}\frac{\vartheta^N}{N!}\,\varphi(sz)^N \\ &= e^{-\vartheta[1-\varphi(sz)]}, \end{aligned} \tag{9}$$

where $\varphi(u) = \mathbb{E}\,e^{-iuY/s}$ is the characteristic function of $Y/s$. For the Gaussian law of (M3), $\varphi(u) = e^{-u^2}$, and Eq. (9) is the single-brane factor (6) with

$$\psi(w) = 1 - \varphi(w) = 1 - e^{-w^2} = \psi_*(w). \tag{10}$$

The response used from Sec. 4.6 on is therefore the response of a brane whose cells shift the proper time by Gaussian amounts, and the strength $\vartheta$ is the mean number of cells that a mode meets, the optical depth of the brane.

All five conditions follow. The exponent $\vartheta\psi$ is entire, so $t^{-1}$ has no zeros, which is (S5); in the model a resonance would need a cell that holds the mode, and (M2) excludes it. Condition (S1) holds because $\varphi(0) = 1$. The law is symmetric, so $\varphi$ is real on the real axis, and for the Gaussian $0 < \varphi \leq 1$ there, which gives $0 \leq \psi < 1$, condition (S2). Passivity is now the statement that the coherent wave can only lose amplitude to the scattered one, $|t| \leq 1$. On the real axis $1 - \psi = e^{-w^2}$ falls faster than any power, which is (S3), and $|e^{-w^2}| = e^{-\mathrm{Re}\,w^2}$ falls in the same way in every closed subsector of $|\arg(\pm w)| < \pi/4$, which is (S4) with $\theta = \pi/4$. For modes much shorter than the brane, $t \to e^{-\vartheta}$, the probability of meeting no cell. Finally, since cells do not reflect and are independent of the cells of other branes, the coherent

[8] With the Euclidean $z = p^2$ of this section, a Lorentzian mode of squared four-momentum $-z$ has the proper-time amplitude $e^{iT(-z+i0)}$, and the propagator is its integral over $T \geq 0$. Whether a cell multiplies the amplitude by $e^{-iYz}$ or $e^{iYz}$ is a convention that the symmetry of the law in (M3) makes irrelevant.

amplitude through several branes is the product of their amplitudes, with no correction: the composition rule (5) holds exactly, with $r_i = 0$.

Other laws for the shift give other responses, $\psi = 1 - \varphi$ with $\varphi$ the characteristic function of a symmetric law, and conditions (S1)–(S5) then ask that $\varphi$ be entire, nonnegative on the real axis, and rapidly decreasing in a sector. A Gaussian whose variance is itself random, between two positive bounds, passes all of them, again with $\theta = \pi/4$. A law with bounded support does not: its characteristic function is entire of exponential type and grows off the real axis, which defeats (S4). Of the admissible exponents of Sec. 4.7, the other two in Table 1 are not of the form $1-\varphi$ at all, since $(1+w^2)e^{-w^2}$ and $e^{-w^4}$ have Fourier transforms that take negative values and are therefore not characteristic functions. We do not derive (M1)–(M3) from a brane action. What the model does is replace five conditions on an unknown function by three statements about the constituents of a brane, each with a physical meaning, and with them the single-brane factor is computed rather than chosen.

4.3. **Sizes and strengths.** A gas is a random set of branes, each with a size and a strength. We impose three requirements on it: branes are placed independently of one another; below the size $S = 1/M^2$ of the largest brane there is no preferred size; and the strength of the gas is fine grained, in a sense made precise below. Together they form assumption A2.

The first requirement makes the set of sizes a Poisson process. A point process without multiple points and without fixed atoms, whose counts in disjoint sets are independent, is a Poisson process with some intensity measure $\nu$ [23, 24].[9] We let the strength depend on the size, $\vartheta = \vartheta(s)$.[10]

The second requirement concerns the quantity that enters the kinetic operator, which is the strength. The expected strength carried by the branes with sizes in a set $A$, $\int_A \vartheta \, \mathrm{d}\nu$, must not change under a dilation $s \to \lambda s$ whenever both $A$ and $\lambda A$ lie in $(0, S]$. In the variable $u = \log s$ this is invariance under translations along the half-line $u \le \log S$, so the measure of an interval depends only on its length; it is additive and monotone in that length, hence linear in it, and $\vartheta \, \mathrm{d}\nu$ is a multiple of Lebesgue measure in $u$.[11] Therefore

$$\vartheta(s)\, \nu(\mathrm{d}s) = n \, \frac{\mathrm{d}s}{s} \quad \text{on } (0, S], \qquad n > 0, \tag{11}$$

[9] This is the theorem of Rényi and Kingman: complete independence plus simplicity characterizes the Poisson process. The intensity measure $\nu(A)$ is the expected number of branes with size in $A$.

[10] Strengths drawn independently from a law that depends on $s$ make the gas a marked Poisson process [23]. The mean and variance formulas below then hold with $\vartheta$ and $\vartheta^2$ replaced by their conditional means at fixed $s$, and the exponential formulas with $e^h$ replaced by its conditional mean.

[11] Additivity gives $\mu(h_1 + h_2) = \mu(h_1) + \mu(h_2)$ for the measure $\mu(h)$ of an interval of length $h$, and a monotone solution of Cauchy's equation is linear. This is the uniqueness of Haar measure on the multiplicative group $(0, \infty)$, restricted to the cut-off range.

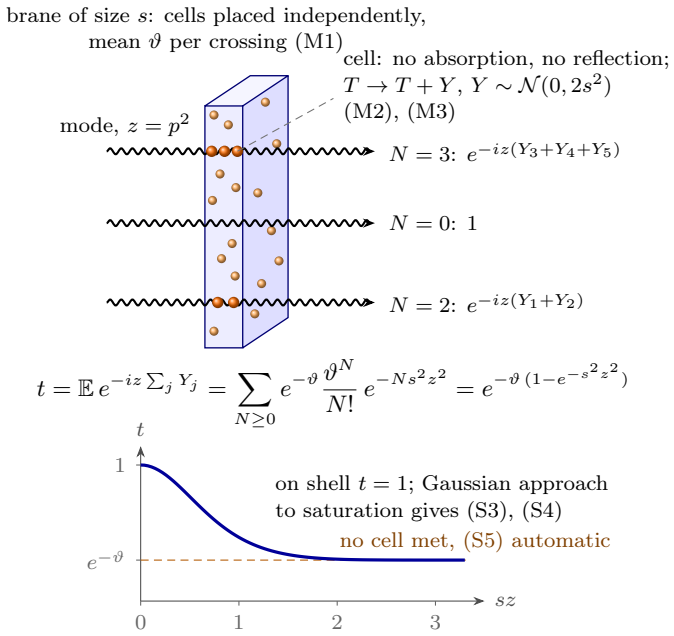



FIGURE 3. A brane made of cells, Sec. 4.2. A mode of squared momentum $z$ crosses a brane of size $s$ at every point of its worldvolume. At each point it meets a Poisson number $N$ of cells with mean $\vartheta$, (M1), each of which shifts its proper time by an independent Gaussian amount $Y_j$ of variance $2s^2$, (M2) and (M3), and so multiplies it by the phase $e^{-iY_jz}$. Three crossing points are drawn, which meet, from bottom to top, two, zero and three cells. The coherent transmitted wave is the average of the phase, Eq. (9), which is the single-brane factor (6) with $\psi = \psi_*$, Eq. (10). Lower panel: $t$ as a function of $sz$ for $\vartheta = 1.2$. It equals 1 on shell and falls to the saturation value $e^{-\vartheta}$, the probability that a crossing meets no cell, as the mode becomes shorter than the brane.

and $n$ is the expected strength per $e$-fold of size. For identical branes, $\vartheta = \vartheta_0$, this says that the number of branes per $e$-fold, $n/\vartheta_0$ on average, is the same at all sizes.

The third requirement is about how that strength is shared. It can be carried by a few strong branes or by many weak ones. Write $\tau = sM^2 \in (0, 1]$ and

$$V(t) = \frac{1}{n}\int_{\tau\leq t} \vartheta^2\, \mathrm{d}\nu = \int_0^t \vartheta(\tau)\,\frac{\mathrm{d}\tau}{\tau}. \tag{12}$$

We show in Sec. 4.4 that $nV(t)$ is the variance of the strength carried by the branes smaller than $t/M^2$, measured from its mean. The requirement is

(G) $V(t) \leq C\, t^\sigma$ for $0 < t \leq 1$, with some $\sigma > 4$.

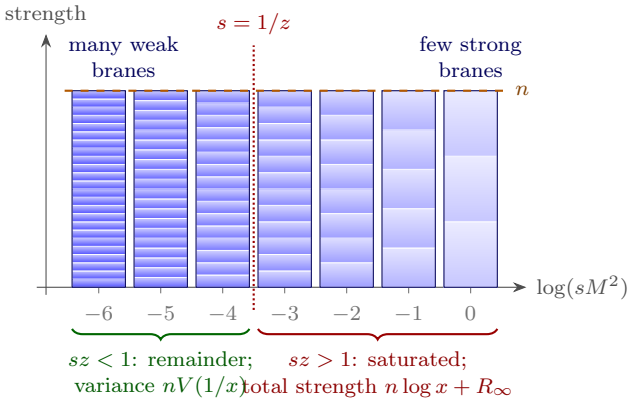



FIGURE 4. How a fine-grained gas shares its strength. Each column is one $e$-fold of size and carries expected strength $n$ (dashed line), Eq. (11). The number of branes in a column, drawn as slabs, grows toward small sizes while each slab gets thinner, as for the power law (13). A mode with $z = p^2$ saturates the branes larger than $1/z$, whose total strength is $n \log x$ plus a fluctuation that converges to $R_\infty$; the branes smaller than $1/z$ act only through a remainder whose variance is $nV(1/x)$ (Sec. 4.8).

In words: the fluctuation of the strength carried by small branes vanishes with their size faster than (size)$^2$. It does not affect the mean, which Eq. (11) fixes. The exponent is not chosen to make the proofs below work. Section 8.2 shows that the zero-point energy of a single realization has a finite-variance subtracted value in the power-law model only if the fluctuation falls at least this fast: when smaller branes are never stronger, it needs $V(t) = o(t^4)$, so (G) is necessary up to factors that vary more slowly than any power of $t$. Sections 4.8–7 show that (G) is also all the field theory needs. The simplest example is a power law,

$$\vartheta(\tau) = \vartheta_0\, \tau^{\zeta}, \qquad \nu(\mathrm{d}\tau) = \frac{n}{\vartheta_0}\, \tau^{-\zeta}\, \frac{\mathrm{d}\tau}{\tau}, \qquad V(t) = \frac{\vartheta_0 t^{\zeta}}{\zeta}, \tag{13}$$

for which (G) holds with $\sigma = \zeta$ when $\zeta > 4$. As the size decreases by one $e$-fold the branes become weaker by a factor $e^{-\zeta}$ and $e^{\zeta}$ times as numerous, so the strength per $e$-fold stays $n$ while the gas becomes ever finer. Any profile that falls faster than every power, such as $\vartheta_0 \exp[-(\log \tau)^2]$, satisfies (G) for every $\sigma$. A gas of identical branes, $\zeta = 0$, has $V \equiv \infty$ and violates (G) completely; Sec. 4.8 shows what goes wrong with it. Figure 5(a) shows the two kinds of gas, and Fig. 4 shows the bookkeeping.

4.4. **Sums and products over the gas.** Because the gas has infinitely many points, we derive the formulas we need instead of quoting them. Let $\Pi$

be a Poisson process with $\sigma$-finite intensity $\nu$ and let $f$ be a complex function with $\int |f|\,\mathrm{d}\nu < \infty$. Then

$$\mathbb{E}\prod_{x\in\Pi}\bigl(1+f(x)\bigr) = \exp\int f\,\mathrm{d}\nu, \tag{14}$$

with the product converging absolutely almost surely. Suppose first that the total mass $\lambda = \nu(\text{all})$ is finite. The number $N$ of points is then Poisson with mean $\lambda$, and given $N$ the points are independent with law $\nu/\lambda$ [24]. Conditioning on $N$,

$$\begin{aligned}\mathbb{E}\prod_{x\in\Pi}(1+f) &= \sum_{N\ge 0} e^{-\lambda}\frac{\lambda^N}{N!}\Bigl(\frac{1}{\lambda}\int(1+f)\,\mathrm{d}\nu\Bigr)^N \\ &= e^{-\lambda}e^{\lambda+\int f\,\mathrm{d}\nu},\end{aligned}$$

which is Eq. (14); the same computation with $|f|$ in place of $f$ gives $\mathbb{E}\prod(1+|f|) = \exp\int|f|\,\mathrm{d}\nu$, so the expectation exists. If $\nu$ has infinite mass, choose sets $A_1 \subset A_2 \subset \cdots$ of finite measure that exhaust the space. Campbell's theorem, $\mathbb{E}\sum_{x\in\Pi} g(x) = \int g\,\mathrm{d}\nu$ for $g \ge 0$ [23], applied to $g = |f|$, shows that $\sum_{x\in\Pi}|f(x)| < \infty$ almost surely, so the product converges absolutely and is the limit of the products over $\Pi\cap A_k$. These partial products are bounded in modulus by $\prod_{x\in\Pi}(1+|f(x)|)$, whose expectation is $\exp\int|f|\,\mathrm{d}\nu$ by monotone convergence and the finite case. Dominated convergence then gives $\mathbb{E}\prod_{x\in\Pi}(1+f) = \lim_k \exp\int_{A_k} f\,\mathrm{d}\nu = \exp\int f\,\mathrm{d}\nu$.

Two consequences are used below. Replace $f$ by $e^{uf}-1$ with real $f$ and real $u$ in the finite case and differentiate twice at $u=0$: the sum $\sum_{x\in\Pi} f(x)$ has mean $\int f\,\mathrm{d}\nu$ and variance $\int f^2\,\mathrm{d}\nu$. For complex $f$, applying this to the real and imaginary parts and to their sum gives

$$\mathbb{E}\Bigl|\sum_{x\in\Pi} f(x) - \int f\,\mathrm{d}\nu\Bigr|^2 = \int |f|^2\,\mathrm{d}\nu. \tag{15}$$

Write the left side as $\mathbb{E}|\int f\,\mathrm{d}(\Pi-\nu)|^2$. If $\int|f|^2\mathrm{d}\nu<\infty$ but $\int|f|\,\mathrm{d}\nu=\infty$, the compensated sums over $\Pi\cap A_k$ still converge: their increments over $A_{k+1}\setminus A_k$ are independent, have mean zero and have summable variances, so they converge almost surely and in mean square by Kolmogorov's criterion [25]. We call the limit $\int f\,\mathrm{d}(\Pi-\nu)$; it has mean zero and obeys Eq. (15). With $f=\vartheta$ on the branes smaller than $t/M^2$ and $f=0$ elsewhere, Eq. (15) shows that $nV(t)$, Eq. (12), is the variance of their total strength measured from its mean. The same exhaustion applied to Eq. (14) with $f = e^h - 1$ gives, for real $h$ with $|h| \le h_{\max}$ and $\int h^2\mathrm{d}\nu<\infty$,

$$\mathbb{E}\exp\int h\,\mathrm{d}(\Pi-\nu) = \exp\int\bigl(e^h-1-h\bigr)\,\mathrm{d}\nu. \tag{16}$$

On each $A_k$ this is Eq. (14) with $f=e^h-1$, multiplied by $\exp(-\int_{A_k} h\,\mathrm{d}\nu)$. The integrand on the right is at most $\frac{1}{2}e^{h_{\max}}h^2$, so the right side is finite, and the same bound with $2h$ in place of $h$ shows that $\mathbb{E}\exp\bigl(2\int_{A_k} h\,\mathrm{d}(\Pi-$

$\nu)\big) \le \exp\big(2e^{2h_{\max}} \int h^2 \mathrm{d}\nu\big)$ uniformly in $k$. The exponentials are therefore uniformly integrable, and their means converge to the mean of the limit.

4.5. **The form factor of one realization.** A realization $\Pi$ of the gas is one draw of the branes, frozen. A graviton crossing it is dressed by the product of the factors (6) of all its branes,

$$\begin{aligned} a_\Pi(z) &= \prod_{s\in\Pi} e^{\vartheta(s)\psi(sz)} = \exp L_\Pi(z), \\ L_\Pi(z) &= \sum_{s\in\Pi} \vartheta(s)\,\psi(sz). \end{aligned} \tag{17}$$

The sum converges. Because $\psi$ is entire and vanishes at the origin, $|\psi(w)| \le A_R|w|$ on the disk $|w| \le R$, and with Eq. (11) Campbell's theorem gives

$$\begin{aligned} \mathbb{E}\sum_{s\in\Pi} \sup_{|z|\le RM^2} \big|\vartheta(s)\psi(sz)\big| &\le n\int_0^1 A_R R\tau\,\frac{\mathrm{d}\tau}{\tau} \\ &= nA_R R < \infty. \end{aligned}$$

Almost surely, then, the sum in Eq. (17) converges absolutely and uniformly on every disk, its limit $L_\Pi$ is an entire function, and $a_\Pi = e^{L_\Pi}$ is an entire function *without zeros*. It satisfies $a_\Pi(0) = 1$, its Taylor coefficients are real, and $a_\Pi \ge 1$ on the whole real axis by (S2). A frozen gas of branes that obey (S5) therefore has no extra poles of any kind, real or complex. Condition (S5) is necessary as well as sufficient for this: if one factor vanished at some $z_0$, the convergent product of the others would be finite there and $a_\Pi(z_0)$ would vanish.

The mean of the exponent is fixed by Eq. (11) alone. By Campbell's theorem, $\mathbb{E}L_\Pi(z) = \int \vartheta\,\psi(sz)\,\mathrm{d}\nu = n\int_0^{1/M^2} \psi(sz)\,\mathrm{d}s/s$, and the substitution $s = \tau/M^2$ gives

$$\begin{aligned} a_\Pi(z) &= e^{n\Phi(z/M^2)}\, e^{R_\Pi(z)}, \\ R_\Pi(z) &= \int \vartheta(s)\,\psi(sz)\,(\Pi-\nu)(\mathrm{d}s), \end{aligned} \tag{18}$$

$$\Phi(w) = \int_0^1 \psi(w\tau)\,\frac{\mathrm{d}\tau}{\tau} = \sum_{k\ge1} \frac{\psi_k}{k}\,w^k, \tag{19}$$

where $\psi(w) = \sum_{k\ge1}\psi_k w^k$. The first factor in Eq. (18) is deterministic. It depends on the gas only through $n$ and not at all on how the strength is shared among branes; we call

$$\hat a(z) = \exp\Big[n\,\Phi\Big(\frac{z}{M^2}\Big)\Big] \tag{20}$$

the skeleton of the form factor. The second factor is random, with $\mathbb{E}R_\Pi(z) = 0$; it is the subject of Sec. 4.8. The skeleton is the exponential of the mean of $\log a_\Pi$. It is not the mean of $a_\Pi$, which for real $z$ and bounded strengths is, by Eq. (16), $\hat a$ times $\exp\int(e^{\vartheta\psi} - 1 - \vartheta\psi)\,\mathrm{d}\nu$, and nothing in this paper

averages the kinetic operator over the gas. The graviton sees one realization, and the theory is that of $a_\Pi$.

The Taylor series of $\psi$ converges uniformly on $\{|w\tau| \le R\}$ and may be integrated term by term, and since the coefficients of $\Phi$ are bounded by those of $\psi$ its radius of convergence is infinite. So $\Phi$ is entire with $\Phi(0) = 0$, and $\hat{a}$ is entire, has no zeros and satisfies $\hat{a}(0) = 1$.

The growth of the skeleton on the real axis follows from (S3). For real $x > 0$, $\Phi(x) = \int_0^x \psi(t)\,\mathrm{d}t/t$. Splitting the range at $t = 1$ and writing $\psi = 1 - (1 - \psi)$ on $[1, x]$,

$$\Phi(x) = \log|x| + C_\pm + \rho_\pm(x), \qquad x \to \pm\infty, \tag{21}$$

with

$$C_\pm = \int_0^1 \psi(\pm t)\frac{\mathrm{d}t}{t} - \int_1^\infty \big(1 - \psi(\pm t)\big)\frac{\mathrm{d}t}{t},$$
$$\rho_\pm(x) = \int_{|x|}^\infty \big(1 - \psi(\pm t)\big)\frac{\mathrm{d}t}{t} = O(|x|^{-N}) \text{ for every } N.$$

The case $x < 0$ is the same after the substitution $t = -x\tau$; both integrals in $C_\pm$ converge by (S1) and (S3). Exponentiating,

$$\hat{a}(z) = e^{nC_\pm}\Big|\frac{z}{M^2}\Big|^n\Big[1 + O\Big(\Big|\frac{M^2}{z}\Big|^N\Big)\Big], \qquad z \to \pm\infty. \tag{22}$$

The power $n$ depends on the gas only through its density of strength and on the shape of $\psi$ not at all. Figure 5(b) shows how this happens. At a given $z$, the branes with $sz \gtrsim 1$ each contribute their full strength to $L_\Pi$ and the others almost nothing, so $L_\Pi$ counts the strength in the $e$-folds of size above $1/z$, of which there are $\log(z/M^2)$, each holding strength $n$ on average.

From here on $n$ is a positive integer. Nothing above needs this, but the local theory of Sec. 7 does, because its kinetic term must be a polynomial in $\Box$.

4.6. **The skeleton in closed form.** For the response (8) everything about the skeleton can be written in closed form. The substitution $u = w^2\tau^2$ in Eq. (19) gives

$$\Phi(w) = \frac{1}{2}\int_0^{w^2} \frac{1 - e^{-u}}{u}\,\mathrm{d}u = \frac{1}{2}\,\mathrm{Ein}(w^2), \tag{23}$$

so that $\hat{a}$ is the function of Eq. (1). The entire function Ein has the series $\mathrm{Ein}(w) = \sum_{k\ge1}(-1)^{k+1}w^k/(k\,k!)$, and we need three facts about it and about the exponential integral $E_1(y) = \int_y^\infty e^{-t}\,\mathrm{d}t/t$ [26].

First, Euler's constant is

$$\gamma_\mathrm{E} = \int_0^1 \frac{1 - e^{-t}}{t}\,\mathrm{d}t - \int_1^\infty \frac{e^{-t}}{t}\,\mathrm{d}t. \tag{24}$$

To see this, start from $\Gamma'(1) = -\gamma_\mathrm{E}$, that is $-\gamma_\mathrm{E} = \int_0^\infty e^{-t}\log t\,\mathrm{d}t$, and integrate by parts on each half of the range: on $[0, 1]$ with the antiderivative

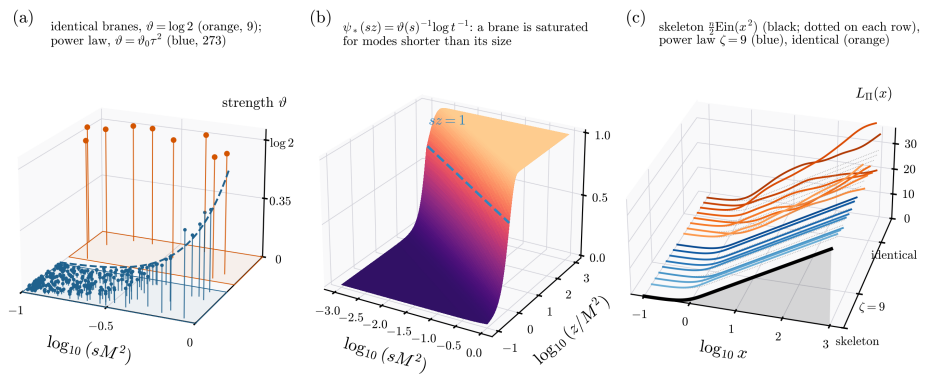



FIGURE 5. Two gases with the same strength per $e$-fold. (a) One realization of each, for $n = 4$ and sizes between $0.1\,M^{-2}$ and $M^{-2}$: identical branes with $\vartheta_0 = \log 2$ (orange) and the fine-grained power law (13) with $\zeta = 2$ (blue). Each brane is a stem at its size, at a random transverse position, with height equal to its strength. At the smallest sizes the fine-grained gas has 100 times more branes per $e$-fold, each 100 times weaker. The dashed curve is the strength $\vartheta_0 \tau^2$ of the power law. The value $\zeta = 2$ keeps the number of stems small enough to draw; condition (G) needs $\zeta > 4$. (b) The exponent of the single-brane factor in units of the strength of the brane, $\psi_*(sz) = \vartheta(s)^{-1} \log t^{-1}$, over the plane of $\log s$ and $\log z$. It switches on across the line $sz = 1$ (dashed), so at fixed $z$ the exponent $L_\Pi$ of Eq. (17) collects the strength of the branes larger than $1/z$, which is the origin of the power law (22). (c) The exponent $L_\Pi(x)$ along the Euclidean axis for eight realizations of each gas, identical branes and the power law with $\zeta = 9$, drawn as a waterfall against the skeleton $n\Phi(x) = \frac{n}{2}\,\mathrm{Ein}(x^2)$ (thick, in front, and dotted on each row). The fine-grained realizations run parallel to the skeleton at large $x$, offset by $R_\infty$; the realizations of identical branes drift away from it like random walks in $\log x$ (Sec. 4.8). For $\zeta = 9$ the branes smaller than $0.3\,M^{-2}$ are represented by the Gaussian limit of their compensated strength, as in Fig. 9.

$1 - e^{-t}$ of $e^{-t}$, so that the boundary term $(1 - e^{-t}) \log t$ vanishes at both ends, and on $[1, \infty)$ with $-e^{-t}$, so that the boundary term vanishes at $t = 1$ and at infinity. The two pieces are $-\int_0^1 (1 - e^{-t})\,\mathrm{d}t/t$ and $\int_1^\infty e^{-t}\,\mathrm{d}t/t$, and their sum is $-\gamma_\mathrm{E}$.

Second, for $y > 0$,

$$\mathrm{Ein}(y) = \log y + \gamma_\mathrm{E} + E_1(y), \qquad 0 < E_1(y) \leq \frac{e^{-y}}{y}. \tag{25}$$

Split $\mathrm{Ein}(y) = \int_0^1 (1-e^{-t})\,\mathrm{d}t/t + \int_1^y \mathrm{d}t/t - \int_1^y e^{-t}\,\mathrm{d}t/t$ and complete the last integral to $\int_1^\infty$; by Eq. (24) the constants combine into $\gamma_\mathrm{E}$, and what is left is $\log y + E_1(y)$. The bound follows from $1/t \le 1/y$ on $[y,\infty)$. Both sides of the identity are analytic in the plane cut along the negative axis, so it continues to $|\arg y| < \pi$ with principal branches. For $\mathrm{Re}\, y > 0$ the integral along the horizontal ray from $y$ gives $|E_1(y)| \le e^{-\mathrm{Re}\, y} \int_0^\infty e^{-u}\,\mathrm{d}u/|y+u| \le e^{-\mathrm{Re}\, y}/|y|$, since $|y+u| \ge |y|$ for $u \ge 0$.

Third, the function $g(y) = \mathrm{Ein}(y) - \log(1+y)$ is nondecreasing on $[0,\infty)$, with $g(0) = 0$ and $g(\infty) = \gamma_\mathrm{E}$. Indeed

$$g'(y) = \frac{1-e^{-y}}{y} - \frac{1}{1+y} = \frac{1-(1+y)e^{-y}}{y(1+y)} \ge 0,$$

because $e^y \ge 1+y$, and by Eq. (25) $g(y) = \gamma_\mathrm{E} + E_1(y) - \log(1+y^{-1}) \to \gamma_\mathrm{E}$.

These three facts give the properties of the skeleton. They are collected in Table 3 and plotted in Figs. 6 and 7; we now derive them one by one. Section 4.8 carries each of them over to a realization.

*Two-sided bound.* Putting $y = x^2$ in the third fact, for every real $z$,

$$\big(1+x^2\big)^{n/2} \le \hat{a}(z) \le \hat{c}\,\big(1+x^2\big)^{n/2}, \qquad \hat{c} = e^{n\gamma_\mathrm{E}/2}. \tag{26}$$

The skeleton is pinched between two polynomials of degree $2n$ in $x$ on the whole real axis, although it has no zero anywhere.

*Large momenta on the real axis.* For $|x| \ge 1$, Eq. (25) gives $\hat{a}(z) = \hat{c}\,|x|^n \exp[\frac{n}{2} E_1(x^2)]$, so $\hat{a} - \hat{c}|x|^n$ is smaller than $\hat{c}|x|^n$ by a factor of order $e^{-x^2}/x^2$.

*Small momenta.* From the series of Ein,

$$\hat{a}(z) = 1 + \frac{n}{2}\,x^2 + \frac{n(n-1)}{8}\,x^4 + O(x^6), \tag{27}$$

with no term linear in $x$, in line with $\psi_*'(0) = 0$.

*Derivatives.* Write $\hat{a} = e^H$ with $H(z) = \frac{n}{2}\,\mathrm{Ein}(z^2/M^4)$. Then

$$H'(z) = \frac{n\big(1-e^{-x^2}\big)}{z},$$

which is entire. For $|x| \ge 1$ it equals $n/z$ minus a term whose derivatives of every order are bounded by $C_j\, e^{-x^2/2} M^{-2-2j}$, so $|H^{(j)}(z)| \le C_j M^{-2j}(1+|x|)^{-j}$ for $j \ge 1$ on the real axis. By Faà di Bruno's formula $\hat{a}^{(j)}$ is $\hat{a}$ times a polynomial in $H', \dots, H^{(j)}$ whose monomials all have total order $j$, so $|\hat{a}^{(j)}| \le C_j M^{-2j}(1+|x|)^{-j}\hat{a}$, and with Eq. (26)

$$\big|\hat{a}^{(j)}(z)\big| \le C_j M^{-2j}(1+|x|)^{n-j}, \qquad z \in \mathbb{R}. \tag{28}$$

*The nonlocal remainder.* The function

$$\hat{\mathcal{G}}(z) = \frac{\hat{a}(z)-1}{z} - \frac{\hat{c}\,z^{n-1}}{M^{2n}} \tag{29}$$

is entire, because $\hat{a} - 1$ vanishes at $z = 0$. It measures how far the nonlocal part of the action would be from a polynomial if the skeleton were the form

factor. On $[0, M^2]$ it is a smooth function on a compact interval, and for $z \geq M^2$,

$$\hat{\mathcal{G}}(z) = -\frac{1}{z} + \frac{\hat{c}\, x^n}{z}\Big(e^{\frac{n}{2}E_1(x^2)} - 1\Big).$$

Since $E_1$ and its derivatives at $y = x^2$ are bounded by polynomials in $1/x$ times $e^{-x^2}$, the second term and all its derivatives are exponentially small, and the derivatives of $-1/z$ give

$$\big|\hat{\mathcal{G}}^{(j)}(z)\big| \leq C_j M^{-2-2j}(1+x)^{-1-j}, \qquad z \geq 0. \tag{30}$$

In the language of pseudodifferential operators $\hat{\mathcal{G}}$ is a symbol of order $-1$. In the same way $\hat{a} - \hat{c}\, x^n = \hat{c}\, x^n(e^{\frac{n}{2}E_1(x^2)} - 1)$ for $x \geq 1$, so $|\frac{\mathrm{d}^j}{\mathrm{d}z^j}(\hat{a} - \hat{c}x^n)| \leq C_j M^{-2j}(1+x)^{-j}$ on $[0,\infty)$: a symbol of order 0. Figure 6(c) shows $z\,\hat{\mathcal{G}}(z)$, which tends to $-1$.

*Complex momenta.* Let $S_\delta = \{z : |\arg z| \leq \pi/4 - \delta\}$ with $0 < \delta < \pi/4$. For $z \in S_\delta$ we have $\operatorname{Re} z^2 \geq |z|^2 \sin 2\delta > 0$, and the continued identity (25) with its bound gives

$$\begin{aligned} \hat{a}(z) &= \hat{c}\Big(\frac{z}{M^2}\Big)^n \Big[1 + \epsilon(z)\Big], \\ |\epsilon(z)| &\leq \exp\Big(\frac{nM^4 e^{-|z|^2 \sin 2\delta/M^4}}{2|z|^2}\Big) - 1. \end{aligned} \tag{31}$$

Because $\hat{a}(-z) = \hat{a}(z)$, the same holds in the reflected cone around the negative axis. Inside the two cones $|\arg(\pm z)| < \pi/4$, which contain the Euclidean and the Lorentzian axes, the skeleton is a monomial up to corrections that fall like a Gaussian. Outside them it is not polynomially bounded. On the imaginary axis $\hat{a}(iy) = \exp[-\frac{n}{2}\int_0^{y^2/M^4}(e^t - 1)\,\mathrm{d}t/t]$ falls faster than any exponential of $y^2$, and in between $|\hat{a}|$ oscillates with growing amplitude (Fig. 7). This rules out a naive Wick rotation, and Sec. 9 explains how the theory is defined instead.

4.7. **Other passive responses.** The closed forms of Sec. 4.6 rest on the choice $\psi = \psi_*$, but the field theory of Secs. 5–9 does not. We call a response *admissible* if its exponent satisfies (S1)–(S4), and show that every admissible response gives a skeleton with the properties used there. Only two numbers change from one response to another, the constant $C_+$ of Eq. (21) and the moment $\mu_\psi = \int_0^\infty t\,(1 - \psi(t))\,\mathrm{d}t$.

*Analyticity and the real axis.* Section 4.5 already shows that $\hat{a} = e^{n\Phi}$ is entire, has no zeros and satisfies $\hat{a}(0) = 1$; its Taylor coefficients are real. By (S2), $\Phi(x) = \int_0^1 \psi(x\tau)\,\mathrm{d}\tau/\tau \geq 0$ for real $x$ of either sign, so $\hat{a} \geq 1$ on the whole real axis. The function $\Phi(x) - \frac{1}{2}\log(1+x^2)$ is continuous on $\mathbb{R}$ and tends to $C_\pm$ as $x \to \pm\infty$, by Eq. (21), so it is bounded, between $m_\psi$ and $M_\psi$ say, and

$$e^{nm_\psi}\big(1+x^2\big)^{n/2} \leq \hat{a}(z) \leq e^{nM_\psi}\big(1+x^2\big)^{n/2}, \qquad z \in \mathbb{R}. \tag{32}$$

For $\psi_*$ this is Eq. (26), with $m_\psi = 0$ and $M_\psi = \gamma_{\mathrm{E}}/2$.

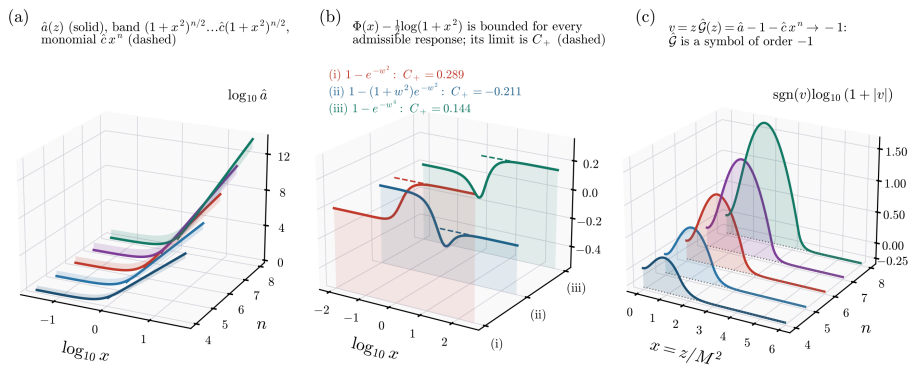



FIGURE 6. The skeleton on the real axis. (a) $\log_{10}\hat a(z)$ for $n=4,\ldots,8$ (solid) inside the band of the two-sided bound (26), $(1+x^2)^{n/2}\le\hat a\le\hat c\,(1+x^2)^{n/2}$ (shaded), with the monomial $\hat c\,x^n$ that $\hat a$ approaches at large $x$ (dashed). (b) $\Phi(x)-\frac12\log(1+x^2)$ for the three admissible responses of Table 1: it is bounded for each of them, as Eq. (32) states, and tends to the constant $C_+$ (dashed) of Eq. (33). (c) $v=z\hat{\mathcal G}(z)=\hat a-1-\hat c\,x^n$ for $n=4,\ldots,8$, plotted as $\mathrm{sgn}(v)\log_{10}(1+|v|)$: it is bounded and tends to $-1$ (dotted), so the nonlocal remainder (29) is a symbol of order $-1$ for every $n$.

*Symbols.* With $H=n\Phi(z/M^2)$ we have $H'(z)=n\psi(x)/z$, and for $|x|\ge1$ this is $n/z$ minus $n(1-\psi(x))/z$, whose derivatives decay faster than any power by (S3). The argument of Sec. 4.6 then gives Eq. (28) with the constant $e^{nM_\psi}$. Put $\hat c=e^{nC_+}$. For $x\ge1$, $\hat a-\hat c\,x^n=\hat c\,x^n(e^{n\rho_+(x)}-1)$, and $\rho_+$ and all its derivatives decay faster than any power, so $\hat a-\hat c x^n$ is a symbol of order 0 and $\hat{\mathcal G}$ is a symbol of order $-1$: Eq. (30) holds as it stands.

*Cones.* Let $z=re^{i\alpha}$ with $|\alpha|\le\theta-\delta$. The function $g(w)=\Phi(w)-\log w$ is analytic in the sector and $g'(w)=-(1-\psi(w))/w$, which by (S4) is bounded by $C_N|w|^{-N-1}$ for every $N$. The integral of $g'$ along the ray from $z$ to infinity therefore converges and is $O(r^{-N})$, so $g(z)$ tends to a limit $L(\alpha)$ along each ray, with an error $O(r^{-N})$. The limit does not depend on $\alpha$, because the integral of $g'$ over an arc of radius $t$ between two rays is at most $\pi t\cdot C_Nt^{-N-1}\to0$. On the positive axis $L(0)=C_+$ by Eq. (21). Hence, in the sector,

$$\hat a(z)=\hat c\Big(\frac{z}{M^2}\Big)^n e^{n\rho(z)},\qquad |\rho(z)|\le C_{N,\delta}\Big|\frac{M^2}{z}\Big|^N, \tag{33}$$

for every $N$, with $\hat c=e^{nC_+}$, and the same holds around the negative axis with $C_-$ and $(-z/M^2)^n$. For $\psi_*$ this is Eq. (31), where the remainder is even Gaussian.

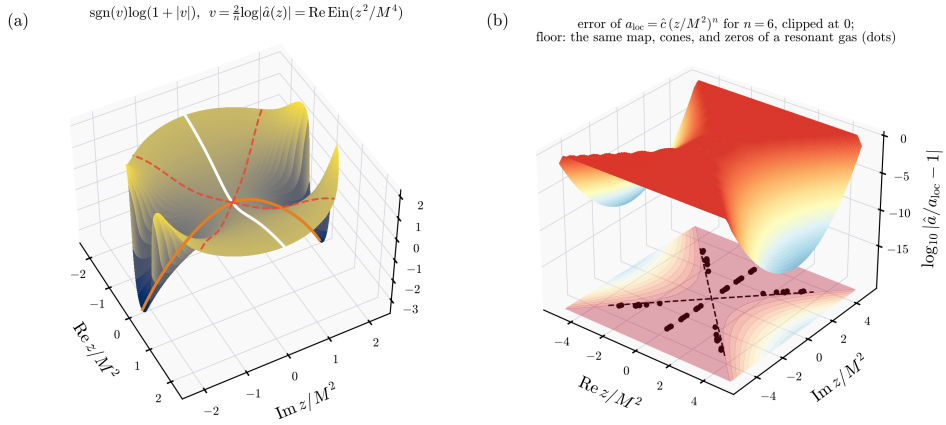



FIGURE 7. The skeleton in the complex $z$ plane. (a) $v = \frac{2}{n}\log|\hat{a}| = \operatorname{Re}\operatorname{Ein}(z^2/M^4)$ over the disk $|z| \le 2.2M^2$, plotted as $\operatorname{sgn}(v)\log(1+|v|)$ to compress its range. Along the real axis (white) it rises like $2\log|z|$; along the imaginary axis (orange) it falls without bound; the dashed red lines are the cone boundaries $\arg z = \pm\pi/4, \pm 3\pi/4$, between which $|\hat{a}|$ oscillates. (b) Accuracy of the local approximation $\hat{c}\,(z/M^2)^n$ for $n = 6$, $\log_{10}|\hat{a}/(\hat{c}z^n/M^{2n}) - 1|$ clipped at 0: the error is exponentially small inside the cones $|\arg(\pm z)| < \pi/4$ (blue), Eq. (31), and of order one outside. The floor repeats the map as a contour plot, with the cone boundaries (dashed) and, as black dots, the zeros that a frozen gas of the resonant factors $2 - e^{-s^2z^2}$ of Eq. (7) would have for sizes in $[0.15, 1]\,M^{-2}$. They all lie outside the cones. A gas of branes obeying (S5) has none.

*The zero-point moment.* Section 8.2 needs the integral of $\Phi(k^2/M^2)$ over four-momentum in dimensional regularization. Its Mellin transform is elementary. For $-2 < \operatorname{Re} s < 0$, exchanging the order of integration in Eq. (19) gives $\int_0^\infty w^{s-1}\Phi(w)\,\mathrm{d}w = -\tilde{\psi}(s)/s$, with $\tilde{\psi}(s) = \int_0^\infty u^{s-1}\psi(u)\,\mathrm{d}u$; the strip is where both integrals converge, since $\psi(u) = O(u^2)$ at small $u$ by passivity and $\psi \to 1$ at large $u$. Splitting $\tilde{\psi}$ at $u = 1$ and writing $\psi = 1 - (1-\psi)$ on $[1,\infty)$ continues it to $\operatorname{Re} s > -2$, $s \neq 0$, and for $\operatorname{Re} s > 0$ the continuation is $\tilde{\psi}(s) = -\int_0^\infty u^{s-1}(1-\psi(u))\,\mathrm{d}u$. At $s = D/2 = 2$,

$$\int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\Phi\Big(\frac{k^2}{M^2}\Big) = \frac{M^4}{32\pi^2}\,\mu_\psi, \qquad \mu_\psi = \int_0^\infty t\,\big(1-\psi(t)\big)\,\mathrm{d}t, \tag{34}$$

| exponent $\psi(w)$ | small $w$ | $C_+$ | $\hat{c}$, $n=4$ | $\mu_\psi$ | $\rho_1/(nM^4)$ | $\theta$ | skeleton exponent $\Phi(x)$ |
|---|---|---|---|---|---|---|---|
| $1-e^{-w^2}$ | $w^2$ | $\gamma_{\mathrm{E}}/2 = 0.2886$ | 3.172 | $\frac{1}{2}$ | $3/(64\pi^2)$ | $\pi/4$ | $\frac{1}{2}\,\mathrm{Ein}(x^2)$ |
| $1-(1+w^2)e^{-w^2}$ | $\frac{1}{2}w^4$ | $(\gamma_{\mathrm{E}}-1)/2 = -0.2114$ | 0.4293 | 1 | $3/(32\pi^2)$ | $\pi/4$ | $\frac{1}{2}[\mathrm{Ein}(x^2)-1+e^{-x^2}]$ |
| $1-e^{-w^4}$ | $w^4$ | $\gamma_{\mathrm{E}}/4 = 0.1443$ | 1.781 | $\frac{\sqrt{\pi}}{4}$ | $3\sqrt{\pi}/(128\pi^2)$ | $\pi/8$ | $\frac{1}{4}\,\mathrm{Ein}(x^4)$ |
| $1-e^{-w}$ | $w$ | fails (S2): unbounded below on the negative axis; the factor decays faster than any power on the timelike axis | | | | | |
| $w^2/(1+w^2)$ | $w^2$ | fails (S1) and (S3): poles at $w=\pm i$, and $1-\psi\sim w^{-2}$, so $G$ and $\Lambda$ run at one loop | | | | | |

TABLE 1. Single-brane responses, given by the exponent $\psi$ of Eq. (6). The first three are admissible; for them $\hat{c}=e^{nC_+}$ is the leading coefficient of the skeleton, $\hat{a}\simeq\hat{c}\,x^n$, $\mu_\psi$ is the moment of Eq. (34), which fixes the mean zero-point energy (89), and $\theta$ is the opening of the sectors of (S4). The last two violate the conditions named in the third column. Values of $\hat{c}$ are for $n=4$; for general $n$, $\hat{c}=e^{nC_+}$. Only the first row arises from the cell model of Sec. 4.2: in the second and third, $1-\psi$ is not a characteristic function.

which is positive and finite by (S2) and (S3).[12] For a single brane of size $\tau/M^2$ no continuation is needed, because the integrand falls off rapidly by (S3) and the integral converges absolutely:

$$\int \frac{\mathrm{d}^4k}{(2\pi)^4}\Big[\psi\Big(\frac{\tau k^2}{M^2}\Big)-1\Big] = -\frac{M^4}{16\pi^2\tau^2}\,\mu_\psi, \tag{35}$$

as follows from the substitution $u=\tau k^2/M^2$, after which the angular integration leaves $\mathrm{d}^4k/(2\pi)^4 \to u\,\mathrm{d}u\,M^4/(16\pi^2\tau^2)$.

*Small momenta.* Passivity gives $\psi(w)=\psi_2 w^2+O(w^3)$ with $\psi_2\geq 0$, so $\hat{a}=1+\frac{n}{2}\psi_2 x^2+O(x^3)$. The leading correction to the kinetic operator is of fourth order in momentum, as in Eq. (27).

Table 1 gives three admissible responses and two that fail, and Fig. 8 shows the sectors of (S4).

4.8. **Fluctuations of a realization.** The skeleton has every property that the field theory needs. We now show that, under (G), almost every realization has them too, with a random constant in place of $\hat{c}$. The two differ by the factor $e^{R_\Pi}$ of Eq. (18), so everything rests on how $R_\Pi$ behaves at large $|z|$.

*The limit at large momentum.* At large real $x$ every brane larger than $1/x$ is saturated and contributes its full strength. This suggests comparing $R_\Pi$

[12] For $\psi_*$, $\Phi(w)=\frac{1}{2}\,\mathrm{Ein}(w^2)$ and $\mu_{\psi_*}=\int_0^\infty te^{-t^2}\mathrm{d}t=\frac{1}{2}$, so the right side is $M^4/(64\pi^2)$. This agrees with Sec. 8.2, where the same integral is done in the variable $k^4/M^4$ with the Mellin transform (87).

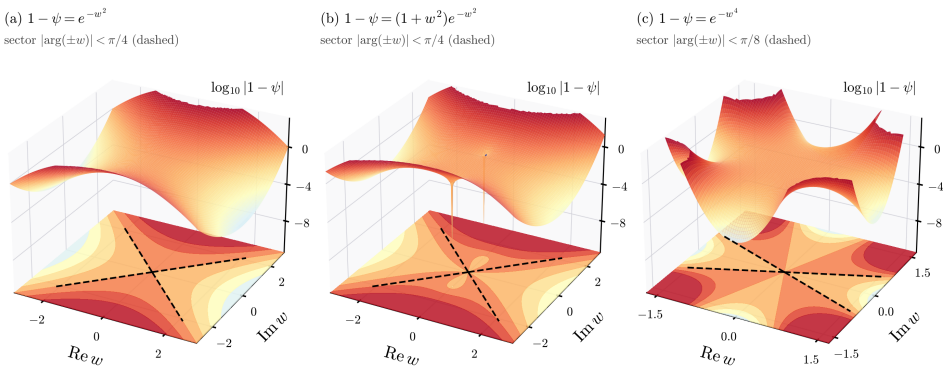



FIGURE 8. The saturation sectors of condition (S4) for the three admissible exponents of Table 1. Each surface is $\log_{10}|1-\psi(w)|$ over the complex $w$ plane, clipped at $-8$ below and $+3$ above, with the floor showing the same quantity as a contour map. Where the surface is low the brane is saturated, $\psi \simeq 1$; the dashed lines are the sector boundaries, $\arg w = \pm\pi/4$ in (a) and (b) and $\arg w = \pm\pi/8$ in (c). The exponent $1-e^{-w^4}$ of panel (c) is also saturated around the imaginary axis, which (S4) does not require.

with the total strength of the gas measured from its mean,

$$\begin{aligned} R_\infty &= \int \vartheta(s)\,(\Pi-\nu)(\mathrm{d}s) \\ &= \lim_{\eta\to 0}\Big[\sum_{s\in\Pi,\ s>\eta/M^2} \vartheta(s) - n\log\frac{1}{\eta}\Big]. \end{aligned} \tag{36}$$

The sum in brackets has mean $\int_{\eta/M^2}^{1/M^2} \vartheta\,\mathrm{d}\nu = n\log(1/\eta)$ by Eq. (11), and by Eq. (15) its fluctuation has variance $n[V(1)-V(\eta)]$. Under (G) $V(1)$ is finite, and Sec. 4.4 gives the limit almost surely, with

$$\mathbb{E}R_\infty = 0, \qquad \mathbb{E}R_\infty^2 = nV(1). \tag{37}$$

The difference

$$R_\Pi(z) - R_\infty = \int \vartheta(s)\big[\psi(sz)-1\big]\,(\Pi-\nu)(\mathrm{d}s) \tag{38}$$

is carried by the branes that are not saturated at $z$.

*Mean-square decay.* By Eq. (15), for complex $z$ and $w = z/M^2$,

$$\mathbb{E}\big|R_\Pi(z)-R_\infty\big|^2 = n\int_0^1 \vartheta(\tau)\,\big|1-\psi(w\tau)\big|^2\,\frac{\mathrm{d}\tau}{\tau}. \tag{39}$$

Let $|w|\ge 1$ and $|\arg(\pm w)|\le\theta-\delta$. For $\tau\le 1/|w|$ the point $w\tau$ lies in the closed unit disk, where $|1-\psi|\le B$, and this part of the integral is at most $nB^2V(1/|w|)\le nB^2C|w|^{-\sigma}$ by (G). For $\tau > 1/|w|$

condition (S4) gives $|1-\psi(w\tau)| \le C_N|w\tau|^{-N}$. Choose $2N > \sigma$ and integrate by parts in $\int_{1/|w|}^{1}\tau^{-2N}\,\mathrm{d}V(\tau)$: the boundary term at $\tau = 1/|w|$ is negative, the one at $\tau = 1$ is $V(1)$, and the remaining integral is $2N\int\tau^{-2N-1}V(\tau)\,\mathrm{d}\tau \le 2NC|w|^{2N-\sigma}/(2N-\sigma)$. Multiplying by $C_N^2|w|^{-2N}$, this part is also $O(|w|^{-\sigma})$, and

$$\begin{aligned}\mathbb{E}\big|R_\Pi(z)-R_\infty\big|^2 &\le K_\delta\Big|\frac{M^2}{z}\Big|^{\sigma},\\ |z|\ge M^2,\quad &|\arg(\pm z)|\le\theta-\delta.\end{aligned}\tag{40}$$

*Almost-sure decay.* Equation (40) controls one point at a time. To control a whole sector we use the subharmonicity of $|f|^2$ for analytic $f$: $|f(w)|^2$ is at most the mean of $|f|^2$ over any disk centered at $w$ on which $f$ is analytic. Put $f(w) = R_\Pi(M^2w) - R_\infty$, an entire function, fix $\delta$, and let $A_k$ be the part of the sector $|\arg w| \le \theta-\delta$ with $2^k \le |w| \le 2^{k+1}$, $k \ge 1$. A disk of radius $r_k = 2^{k-1}\sin(\delta/2)$ around a point of $A_k$ lies in the region $A_k'$ where $|\arg w| \le \theta-\delta/2$ and $2^{k-1} \le |w| \le 2^{k+2}$. Hence

$$\sup_{A_k}|f|^2 \le \frac{1}{\pi r_k^2}\int_{A_k'}|f|^2\,\mathrm{d}A,$$

and taking expectations with Eq. (40) in the wider sector,

$$\mathbb{E}\sup_{A_k}|f|^2 \le \frac{\operatorname{area}(A_k')}{\pi r_k^2}\,K_{\delta/2}\,2^{-(k-1)\sigma} \le K'\,2^{-k\sigma},$$

since the ratio of the area of $A_k'$ to $r_k^2$ does not depend on $k$. Chebyshev's inequality gives $\mathrm{P}\big(\sup_{A_k}|f| > 2^{-k(\sigma/2-\varepsilon)}\big) \le K'2^{-2k\varepsilon}$, which is summable in $k$ for every $\varepsilon > 0$. By the Borel–Cantelli lemma, almost surely $\sup_{A_k}|f| \le 2^{-k(\sigma/2-\varepsilon)}$ for all but finitely many $k$, and on the remaining annuli $f$ is bounded because it is continuous. Taking $\delta$ and $\varepsilon$ rational, and treating the sectors around the negative axis in the same way, we conclude that almost surely

$$\begin{aligned}\big|R_\Pi(z)-R_\infty\big| &\le C_\Pi\Big|\frac{M^2}{z}\Big|^{\sigma/2-\varepsilon},\\ |z|\ge M^2,\quad &|\arg(\pm z)|\le\theta-\delta,\end{aligned}\tag{41}$$

for every $\delta \in (0,\theta)$ and every $\varepsilon > 0$, with $C_\Pi$ finite and depending on the realization, on $\delta$ and on $\varepsilon$. For the power law below $\sigma = \zeta$, and the exponent matches, up to $\varepsilon$, the decay of the standard deviation of $R_\Pi(z) - R_\infty$, which falls like $|z|^{-\zeta/2}$, Eq. (49).

*The form factor of a realization.* With Eqs. (18) and (33), Eq. (41) says that almost surely, in the same sectors,

$$\begin{aligned}a_\Pi(z) = c_\Pi\Big(\frac{\pm z}{M^2}\Big)^n e^{\eta_\Pi(z)},\qquad c_\Pi = e^{nC_\pm + R_\infty},\\ |\eta_\Pi(z)| \le C_\Pi'\Big|\frac{M^2}{z}\Big|^{\sigma/2-\varepsilon}.\end{aligned}\tag{42}$$

A realization has the asymptotics of the skeleton, with a random coefficient. Cauchy's estimate on a disk of radius $\frac{1}{2}|z|\sin(\delta/2)$ around $z$, which stays inside the wider sector, bounds the derivatives: $|\eta_\Pi^{(j)}(z)| \le C_j|z|^{-j}|M^2/z|^{\sigma/2-\varepsilon}$. On the real axis $R_\Pi$ is continuous and tends to $R_\infty$ at both ends, so $\|R_\Pi\| = \sup_{\mathbb{R}}|R_\Pi|$ is finite, and Eq. (32) gives

$$e^{nm_\psi - \|R_\Pi\|}\big(1+x^2\big)^{n/2} \le a_\Pi(z) \le e^{nM_\psi + \|R_\Pi\|}\big(1+x^2\big)^{n/2} \tag{43}$$

for every real $z$.

Now let $\sigma > 4$, as (G) requires, fix $\varepsilon$ with $0 < \varepsilon < \sigma/2 - 2$, and put

$$m_\sigma = \max\big(0,\, n - \sigma/2 + \varepsilon\big), \qquad 0 \le m_\sigma < n-2. \tag{44}$$

Then for $|x| \ge 1$

$$a_\Pi(z) - c_\Pi|x|^n = c_\Pi|x|^n\big(e^{\eta_\Pi} - 1\big) = O\big(|x|^{\,n-\sigma/2+\varepsilon}\big),$$

and each derivative falls by one more power of $x$; so $a_\Pi - c_\Pi x^n$ is a symbol of order $m_\sigma$, and $a_\Pi$ obeys the derivative bounds (28) of the skeleton. When $\sigma > 2n$ one may take $m_\sigma = 0$, and $a_\Pi - c_\Pi x^n$ stays bounded, as it does for the skeleton; for $4 < \sigma \le 2n$ it grows, but more slowly than $x^{n-2}$. The nonlocal remainder of the realization,

$$\mathcal{G}(z) = \frac{a_\Pi(z) - 1}{z} - \frac{c_\Pi\, z^{n-1}}{M^{2n}}, \tag{45}$$

is entire, equals $-1/z + c_\Pi x^n(e^{\eta_\Pi} - 1)/z$ for $x \ge 1$, and obeys

$$\big|\mathcal{G}^{(j)}(z)\big| \le C_j M^{-2-2j}(1+x)^{m_\sigma - 1 - j}, \qquad z \ge 0, \tag{46}$$

almost surely; the same bounds hold with $|z|$ in place of $z$ in every closed subsector of the sectors of (S4). So $\mathcal{G}$ is a symbol of order $m_\sigma - 1$; for the skeleton it is a symbol of order $-1$, with corrections that fall faster than any power. The field theory of Secs. 6–9 uses only that the order is below $n-3$, and this is where the exponent 4 in (G) comes from. The fluctuation of the gas at momentum $p$ must fall faster than $p^{-4}$, so that the one-loop integrals of Secs. 7.2 and 8.2 converge; the growth that $a_\Pi - c_\Pi x^n$ may then have is absorbed by the vertex bound of Sec. 6.3.

*Identical branes.* Without (G) the conclusion fails, and it fails for the most obvious gas. Let all branes have the same strength $\vartheta_0$, so that $\nu = (n/\vartheta_0)\,\mathrm{d}s/s$. For real $x$ the sum $R_\Pi(x)$ is an ordinary compensated sum, and Eq. (14) with $f = e^{iu\vartheta_0\psi} - 1$ gives its characteristic function,

$$\Big|\mathbb{E}\, e^{iuR_\Pi(x)}\Big| = \exp\Big[\frac{n}{\vartheta_0}\int_0^x \big(\cos(u\vartheta_0\psi(t)) - 1\big)\frac{\mathrm{d}t}{t}\Big].$$

As $x \to \infty$ the integrand tends to $\cos(u\vartheta_0) - 1$, which is negative unless $u\vartheta_0 \in 2\pi\mathbb{Z}$, and $\int \mathrm{d}t/t$ diverges, so the right side tends to 0 for all such $u$. If $R_\Pi(x)$ converged in distribution, its characteristic function would converge to a function continuous at $u = 0$ [25]; the limit here is 1 at $u = 0$ and 0 around it. So $R_\Pi(x)$ has no limit in distribution, and a fortiori no limit almost surely: $a_\Pi(x)/x^n$ does not converge, and no constant $c$ makes $a_\Pi - c\,x^n$ a

symbol of order 0. For $c > 0$ this would force $R_\Pi(x)$ to converge. For $c \le 0$ it would force $a_\Pi(x)$ to stay bounded, hence $R_\Pi(x) \le C - n \log x$ for large $x$, so that only a bounded number of branes could have $sx$ above a fixed $t_0$ beyond which $\psi \ge \frac{1}{2}$; their number is Poisson with mean $(n/\vartheta_0)\log(x/t_0)$, and the probability of that event tends to zero. The variance shows the mechanism,

$$\begin{aligned} \mathbb{E}R_\Pi(x)^2 &= n\vartheta_0 \Psi_*(x), \\ \Psi_*(x) &= \int_0^1 \psi_*(x\tau)^2 \,\frac{\mathrm{d}\tau}{\tau} = \mathrm{Ein}(x^2) - \tfrac{1}{2}\,\mathrm{Ein}(2x^2) \\ &= \log|x| + \tfrac{1}{2}(\gamma_{\mathrm{E}} - \log 2) + O\big(e^{-x^2}\big), \end{aligned} \tag{47}$$

from $(1-e^{-u})^2 = 2(1-e^{-u}) - (1-e^{-2u})$, the substitution $u = x^2\tau^2$ and Eq. (25). The saturated strength performs a random walk in $\log x$, one step of size $\vartheta_0$ for each of the roughly $(n/\vartheta_0)\log x$ branes larger than $1/x$. Such a gas still gives an entire form factor without zeros, by (S5), and $\log a_\Pi(x)/\log x \to n$ almost surely by the strong law of large numbers. But the local split of Sec. 7.1, on which the one-loop analysis rests, does not exist for it.

*Statistics of a realization.* For the power law (13) and $\psi_*$ the moments on the real axis are incomplete gamma functions. With $\gamma(b,y) = \int_0^y t^{b-1}e^{-t}\mathrm{d}t$ and $(1-e^{-u})^2 = 1 - 2e^{-u} + e^{-2u}$,

$$\mathbb{E}R_\Pi(x)^2 = n\vartheta_0\Big[\frac{1}{\zeta} - \frac{\gamma(\frac{\zeta}{2},x^2)}{x^\zeta} + \frac{\gamma(\frac{\zeta}{2},2x^2)}{2\,(2x^2)^{\zeta/2}}\Big], \tag{48}$$

$$\mathbb{E}\big|R_\Pi(x) - R_\infty\big|^2 = \frac{n\vartheta_0}{2}\,\frac{\gamma(\frac{\zeta}{2},2x^2)}{(2x^2)^{\zeta/2}} \le \frac{n\vartheta_0\,\Gamma(\frac{\zeta}{2})}{2\,(2x^2)^{\zeta/2}}, \tag{49}$$

and $\mathbb{E}R_\infty^2 = n\vartheta_0/\zeta$. The leading coefficient $c_\Pi = \hat{c}\,e^{R_\infty}$ has $\mathbb{E}\log c_\Pi = \log\hat{c}$, and by Eq. (16), with the substitution $t = \vartheta_0\tau^\zeta$,

$$\mathbb{E}\,c_\Pi = \hat{c}\,\exp\Big[\frac{n}{\zeta}\int_0^{\vartheta_0}\frac{e^t - 1 - t}{t^2}\,\mathrm{d}t\Big]. \tag{50}$$

The image of $\nu$ under $s \mapsto \vartheta(s)$ is $(n/\zeta)\,\mathrm{d}t/t^2$ on $(0,\vartheta_0]$, so $R_\infty$ is a centered, totally skewed variable whose Lévy measure is that of a stable law of index one, cut off at $\vartheta_0$. Its jumps are positive, and since $e^{-\lambda t} - 1 + \lambda t \le \frac{1}{2}\lambda^2 t^2$ for $t \ge 0$, Eq. (16) gives $\mathbb{E}e^{-\lambda R_\infty} \le e^{\lambda^2 nV(1)/2}$ and hence, for any gas with bounded strengths that obeys (G),

$$\mathrm{P}\big(c_\Pi < \hat{c}\,e^{-y}\big) \le \exp\Big[-\frac{y^2}{2nV(1)}\Big]. \tag{51}$$

The coefficient of a realization is bounded away from zero except with Gaussian probability. Table 2 gives the numbers for $\vartheta_0 = \log 2$, a brane at the top of the range doubling the kinetic operator when saturated, and $\zeta = 2n+1$, the smallest integer for which $a_\Pi - c_\Pi x^n$ stays bounded, and in its first row for $\zeta = 5$, the smallest integer allowed by (G). Figure 9 shows sample paths

| $n$ | $\zeta$ | $(n\vartheta_0/\zeta)^{1/2}$ | $\mathbb{E}\, e^{R_\infty}$ | 95% bound | $x_*$ | $\delta\rho/\rho_1$ |
|---|---|---|---|---|---|---|
| 4 | 5 | 0.745 | 1.368 | 0.162 | 12.66 | 0.833 |
| 4 | 9 | 0.555 | 1.190 | 0.257 | 4.47 | 0.372 |
| 5 | 11 | 0.561 | 1.195 | 0.253 | 3.74 | 0.281 |
| 6 | 13 | 0.566 | 1.198 | 0.250 | 3.35 | 0.227 |
| 7 | 15 | 0.569 | 1.201 | 0.249 | 3.11 | 0.190 |
| 8 | 17 | 0.571 | 1.202 | 0.247 | 2.97 | 0.163 |

TABLE 2. Fluctuations of a realization for the power law (13) with $\zeta = 5$ (first row) and $\zeta = 2n+1$, $\vartheta_0 = \log 2$ and $\psi = \psi_*$. Columns: standard deviation of $\log(c_\Pi/\hat{c}) = R_\infty$; mean of $c_\Pi/\hat{c}$, Eq. (50); lower bound on $c_\Pi/\hat{c}$ that holds with probability at least 0.95, from Eq. (51); the value $x_*$ beyond which the root-mean-square remainder (49) is below $10^{-3}$; and the relative standard deviation of the zero-point energy, Eq. (90). All entries are closed-form expressions evaluated numerically.

of $R_\Pi$ for the two kinds of gas, the root-mean-square remainder (49) as a function of $x$ and $\zeta$, and the remainder of one realization over the complex plane.

4.9. **A unitary realization of the coherent multiplier.** The scalar transmission factor can be computed as a matrix coefficient of an explicit scattering channel. Fix a size $s > 0$ and a mean count $\vartheta > 0$. Take the auxiliary Hilbert space to be the symmetric Fock space

$$\mathcal{H} = \bigoplus_{m\geq 0} L^2_{\mathrm{sym}}(\mathbb{R}^m, \mathrm{d}^m y), \qquad \phi_s(y) = (4\pi s^2)^{-1/4} e^{-y^2/(8s^2)}.$$

The zero-particle sector is $\mathbb{C}$. In the $m$-particle sector let $D$ be multiplication by $y_1 + \cdots + y_m$, with $D = 0$ in the zero-particle sector. The direct sum is self-adjoint on its natural multiplication domain. The vector with components

$$\Omega_m = e^{-\vartheta/2} \frac{\vartheta^{m/2}}{\sqrt{m!}}\, \phi_s^{\otimes m}$$

has squared norm one, since $\|\phi_s\|_2 = 1$ and the squared sector weights sum to one. For real $z$, the operator $U(z) = e^{-izD}$ is unitary. Gaussian integration gives

$$\int_{\mathbb{R}} |\phi_s(y)|^2 e^{-izy}\, \mathrm{d}y = e^{-s^2 z^2}.$$

For example differentiation of the integral in $z$ and integration by parts in $y$ give $f'(z) = -2s^2 z f(z)$ with $f(0) = 1$. Therefore

$$\langle \Omega, U(z)\Omega \rangle = e^{-\vartheta} \sum_{m\geq 0} \frac{\vartheta^m}{m!} e^{-m s^2 z^2} = \exp[-\vartheta(1 - e^{-s^2 z^2})]. \tag{52}$$

The series converges normally on every complex disk and gives the entire extension of the matrix coefficient. With $P = |\Omega\rangle\langle\Omega|$, the forward channel

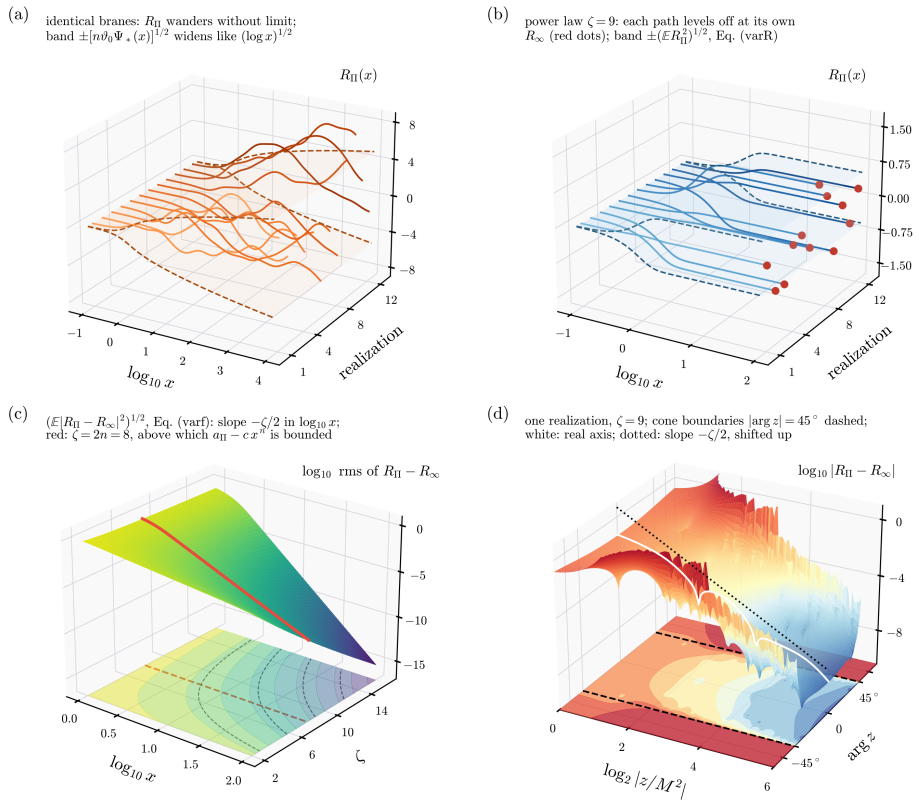



FIGURE 9. Frozen-gas fluctuations for $n = 4$ and $\vartheta_0 = \log 2$. (a) Identical branes and the $(\log x)^{1/2}$ fluctuation scale. (b) Fine graining with $\zeta = 9$ and the limit $R_\infty$. (c) The root-mean-square remainder of Eq. (49), with slope $-\zeta/2$ on logarithmic axes. (d) A complex realization showing the sector of decay. In (b) and (d), branes below $0.3M^{-2}$ are simulated by their Gaussian limit.

is $PU(z)P = t(z)P$, and $\|(I-P)U(z)\Omega\|^2 = 1 - |t(z)|^2$. This accounts for the coherent attenuation by leakage into orthogonal auxiliary states. For finitely many independently prepared auxiliary spaces, the product unitary and product vector give precisely the product of the single-brane coefficients.

4.10. **Absolute convergence for one frozen gas.** Write $S = M^{-2}$ for the largest allowed size. Condition on a realization of the brane sizes and strengths $\{(s_i, \vartheta_i)\}_i$. Independently for each brane take $N_i \sim \text{Poisson}(\vartheta_i)$ and Gaussian shifts $Y_{ij}$ with variance $2s_i^2$. The expected sum of the absolute shifts conditional on the brane realization is

$$\mathbb{E}\left[\sum_i \sum_{j=1}^{N_i} |Y_{ij}| \,\middle|\, \Pi\right] = \frac{2}{\sqrt{\pi}} \sum_i \vartheta_i s_i.$$

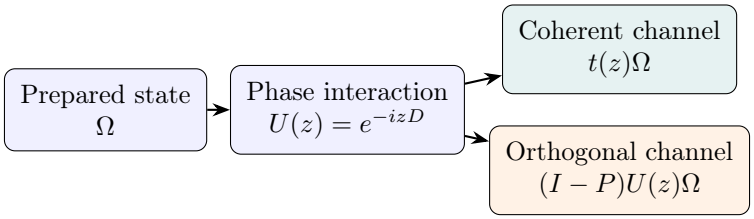



FIGURE 10. The same unitary interaction gives the coherent transmission amplitude and its complementary probability.

By the intensity identity (11), the expectation of the last sum over brane realizations is $\int_0^S ns\,\mathrm{d}s/s = nS$. Tonelli's theorem therefore gives absolute convergence of the full shift $X = \sum_i \sum_{j=1}^{N_i} Y_{ij}$ almost surely. For almost every frozen realization, $\sum_i \vartheta_i s_i^2 \le S \sum_i \vartheta_i s_i < \infty$. For real $b$, the exponential moment of the absolute Gaussian shift at size $s_i$ is $1+O_{b,S}(s_i)$ when $b \ge 0$. Indeed $|Y_{ij}|$ has the law $s_i|Z|$, where $Z \sim \mathcal{N}(0,2)$, and $e^{bs_i|Z|}-1 \le bs_i|Z|e^{bS|Z|}$. Applying the Poisson generating formula and then monotone convergence shows that $\sum_{i,j} |Y_{ij}|$ has every positive exponential moment conditional on the same realization. Dominated convergence and Eq. (52) now give, for complex $z$,

$$\mathbb{E}[e^{-izX} \mid \Pi] = \exp\left[-\sum_i \vartheta_i (1 - e^{-s_i^2 z^2})\right] = a_\Pi(z)^{-1}. \tag{53}$$

The series in the exponent converges normally because $1 - e^{-s_i^2 z^2} = O(s_i^2)$ uniformly on each disk. This realizes the frozen form factor as a coherent channel with a convergent total shift. The logarithmic size law and the fine-graining assumptions still govern its high-momentum power and remainder.

The deterministic skeleton has a separate realization as a Poisson sum of Gaussian jumps with intensity $\int_0^S n\,\mathcal{N}(0, 2s^2)(\mathrm{d}y)\,\mathrm{d}s/s$. For $y \ne 0$ its density is

$$\frac{n}{2\sqrt{\pi}} \int_0^S s^{-2} e^{-y^2/(4s^2)}\,\mathrm{d}s = \frac{n}{2|y|}\,\mathrm{erfc}\left(\frac{|y|}{2S}\right).$$

The substitution $u = |y|/(2s)$ gives the equality. The integral of $|y|$ against this intensity is $2nS/\sqrt{\pi}$, obtained by integrating the absolute Gaussian moment over size. Thus the jumps sum absolutely almost surely. The Poisson exponential formula, first above a size cutoff and then by dominated convergence, gives the characteristic function

$$\exp\left[-n \int_0^S (1 - e^{-s^2 z^2}) \frac{\mathrm{d}s}{s}\right] = \exp[-(n/2)\,\mathrm{Ein}(S^2 z^2)].$$

The skeleton results from the mean logarithm. It is distinct from averaging the inverse multiplier over frozen realizations, as computed in Sec. 4.12.

4.11. **A spectral support obstruction for the shift generator.** In any nonempty realization, choose one brane of positive strength and condition on its having exactly one cell. This event has positive probability, and its shift is a nondegenerate Gaussian independent of the sum of all other shifts. For every nonempty open interval $I$, the conditional probability that the total shift belongs to $I$, given the other shifts and this event, is strictly positive. Thus the conditional law of $X$ given the brane configuration has support all of $\mathbb{R}$. The same proof applies to the single-brane multiplier. For the skeleton, split off any size interval strictly away from zero. Its positive finite jump intensity gives an event with one Gaussian jump there, and the other size intervals are independent.

For a self-adjoint $D_0$ and a unit vector $\Omega_0$, the spectral theorem expresses $\langle \Omega_0, e^{-izD_0}\Omega_0\rangle$ as the Fourier transform of its probability spectral measure. Uniqueness of Fourier transforms for finite measures identifies that measure with the shift law if the coefficient equals Eq. (53) for real $z$.[13] No such $D_0$ can be bounded below or above. Multiplying the coefficient by a deterministic phase $e^{-idz}$ merely translates its spectral measure by $d$ and does not change this conclusion. The generator here is conjugate to virtuality $z$, and the conclusion is about the support of proper-time shifts. Reflection positivity of the gravitational curvature is a separate condition tested in Sec. 9.6.

4.12. **Exact averaged inverse moments for power-law fine graining.** For the response $\psi_*(y) = 1 - e^{-y^2}$, take the explicit size law (13) with $\zeta > 0$. The integer $n$ need not be at least four for this subsection. Let $r > 0$ and set $G_r(z) = \mathbb{E}[a_\Pi(z)^{-r}]$, where the complex power means $\exp(-r \log a_\Pi)$ with the logarithm supplied by the gas and normalized to zero at the origin. The Laplace functional gives

$$G_r(z) = \hat{a}(z)^{-r} \exp J_r(z), \tag{54}$$

$$J_r(z) = \int_0^1 \Big( e^{-r\vartheta(\tau)\psi_*(\tau z/M^2)} - 1 + r\vartheta(\tau)\psi_*(\tau z/M^2) \Big)\, \nu(\mathrm{d}\tau). \tag{55}$$

In particular $G_r$ is entire and zero-free, $G_r(0) = 1$, and the averaged scalar covariance $G_1(z)/z$ has the single massless pole with unit residue.

Here is a justification for the complex Laplace functional with infinitely many branes. On any disk $|z| \le A$, $\big|\psi_*(\tau z/M^2)\big| \le C_A \tau^2$. The random majorant $Y = \int \vartheta(\tau)\tau^2 \Pi(\mathrm{d}\tau)$ has exponential moments of every positive order. Indeed $\vartheta\tau^2 \le \vartheta_0$ and

$$\int_0^1 (e^{b\vartheta\tau^2} - 1)\, \nu(\mathrm{d}\tau) \le b e^{b\vartheta_0} \int_0^1 \vartheta\tau^2\, \nu(\mathrm{d}\tau) = \tfrac{1}{2} n b e^{b\vartheta_0}.$$

The Laplace functional first applies to $\tau \ge \epsilon$, and monotone convergence proves the exponential-moment estimate at $\epsilon = 0$. Dominated convergence

[13] One proof of uniqueness multiplies the two transforms by the Fourier transform of a Gaussian. Fourier inversion identifies the smoothed densities, and shrinking the Gaussian gives equality of the original measures.

then gives $G_r$ on each disk. The deterministic exponent on the right converges normally on disks, by the same bounds and $|e^{-u}-1+u| \le |u|^2 e^{|u|}/2$. Its exponential gives the entire zero-free extension.

For $x = z/M^2 \ge 0$, the real function $b(u) = e^{-u} - 1 + u$ is increasing on $u \ge 0$ and $0 \le b'(u) \le u$. Therefore

$$0 \le J_{r,\infty} - J_r(M^2 x) \le r^2 n \vartheta_0 \int_0^1 \tau^{\zeta-1} e^{-\tau^2 x^2} \mathrm{d}\tau \le \frac{r^2 n \vartheta_0}{2} \Gamma(\zeta/2) x^{-\zeta}, \quad x > 0,$$

where $J_{r,\infty} = \int_0^1 b(r\vartheta)\,\mathrm{d}\nu \le r^2 n \vartheta_0/(2\zeta)$. For example the middle inequality follows by integrating $b'$ from $r\vartheta\psi_*$ to $r\vartheta$ and replacing it by $r\vartheta$. Consequently

$$G_r(M^2 x) = e^{-rn\gamma_{\mathrm{E}}/2 + J_{r,\infty}} x^{-rn} \big(1 + O(x^{-\zeta})\big), \qquad x \to +\infty. \tag{56}$$

This result controls annealed inverse moments without interchanging an almost-sure asymptotic limit with expectation. It concerns the quadratic law. Averaging interacting amplitudes or replacing $a_\Pi^{-1}$ by its mean inside every Feynman graph would require a different calculation. For instance two factors at $z, w > 0$ have the joint Laplace exponent with $r\vartheta\psi_*(\tau z/M^2)$ replaced by the sum of the two exponents. Unless one exponent vanishes, the joint expectation is strictly greater than the product, since the difference of the logarithms is the integral of $(1-e^{-u})(1-e^{-v})$.

4.13. **What the field theory uses.** From here on the form factor is that of one realization, and we drop the subscript: $a = a_\Pi$ and $c = c_\Pi$, with the remainder $\mathcal{G}$ of Eq. (45). Sections 5–9 use $a$ only through the following properties, each proved above for almost every realization of a gas obeying A1 and A2: $a$ is entire, has no zeros, satisfies $a(0) = 1$ and has real Taylor coefficients (Sec. 4.5); on the real axis it obeys the two-sided bound (43) and the derivative bounds (28); $\mathcal{G}$ is a symbol of order $m_\sigma - 1$ and $a - cx^n$ a symbol of order $m_\sigma$, with $0 \le m_\sigma < n - 2$, Eqs. (44) and (46); and in the cones of (S4) the asymptotics (42) hold, with $|a(z)| \ge \frac{c}{2}|z/M^2|^n$ for large $|z|$. Where a statement depends on the realization, the dependence is through $c$, through the bound $C_\Pi$ of Eq. (41), or through $\|R_\Pi\|$; we point it out where it matters. The skeleton enters directly in two places only, the zero-point energy of Sec. 8.2, whose mean it gives, and the Newtonian numbers of Sec. 10. Table 3 collects the properties of the skeleton and of a realization side by side.

The two premises are the only places where the brane picture enters. Assumption A1 is a statement about a single brane, (S1)–(S5), which Sec. 4.2 derives from the cell model (M1)–(M3); the model itself is not derived from a brane action. Assumption A2 is a statement about the gas: independence, the scale-invariant strength (11) and the fine graining (G). No average over realizations is taken anywhere, and no limit of many regions or many samples is needed: the field theory below is that of one frozen gas. Table 4 records where each input is used.

| property | skeleton $\hat{a}$ | realization $a = \hat{a}\, e^{R_\Pi}$ | used in |
|---|---|---|---|
| analyticity | entire, no zeros, $\hat{a}(0) = 1$, $\hat{a}(-z) = \hat{a}(z)$, real coefficients | entire, no zeros by (S5), $a(0) = 1$, $a(-z) = a(z)$, real coefficients | pole content, Sec. 5.3 |
| real axis | $(1+x^2)^{n/2} \le \hat{a} \le \hat{c}\,(1+x^2)^{n/2}$ | same, with factors $e^{\mp\|R_\Pi\|}$, Eq. (43) | propagator bound (67) |
| large $\|x\|$, real | $\hat{a} = \hat{c}\,\|x\|^n \exp[\frac{n}{2}E_1(x^2)]$ | $a = c\,\|x\|^n[1 + O(\|x\|^{-\sigma/2+\varepsilon})]$, $c = \hat{c}\,e^{R_\infty}$ | local theory, Sec. 7.1 |
| small $x$ | $\hat{a} = 1 + \frac{n}{2}x^2 + \frac{n(n-1)}{8}x^4 + O(x^6)$ | $a = 1 + b_\Pi x^2 + O(x^4)$, $b_\Pi = \sum \vartheta(s) s^2 M^4$, $\mathbb{E}\, b_\Pi = \frac{n}{2}$ | Newtonian limit, Sec. 10 |
| remainder | $\hat{\mathcal{G}}$ of order $-1$; $\hat{a} - \hat{c}x^n$ exponentially small | $\mathcal{G}$ of order $m_\sigma - 1$; $a - cx^n$ of order $m_\sigma < n-2$, Eq. (44) | vertex bound, Sec. 6.3 |
| cones | $\hat{a} = \hat{c}\,(\pm z/M^2)^n[1+\epsilon]$, $\|\epsilon\| \lesssim e^{-\mathrm{Re}\, z^2/M^4}$ | $a = c\,(\pm z/M^2)^n e^{\eta_\Pi}$, $\|\eta_\Pi\| \le C\|z/M^2\|^{-\sigma/2+\varepsilon}$ | unitarity, Sec. 9 |
| imaginary axis | $\hat{a}(iy) \to 0$ faster than any exponential | not controlled, not used | Sec. 9 |
| constant | $\hat{c} = 3.172, 4.234, 5.650, 7.540, 10.063$ for $n = 4, \dots, 8$ | $\log c$ has mean $\log \hat{c}$, variance $nV(1)$ | one-loop tuning, Sec. 7.4 |
| other responses | same with $\hat{c} = e^{nC_+}$, Sec. 4.7 | same with $c = e^{nC_+ + R_\infty}$ | Sec. 4.7 |

TABLE 3. Properties of the skeleton $\hat{a} = \exp[\frac{n}{2}\,\mathrm{Ein}(x^2)]$ for $\psi_*$, with $\hat{c} = e^{n\gamma_E/2}$, and of the form factor $a$ of a realization, which hold almost surely for a gas obeying (G). Here $x = z/M^2$, $\varepsilon > 0$ is arbitrary and $C$ depends on the realization. All entries are derived in Secs. 4.5–4.8; the last column says where each is used.

## 5. Action, propagator and spectrum

5.1. **The action.** Let $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$ with $\kappa^2 = 32\pi G$, and $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$. The Euclidean action is

$$S = -\frac{2}{\kappa^2}\int \mathrm{d}^4x\sqrt{g}\,\big[R + G_{\mu\nu}F(\Box)R^{\mu\nu}\big] + S_{\mathrm{q}},$$

$$F(\Box) = \frac{a(-\Box) - 1}{\Box}, \quad (57)$$

where $S_{\mathrm{q}}$ collects three operators quartic in curvature, introduced in Sec. 7.4. Being quartic, they contribute nothing to the quadratic or cubic terms of the action around flat space. Since $a(0) = 1$, $F$ is an entire function of $\Box$. The combination $G_{\mu\nu}F(\Box)R^{\mu\nu}$ is the one used in Refs. [8, 11], and the reason for it appears below: around flat space it multiplies the whole Einstein operator by one function.

5.2. **Linearized curvature.** We use the spin projectors of Barnes and Rivers [27, 28]. With $\theta_{\mu\nu} = \delta_{\mu\nu} - p_\mu p_\nu/p^2$ and $\omega_{\mu\nu} = p_\mu p_\nu/p^2$, in $D$ dimensions,

$$P^{(2)}_{\mu\nu\alpha\beta} = \tfrac{1}{2}(\theta_{\mu\alpha}\theta_{\nu\beta} + \theta_{\mu\beta}\theta_{\nu\alpha}) - \tfrac{1}{D-1}\theta_{\mu\nu}\theta_{\alpha\beta},$$
$$(P^{(0)}_s)_{\mu\nu\alpha\beta} = \tfrac{1}{D-1}\theta_{\mu\nu}\theta_{\alpha\beta}.$$

They are orthogonal idempotents, and both are annihilated by contraction with $p^\mu$. For a metric perturbation $h_{\mu\nu}$ with Fourier transform $h(p)$, the

| input | used for | if dropped |
|---|---|---|
| cells, (M1)–(M3) | derive (S1)–(S5) and $\psi = \psi_*$ | A1 must be assumed |
| A1, (S1) | $a$ entire with $a(0) = 1$ | $F(\Box)$ not entire |
| A1, (S2) | $a \geq 1$ on both half-axes | no polynomial bound on one half-axis |
| A1, (S3) | skeleton corrections beyond all powers | $G$ and $\Lambda$ run at one loop |
| A1, (S4) | cone asymptotics (33), (42) | cutting rules not established |
| A1, (S5) | no zeros in any realization | complex pole pairs, Eq. (7) |
| A2, independence | Poisson gas | formulas of Sec. 4.4 lost |
| A2, Eq. (11) | growth $\lvert x\rvert^n$ with integer $n$ | no fixed power |
| A2, (G) | $a - cx^n = O(x^{m_\sigma})$, Eq. (46); finite $\delta\rho_\Pi$ | $\zeta \leq 4$: $\delta\rho_\Pi$ diverges; identical branes: no limit of $a/x^n$ |
| $\psi = \psi_*$ | closed forms; numbers of Sec. 10 | other constants, Table 1 |
| $n$ integer | local theory polynomial | no local split |
| $n \geq 4$ | every graph with $L \geq 2$ converges | $n = 3$: $\bar{\omega}(2) = 0$ |

TABLE 4. Inputs of the paper, what each is used for, and what fails without it. Everything not listed here is derived, for almost every realization of the gas.

linearized Ricci tensor is

$$R^{(1)}_{\mu\nu} = \tfrac{1}{2}\big(-p_\rho p_\mu h^\rho{}_\nu - p_\rho p_\nu h^\rho{}_\mu + p^2 h_{\mu\nu} + p_\mu p_\nu h\big),$$

with $h = h^\mu{}_\mu$. Substituting a gauge mode $h_{\mu\nu} = p_\mu\xi_\nu + p_\nu\xi_\mu$, the terms cancel in pairs, so $R^{(1)}$ depends only on the transverse part $(P^{(2)} + P^{(0)}_s)h$; every symmetric tensor is its transverse part plus a gauge mode, because $1 - P^{(2)} - P^{(0)}_s$ projects onto tensors of the form $p_\mu\xi_\nu + p_\nu\xi_\mu$. For transverse $h$ the formula reduces to $\frac{1}{2}(p^2 h_{\mu\nu} + p_\mu p_\nu h)$, and writing $h_{\mu\nu} = (P^{(2)}h)_{\mu\nu} + (\theta\cdot h)\,\theta_{\mu\nu}/(D-1)$, with $\theta\cdot h = \theta^{\mu\nu}h_{\mu\nu}$, we get

$$R^{(1)}_{\mu\nu} = \frac{p^2}{2}\Big[(P^{(2)}h)_{\mu\nu} + (\theta\cdot h)\,T_{\mu\nu}\Big], \qquad T_{\mu\nu} = \frac{\theta_{\mu\nu}}{D-1} + \omega_{\mu\nu}. \tag{58}$$

Its trace is $R^{(1)} = p^2\,\theta\cdot h$. Squaring Eq. (58), the cross terms vanish because $P^{(2)}h$ is transverse and traceless, $T\cdot T = D/(D-1)$, and $(\theta\cdot h)^2 = (D-1)\,hP_s^{(0)}h$, where $hPh$ stands for $h^{\mu\nu}P_{\mu\nu\alpha\beta}h^{\alpha\beta}$ and products are understood as $h(-p)\cdots h(p)$. This gives

$$R^{(1)}_{\mu\nu}R^{(1)\mu\nu} = \tfrac{p^4}{4}\big[hP^{(2)}h + D\,hP_s^{(0)}h\big], \tag{59}$$

$$\big(R^{(1)}\big)^2 = p^4(D-1)\,hP_s^{(0)}h, \tag{60}$$

$$G^{(1)}_{\mu\nu}R^{(1)\mu\nu} = \tfrac{p^4}{4}\big[hP^{(2)}h - (D-2)\,hP_s^{(0)}h\big], \tag{61}$$

$$h^{\mu\nu}G^{(1)}_{\mu\nu} = \tfrac{p^2}{2}\big[hP^{(2)}h - (D-2)\,hP_s^{(0)}h\big]. \tag{62}$$

Equation (61) is Eq. (59) minus half of Eq. (60), and Eq. (62) follows from $G^{(1)} = \frac{p^2}{2}[P^{(2)}h - \frac{D-2}{D-1}(\theta\cdot h)\theta]$.

5.3. **Quadratic action and propagator.** Since $\delta(\sqrt{g}R) = -\sqrt{g}\,G^{\mu\nu}\delta g_{\mu\nu}$ up to a total derivative and $G_{\mu\nu}$ vanishes on flat space, the second-order term of $\int\sqrt{g}R$ is $-\frac{1}{2}\int\delta g^{\mu\nu}G^{(1)}_{\mu\nu}[\delta g]$, which for $\delta g = \kappa h$ is $-\frac{\kappa^2}{2}\int h\,G^{(1)}[h]$. The nonlocal term is already quadratic in curvature, so its second-order part is $\kappa^2\int G^{(1)}F(\Box)R^{(1)}$ with $\Box\to -p^2$. With the prefactor $-2/\kappa^2$ and Eqs. (61) and (62) at $D=4$, the quadratic action is $\frac{1}{2}p^2[1-p^2F(-p^2)]\,h[P^{(2)}-2P_s^{(0)}]h$, and $1-p^2F(-p^2) = 1+(a(p^2)-1) = a(p^2)$. Hence

$$S^{(2)} = \frac{1}{2}\int\frac{\mathrm{d}^4p}{(2\pi)^4}\,p^2a(p^2)\;h(-p)\big[P^{(2)} - 2P_s^{(0)}\big]h(p), \tag{63}$$

the Einstein–Hilbert quadratic action multiplied by $a(p^2)$. On the gauge-invariant subspace it is inverted by

$$\mathcal{D}(p) = \frac{P^{(2)} - \frac{1}{2}P_s^{(0)}}{p^2\,a(p^2)}, \tag{64}$$

the propagator of general relativity divided by $a$.

As a function of $z = p^2\in\mathbb{C}$, Eq. (64) is analytic except at $z=0$: $a$ is entire and zero-free, so $1/a$ is entire, and $1/(za)$ has a simple pole at the origin with residue $1/a(0) = 1$; the projectors do not depend on $|p|$. The residue is that of general relativity, so the exchange between conserved sources at long distances is Einstein's and carries the same positive norm. In Lorentzian signature the only on-shell condition is $p^2 = 0$, and because the form factor multiplies the spin-two projector by a scalar, both helicities have the dispersion relation $\omega^2 = k^2$ at every wavenumber. There are no massive states, ghosts or tachyons in the tree-level spectrum, and the speed of gravitational waves equals that of light identically.[14]

[14]The bound from GW170817 and GRB 170817A is $-3\times10^{-15} < (c_g - c)/c < 7\times10^{-16}$ [29]. Theories with a momentum-dependent spin-two form factor that is not shared by the whole Einstein operator generically violate it at some wavenumber; here the factorized structure (63) prevents that.

For comparison, a polynomial form factor of degree $N$ has $N$ extra poles whose residues add up to $-1$, by the residue theorem applied to $1/(za)$ [4]; Stelle's theory is $N = 1$, with residue exactly $-1$ [3]. A frozen gas of resonant branes would have infinitely many complex pairs, Eq. (7); a realization of the gas of Sec. 4, whose branes have no resonances, has no extra pole at all.

## 6. Ultraviolet power counting

6.1. **Gauge fixing and ghosts.** We use the background-field method [30], $g = \bar{g} + \kappa h$, with the background-covariant de Donder condition $\chi_\mu = \bar{\nabla}^\nu h_{\mu\nu} - \frac{1}{2}\bar{\nabla}_\mu h$ and the weighted gauge-fixing term

$$S_{\rm gf} = \int \mathrm{d}^4x\sqrt{\bar{g}}\;\chi_\mu\, a(-\bar{\Box})\,\chi^\mu. \tag{65}$$

For $a = 1$ this is the Feynman–de Donder term, which added to the Einstein part of Eq. (63) gives the operator $p^2(\mathbf{1} - \frac{1}{2}\delta\otimes\delta)$, with $\mathbf{1}$ the identity on symmetric tensors.[15] Around flat space both the gauge-invariant part and the gauge-fixing part are those of general relativity multiplied by $a(p^2)$, so the full propagator is

$$\mathcal{D}_{\mu\nu\alpha\beta}(p) = \frac{\frac{1}{2}(\delta_{\mu\alpha}\delta_{\nu\beta} + \delta_{\mu\beta}\delta_{\nu\alpha}) - \frac{1}{2}\delta_{\mu\nu}\delta_{\alpha\beta}}{p^2\, a(p^2)}. \tag{66}$$

The weight $Y = a(-\bar{\Box})$ is handled in the standard way [31, 32]. Averaging the gauge condition with the Gaussian weight $e^{-\int fYf}$ requires the normalization $(\det Y)^{1/2}$; combined with the Faddeev–Popov determinant $\det Q$, where $Q$ is the usual second-order ghost operator, this is $\det(YQ)\,(\det Y)^{-1/2}$. The ghosts therefore get the action $\bar{c}^\mu Y Q_{\mu\nu} c^\nu$, and a factor $(\det Y)^{-1/2}$ depending on the background alone compensates; it contributes $+\frac{1}{2}\operatorname{Tr}\log a(-\bar{\Box})$ to the effective action and has no vertices with quantum fields. The ghost propagator is $\delta_{\mu\nu}/(p^2 a(p^2))$. Every propagator of the theory, graviton or ghost, is then bounded by

$$|\mathcal{D}(p)| \le \frac{C}{p^2(1 + p^2/M^2)^n}, \tag{67}$$

by the lower bound in Eq. (43) and $1 + x^2 \ge \frac{1}{2}(1+x)^2$; the constant $C$ depends on the realization through $\|R_\Pi\|$.

6.2. **Functions of the d'Alembertian in perturbation theory.** The vertices of a nonlocal theory come from expanding $F(\Box)$, $a(-\bar{\Box})$ and the like in powers of the fields, and a standard formula for functions of perturbed

[15] In four dimensions $(\mathbf{1} - \frac{1}{2}\delta\otimes\delta)^2 = \mathbf{1} - \delta\otimes\delta + \frac{1}{4}(\delta\cdot\delta)\,\delta\otimes\delta = \mathbf{1}$, since $\delta\cdot\delta = 4$. The normalization of Eq. (65), coefficient one in front of $\chi_\mu\chi^\mu$ with the action normalized as in Eq. (63), is the one that makes the trace and traceless parts combine in this way.

operators organizes this [33]. For points $z_0, \dots, z_m$ the divided difference $f[z_0, \dots, z_m]$ is defined by $f[z_0] = f(z_0)$ and

$$f[z_0, \dots, z_m] = \frac{f[z_1, \dots, z_m] - f[z_0, \dots, z_{m-1}]}{z_m - z_0},$$

extended to coincident points by continuity. It is symmetric in its arguments and obeys the Hermite–Genocchi formula [34]

$$f[z_0, \dots, z_m] = \int_{\Sigma_m} f^{(m)}\Big(\sum_{i=0}^{m} t_i z_i\Big)\,\mathrm{d}t, \tag{68}$$

where $\Sigma_m = \{t_i \geq 0, \sum t_i = 1\}$ carries Lebesgue measure of total mass $1/m!$.

Let $A + B$ act on plane waves, with $A$ diagonal, $A e^{iq\cdot x} = z(q) e^{iq\cdot x}$. For entire $f$, the term of $f(A+B)$ of order $m$ in $B$, acting on $e^{iq_0\cdot x}$ and projected on $e^{iq_m\cdot x}$, is

$$\sum_{q_1, \dots, q_{m-1}} f[z_0, \dots, z_m]\; B_{q_m q_{m-1}} \cdots B_{q_1 q_0}, \qquad z_i = z(q_i), \tag{69}$$

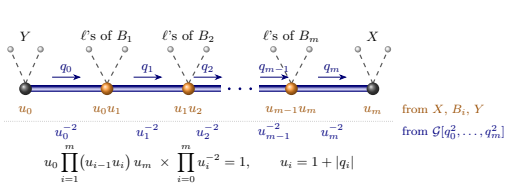
summed over the intermediate momenta that $B$ allows. Appendix C gives the short proof. In perturbation theory $A = -\Box_0$ with $z(q) = q^2$, and $B = -(\Box - \Box_0)$ collects the terms of the covariant d'Alembertian with at least one power of the fields. Each insertion of $B$ is a polynomial in the fields with at most two derivatives, which act either on the fields of the insertion or on the operand. Figure 11 shows the structure.

The divided differences that appear are those of $\mathcal{G}$, of $a - cx^n$ and of polynomials. For the first two we need a bound that controls mixed configurations, in which some of the $z_i$ are large and others small. Let $f$ be smooth on $[0, \infty)$ with $|f^{(j)}(z)| \leq C_j (1 + z)^{\mu - j}$ for all $j$, for some real $\mu$, and write $\mu_- = \min(\mu, 0)$ and $\mu_+ = \max(\mu, 0)$. Then for every $m$ there is a constant $K_m$ such that, for all $z_0, \dots, z_m \geq 0$,

$$\big|f[z_0, \dots, z_m]\big| \leq K_m\, (1 + z_{\min})^{\mu_-} (1 + z_{\max})^{\mu_+} \prod_{i \neq i_{\min}} (1 + z_i)^{-1}, \tag{70}$$

where $z_{\min} = z_{i_{\min}}$ is a smallest argument and $z_{\max}$ a largest one. The proof, by induction on $m$ using Eq. (68) when the arguments are comparable and the recursion when they are not, is in Appendix C. The bound says that a divided difference of a symbol of order $\mu \leq 0$ decays like the symbol in its smallest argument and like an inverse power in each of the others; for $\mu > 0$ the growth of the symbol is charged to the largest argument.

6.3. **Vertex bound.** Consider any vertex of the gauge-fixed action, ghost vertices and the quartic operators $S_{\mathrm{q}}$ included, with quantum legs carrying Euclidean momenta $\ell_1, \dots, \ell_N$ and background legs carrying momenta in a



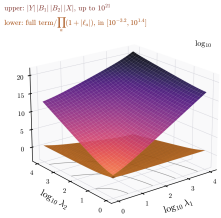




FIGURE 11. The chain behind the vertex bound. (a) A function of the covariant d'Alembertian between two field monomials, $X\,\mathcal{G}(-\Box)\,Y$, expanded as in Eq. (69). Each insertion $B_i$ of the field-dependent part of $\Box$ attaches legs (dashed, gray balls) and changes the running momentum from $q_{i-1}$ to $q_i$; the function enters only through the divided difference $\mathcal{G}[q_0^2,\dots,q_m^2]$. By Eq. (72) each insertion costs one power of the running momentum on each side of it (orange), and $Y$ and $X$ one power each; by Eq. (70) the divided difference returns $u_i^{-2}$ for every segment (blue). The running momenta cancel exactly, and only the legs are left. (b) A numerical check of this bookkeeping for a chain with two insertions and $n = 4$, in units $M = 1$. The running momenta are $q_0 = \ell_Y$ with $|q_0| = 1$, $q_1 = q_0 + \ell_1$ with $|q_1| \simeq \lambda_1$, and $q_2 = q_1 + \ell_2$ with $|q_2| \simeq \lambda_2$; each insertion is given its largest size $(|q_{i-1}| + |\ell_i|)^2$ and $X$, $Y$ the sizes $u_2$, $u_0$. The upper surface is $\log_{10}$ of the product of these sizes, which grows like the fifth power of the momenta, to $10^{21}$; the lower surface is $\log_{10}$ of the full term, with the exact divided difference of the skeleton's $\hat{\mathcal{G}}$, divided by $(1 + |\ell_1|)(1 + |\ell_2|)$. It stays between $10^{-3.2}$ and $10^{1.4}$ over the four decades in $\lambda_1$ and $\lambda_2$.

fixed bounded set. We show that, for every $n \geq 1$,

$$|V| \leq C\left[\left(1+\frac{|\ell|_{\max}}{M}\right)^{2n+2} + \left(1+\frac{|\ell|_{\max}}{M}\right)^{2m_\sigma}\prod_{j=1}^{N}\left(1+\frac{|\ell_j|}{M}\right)\right], \quad (71)$$

with $|\ell|_{\max}$ the largest $|\ell_j|$, $m_\sigma$ the order of Eq. (44), and $C$ independent of the momenta. The first term bounds the local part of the vertex and the second bounds everything else. When all momenta at a vertex are of order $\lambda$ the first term grows like $\lambda^{2n+2}$, as dimensional analysis requires. The second term grows by one power of $\lambda$ for each hard leg, however many there are, times the fixed power $\lambda^{2m_\sigma}$, which is absent when $\sigma > 2n$.

Set $M = 1$. Background fields never run in loops, so their momenta are external and bounded in every graph, by $P$ say. By Eq. (45), $F(\Box) =$

$-c(-\Box)^{n-1} - \mathcal{G}(-\Box)$, and we also split $a(-\bar{\Box}) = c(-\bar{\Box})^n + [a - cx^n](-\bar{\Box})$. Every vertex is then a sum of terms of three kinds.

The first kind are local, and there are three sources of them: the operator $c\,G_{\mu\nu}(-\Box)^{n-1}R^{\mu\nu}$, the quartic action $S_{\rm q}$, and the local parts $c(-\bar{\Box})^n$ of the gauge-fixing and ghost actions. Each is a polynomial in the momenta of total degree at most $2n+2$, whatever the number of legs, so it is bounded by $C(\sum_j |\ell_j| + P)^{2n+2} \leq C(1 + |\ell|_{\max})^{2n+2}$.

The second kind contain no function of the full d'Alembertian. They come from the Einstein–Hilbert term, which has two derivatives, and from the remainders $[a - cx^n](-\bar{\Box})$ in the gauge-fixing and ghost actions. In those two actions the weight stands between two factors $\chi_\mu$ that are linear in the quantum field, or acts on $Q_{\mu\nu}c^\nu$, where $Q$ has two derivatives, so the quantum legs enter through at most two derivatives. The d'Alembertian inside the weight is the background one. Expanding it by Eq. (69) produces only background insertions, each of which multiplies the operand by at most $C(1+|q|)^2$, with $q$ the running momentum. Since background momenta are bounded, all running momenta of such a chain lie within a fixed distance of one another, and by Eq. (70) the divided difference of the symbol $a - cx^n$, of order $m_\sigma$, is at most $C(1+q^2)^{m_\sigma}$ times a factor $(1+q^2)^{-1}$ for every insertion. Every term of the second kind is therefore bounded by $C(1 + |\ell|_{\max})^{2+2m_\sigma}$. Momentum conservation gives $|\ell|_{\max} \leq (N-1)|\ell_{(2)}| + P$, with $\ell_{(2)}$ the second largest quantum momentum, so $(1 + |\ell|_{\max})^2 \leq C(1 + |\ell_{(1)}|)(1 + |\ell_{(2)}|)$, and the term lies within the second term of Eq. (71). (For $N = 1$ the single quantum momentum is bounded by $P$, and there is nothing to prove.)

The third kind contain the full d'Alembertian, through $\mathcal{G}(-\Box)$ in the form $X\,\mathcal{G}(-\Box)\,Y$, where $X$ comes from $\sqrt{g}\,G_{\mu\nu}$ and $Y$ from $R^{\mu\nu}$. Here the insertions carry quantum legs, which may be hard, and the estimate has to be made with some care. By Eq. (69) the term of order $m$ is $\mathcal{G}[z_0, \ldots, z_m]\,B_m \cdots B_1$, with $z_i = q_i^2$. The running momentum $q_0$ is the total momentum of the legs of $Y$, and $q_i = q_{i-1} + \sum_{a\in i} \ell_a$, the sum running over the legs of the $i$th insertion. Write $u_i = 1 + |q_i|$ and $s_i = \sum_{a\in i} |\ell_a|$. An insertion has at most two derivatives, acting on its own legs or on the operand, so $|B_i| \leq C(|q_{i-1}| + s_i)^2$; in the same way $|Y| \leq C s_Y^2$ and $|X| \leq C s_X^2$. The estimate that does the work, proved in Appendix C, is

$$\big(|q| + s\big)^2 \leq C_r\,(1 + |q|)(1 + |q'|) \prod_{a=1}^{r} \big(1 + |\ell_a|\big), \tag{72}$$

for any $q$ and legs $\ell_1, \ldots, \ell_r$, with $q' = q + \sum_a \ell_a$ and $s = \sum_a |\ell_a|$. An insertion costs one power of each of the two running momenta next to it and one power of each of its own legs. Applying Eq. (72) to every insertion, to $Y$ (with $q = 0$, $q' = q_0$) and to $X$ (with $q = 0$, $q' = -q_m$, since the legs of $X$ carry away the momentum $q_m$), the running momenta enter through

$$u_0\,u_m \prod_{i=1}^{m} u_{i-1}u_i = \prod_{i=0}^{m} u_i^2.$$

On the other side, $\mathcal{G}$ is a symbol of order $m_\sigma - 1$ by Eq. (46), and Eq. (70) with $\mu = m_\sigma - 1$ gives

$$|\mathcal{G}[z_0, \dots, z_m]| \le K(1 + z_{\max})^{m_\sigma} \prod_{i=0}^{m} (1 + z_i)^{-1}$$
$$\le 2^{m+1} K(1 + z_{\max})^{m_\sigma} \prod_{i=0}^{m} u_i^{-2}$$

whether $\mu$ is negative or positive.[16] The running momenta cancel exactly. Every running momentum is a sum of leg momenta and bounded background momenta, so $1 + z_{\max} \le C(1 + |\ell|_{\max})^2$, and the term is bounded by $C(1 + |\ell|_{\max})^{2m_\sigma} \prod_a (1 + |\ell_a|)$ over all its legs. The sum over intermediate momenta in Eq. (69) is a finite sum over the ways of distributing the legs among the insertions, which proves Eq. (71).[17]

6.4. **Superficial degree.** Let $\gamma$ be a connected subgraph with $L$ loops of some graph of the theory, and consider the region in which the loop momenta of $\gamma$ are of order $\lambda \to \infty$ while all other momenta are fixed. Let $\gamma$ have $I$ internal lines and $V$ vertices, so that $I - V = L - 1$, and let $h_v$ be the number of lines of $\gamma$ at the vertex $v$, so that $\sum_v h_v = 2I$. Every vertex of $\gamma$ has $h_v \ge 2$, because a line of $\gamma$ carries loop momentum and momentum conservation at its ends needs a second line that does. Each line contributes at most $C\lambda^{-2n-2}$ by Eq. (67). At a vertex of $\gamma$ the legs outside $\gamma$ carry fixed momenta, so Eq. (71) contributes at most $C\lambda^{\max(2n+2,\, h_v + 2m_\sigma)}$. The measure contributes $\lambda^{4L}$. With $x_+ = \max(x, 0)$ and $k_\sigma = 2n + 2 - 2m_\sigma$, which exceeds 6, the superficial degree is therefore at most

$$4L - (2n+2)I + \sum_v \max(2n + 2, h_v + 2m_\sigma)$$
$$= 4L - (2n+2)(L-1) + \sum_v \big(h_v - k_\sigma\big)_+ .$$

The last sum is largest when all the lines beyond the minimum meet at one vertex. Indeed $(a-k)_+ + (b-k)_+ \le (a+b-2-k)_+$ whenever $a, b \ge 2$ and $k \ge 2$,[18] so merging the vertices one pair at a time gives $\sum_v (h_v - k_\sigma)_+ \le$

[16] For $\mu \le 0$, Eq. (70) gives $(1 + z_{\min})^{m_\sigma - 1} \prod_{i \ne i_{\min}} (1 + z_i)^{-1}$, and $(1 + z_{\min})^{m_\sigma} \le (1 + z_{\max})^{m_\sigma}$. For $\mu > 0$ it gives $(1 + z_{\max})^{m_\sigma - 1} \prod_{i \ne i_{\min}} (1 + z_i)^{-1}$, and $1 + z_{\min} \le 1 + z_{\max}$. For the skeleton, and whenever $\sigma > 2n$, $m_\sigma = 0$ and the bound is $K \prod_i (1 + z_i)^{-1}$.

[17] The cancellation is not an artifact of the bounds. When the running momentum rises from soft to hard at one insertion and falls back at the next, the divided difference is of order $\lambda^{-2}$ and the two insertions of order $\lambda^2$ each, so the term grows like $\lambda^2$ while two hard legs are attached: one power per hard leg, as Eq. (71) says. Appendix D lists a numerical check of the bound on random chains with up to three insertions.

[18] If $a$ and $b$ both exceed $k$, the left side is $a + b - 2k \le a + b - 2 - k$ because $k \ge 2$. If only $a$ does, it is $a - k \le a + b - 2 - k$ because $b \ge 2$. If neither does, it is zero.

$(\sum_v h_v - 2(V-1) - k_\sigma)_+ = (2L - k_\sigma)_+$. Hence the superficial degree of an $L$-loop subgraph is at most

$$\bar{\omega}(L) = 4L - (2n+2)(L-1) + 2\,(L - n - 1 + m_\sigma)_+. \tag{73}$$

For $m_\sigma = 0$ and $L \le n+1$ this is the standard degree $2n + 2 + (2-2n)L$ of local higher-derivative gravity [11, 35], and for $L > n+1$ it is $(4-2n)L$. At one loop $\bar{\omega}(1) = 4$ for every $n$. At two loops $\bar{\omega}(2) = 6 - 2n$, because $m_\sigma < n-2$ keeps the last term at zero. For $L \ge 3$ the bound grows with $m_\sigma$, and as $m_\sigma \to n-2$, the weakest fine graining that (G) allows, it tends to $4L - (2n+2)(L-1) + 2(L-3) = -(2n-4)(L-1)$. So $\bar{\omega}(L) < 0$ for every $L \ge 2$ and every $\sigma > 4$ exactly when $n \ge 4$. Table 5 and Fig. 13 give the numbers.

We give the hard-subspace check explicitly. Fix an orientation and a momentum routing for the graph. At fixed external momenta, the linear part of the internal momenta is a divergence-free flow, one copy for each of the four momentum coordinates. Given a linear subspace $U$ of the loop variables, retain precisely the edges on which some routing functional is nonzero on $U$. Call the resulting graph $H(U)$. Conservation implies that each retained vertex has degree at least two and that $H(U)$ has no bridges: summing the flow divergence on one side of a bridge would force the flow on the bridge to vanish for every vector in $U$. The routing map is injective on $U$, and maps it into the cycle space of $H(U)$ with four components per edge. If the connected components of $H(U)$ have loop numbers $L_j$, then

$$\dim U \le 4 \sum_j L_j.$$

Equality holds for the saturated flow subspace on these edges. The powers from denominators and numerator bounds are the same on the original subspace and its saturation, since they depend only on which momentum forms vanish identically. External affine shifts do not change this degree.

For fixed $\mu > 0$ in an infrared-regularized propagator, put $\langle p \rangle = 1 + |p|$. Expanding the finite sums in the vertex estimate (71) gives a finite sum of products of powers of brackets of affine momentum forms that dominate the absolute graph integrand. On each such term, the degree on $U$, including its measure, is bounded by

$$\dim U - 4 \sum_j L_j + \sum_j \bar{\omega}(L_j).$$

This follows by applying the hard-leg estimate on each component, using $\dim U$ in place of the saturated measure dimension. At a vertex the bracket of a soft leg has degree zero. The estimate includes disconnected supports and proper subspaces of saturated flows.[19] Weinberg's convergence theorem

[19] For a real line of loop variables contained in the kernel of one edge map, the same line can make several other edges hard. Testing single loop variables in a chosen routing would not detect all such regions.

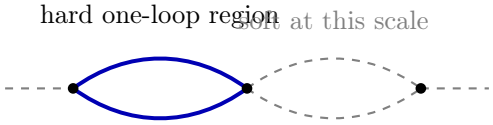



FIGURE 12. A two-loop graph with a one-loop hard subspace. The negative degree under common scaling does not control the one-loop subspace.

for products of powers of affine momentum forms [36] therefore proves absolute ultraviolet convergence of any graph for which every nonempty hard support has strictly negative degree. In particular, when $n \geq 4$ every primitive multi-loop graph with no one-loop hard component requiring subtraction passes this test. A connected multi-loop hard component has negative degree. A graph of total loop number at least two can still have a one-loop hard component. Its integral can require a subdivergence subtraction before it is finite.

The distinction can be seen even in a scalar bracket model. In four dimensions a factor $(1+|k|^2)^{-2}$ has logarithmically divergent integral, while $(1+|\ell|^2)^{-5}$ is integrable. The product has degree $8-4-10=-6$ under common scaling of $(k,\ell)$. Its $k$-only region has degree zero, and the double integral is infinite by Tonelli's theorem. In the gravitational calculation the corresponding regions are the ones that must be handled by the one-loop forest operation.

The threshold $n \geq 4$ is the one found in the literature by bounding every vertex by $\lambda^{2n+2}$, which is legitimate when all momenta at a vertex are comparable [11, 35]. The vertex bound shows that vertices with many hard legs do not spoil it: the nonlocal remainder gains at most one power of $\lambda$ per hard leg, and a fixed power below $2n-4$ in all, while each hard leg brings a propagator that falls like $\lambda^{-2n-2}$. For $n \leq 3$ the two-loop degree is not negative and the argument gives nothing. From now on $n \geq 4$.

## 7. One loop

7.1. **The local theory.** Split the action into a local part and a remainder,

$$S_{\mathrm{loc}} = \frac{2c}{\kappa^2 M^{2n}} \int \sqrt{g}\, G_{\mu\nu}(-\Box)^{n-1} R^{\mu\nu} + S_{\mathrm{q}}, \tag{74}$$

$$S_{\mathrm{rest}} = -\frac{2}{\kappa^2} \int \sqrt{g}\, \big[R - G_{\mu\nu}\, \mathcal{G}(-\Box)\, R^{\mu\nu}\big],$$

so that $S = S_{\mathrm{loc}} + S_{\mathrm{rest}}$ by Eq. (45). Split the gauge-fixing weight in the same way, $a(-\bar{\Box}) = c(-\bar{\Box}/M^2)^n + [a - cx^n](-\bar{\Box})$. Both parts of the action are separately invariant under diffeomorphisms, and both parts of the gauge-fixing term are background covariant. The local theory, with action $S_{\mathrm{loc}}$, gauge-fixing weight $c(-\bar{\Box}/M^2)^n$ and the corresponding ghosts, is therefore

| $n$ | $L=1$ | $L=2$ | $L=3$ | $L=4$ | $L=5$ |
|---|---|---|---|---|---|
| 3 | 4 | 0 | $-4$ | $-8$ | $-10$ |
| 4 | 4 | $-2$ | $-8$ | $-14$ | $-20$ |
| 5 | 4 | $-4$ | $-12$ | $-20$ | $-28$ |
| 6 | 4 | $-6$ | $-16$ | $-26$ | $-36$ |
| 7 | 4 | $-8$ | $-20$ | $-32$ | $-44$ |
| 8 | 4 | $-10$ | $-24$ | $-38$ | $-52$ |
| $m_\sigma \to n-2$ $(\sigma \to 4)$ | | | | | |
| 3 | 4 | 0 | $-4$ | $-6$ | $-8$ |
| 4 | 4 | $-2$ | $-8$ | $-12$ | $-16$ |
| 5 | 4 | $-4$ | $-12$ | $-18$ | $-24$ |
| 6 | 4 | $-6$ | $-16$ | $-24$ | $-32$ |
| 7 | 4 | $-8$ | $-20$ | $-30$ | $-40$ |
| 8 | 4 | $-10$ | $-24$ | $-36$ | $-48$ |

TABLE 5. Superficial degree $\bar{\omega}(L) = 4L - (2n+2)(L-1) + 2(L-n-1+m_\sigma)_+$ from the vertex bound (71), for $L = 1, \dots, 5$. Upper block: $m_\sigma = 0$, which holds for the skeleton and whenever $\sigma > 2n$; for $L \le n+1$ the entries coincide with the standard estimate $2n+2+(2-2n)L$, and the first entry that differs is $n=3$, $L=5$. Lower block: the limit $m_\sigma \to n-2$, the weakest fine graining allowed by (G), which bounds the degree for every $\sigma > 4$. Negative entries are convergent.

a consistent background-gauge-fixed theory on its own. Its flat-space propagator is $M^{2n}(P^{(2)} - \frac{1}{2}P_s^{(0)})/(c\,p^{2n+2})$ on the gauge-invariant subspace, from Eq. (64) with $a$ replaced by $cx^n$.

The local theory is badly behaved in the infrared, since its propagator falls like $p^{-2n-2}$ at small $p$ as well as at large $p$. This does not matter for what follows, because we compare the two theories only at large loop momentum. We define the ultraviolet pole of a one-loop integral as the pole, in dimensional regularization with $D = 4-2\epsilon$, of its integral over the region $|k| \ge M$ of the loop momentum.[20] With this definition the ultraviolet pole of the full theory and that of the local theory can be compared graph by graph.

7.2. **One-loop poles are those of the local theory.** Let $n \ge 4$, and consider a one-particle-irreducible one-loop graph of the full theory with external legs, background or quantum, of bounded momenta. Write each

[20]For a one-loop integrand $I(k,p)$ with bounded external momenta $p$, the region $|k| \ge M$ can be split into $M \le |k| \le K$, which is compact and gives no pole, and $|k| \ge K$ with $K$ large compared with the external momenta, where $I$ may be expanded in $p$. Only the terms of the expansion that are homogeneous of degree $-D$ in $k$ produce a pole, and their coefficients are polynomials in $p$. The pole is therefore local and independent of the choice of $M$ and $K$. With this definition $\int_{|k|\ge M} \mathrm{d}^D k\,(2\pi)^{-D} k^{-4} = 1/(16\pi^2\epsilon) + O(1)$.

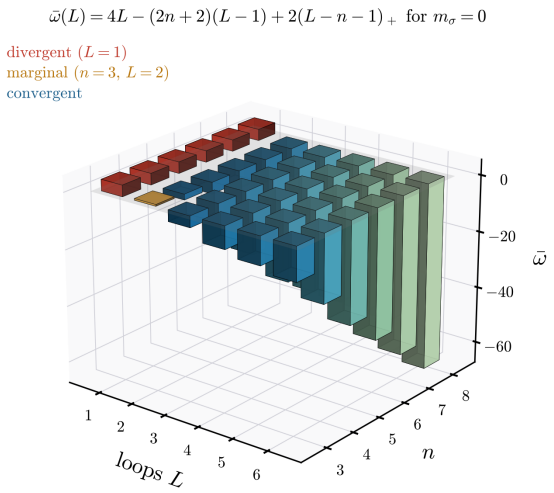



FIGURE 13. Upper bound $\bar\omega(L)$ on the superficial degree of an $L$-loop graph, Eq. (73), for $n = 3, \dots, 8$ and $L \le 6$, drawn as bars hanging from the plane $\bar\omega = 0$. One-loop graphs have degree 4 for every $n$ (red). For $n = 3$ the two-loop bound is 0 (amber), so power counting alone leaves a possible logarithm; for $n \ge 4$ every graph with $L \ge 2$ has negative degree (blue to green), and the degree falls without bound as $L$ grows. The bars are drawn for $m_\sigma = 0$, which holds for the skeleton and whenever $\sigma > 2n$; the lower block of Table 5 gives the bound for the weakest fine graining, $m_\sigma \to n - 2$, which is negative for $L \ge 2$ as well.

vertex as $V = V_{\rm loc} + V_{\rm rest}$ and each propagator as $\mathcal{D} = \mathcal{D}_{\rm loc} + \mathcal{D}_{\rm rest}$, where the local pieces are those of the local theory. For loop momentum $|k| \ge M$ we have $|V_{\rm loc}| \le C|k|^{2n+2}$ and $|\mathcal{D}_{\rm loc}| \le C|k|^{-2n-2}$. The remainders are smaller. A remainder vertex consists of terms of the second and third kinds of Sec. 6.3, which are bounded by the second term of Eq. (71); two of its legs carry the loop momentum, so $|V_{\rm rest}| \le C|k|^{2+2m_\sigma}$. And $|\mathcal{D}_{\rm rest}| \le C|k|^{-2n-2-\sigma+2\varepsilon}$, because $1/a - 1/(cx^n) = (cx^n)^{-1}(e^{-\eta_\Pi} - 1)$ with $\eta_\Pi$ bounded by Eq. (42). Expanding the product around the loop, the term with only local factors is the integrand of the local theory. Every other term contains at least one $V_{\rm rest}$ or one $\mathcal{D}_{\rm rest}$. For a loop with $r$ vertices it is bounded for $|k| \ge M$ either by $C|k|^{(2n+2)(r-1)+2+2m_\sigma-(2n+2)r} = C|k|^{2m_\sigma-2n}$, if it contains a remainder vertex, or by $C|k|^{-\sigma+2\varepsilon}$, if it contains a remainder propagator. Both exponents are below $-4$, because $m_\sigma < n - 2$ and $\sigma - 2\varepsilon > 4$. Its integral over $|k| \ge M$ therefore converges in a range of $D$ that contains $D = 4$, and has no pole there. The ultraviolet pole of the full graph is that of the local graph.

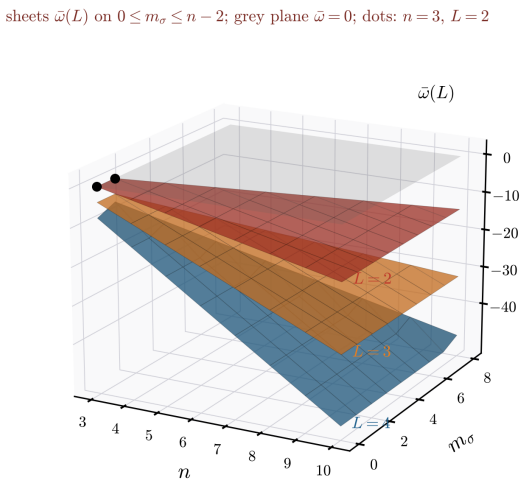



FIGURE 14. The bound $\bar\omega(L)$ of Eq. (73) over the plane of the strength $n$ and the fine-graining exponent $m_\sigma = \max(0, n - \sigma/2 + \varepsilon)$, on the range $0 \le m_\sigma \le n-2$ that (G) allows, as one sheet for each of $L = 2, 3, 4$; the grey plane is $\bar\omega = 0$. The sheet $L = 2$ does not depend on $m_\sigma$ and touches the plane only along $n = 3$ (black dots), where the two-loop degree is 0; every sheet lies below the plane for $n \ge 4$, and the sheets for $L \ge 3$ rise with $m_\sigma$ without reaching it, their largest value being $-(2n-4)(L-1)$ at $m_\sigma = n-2$. The surfaces are drawn through the integer points of the table, where the bound is defined.

The compensating factor of Sec. 6.1 contributes $\frac{1}{2}\operatorname{Tr}\log a(-\bar\Box)$, which at large Euclidean momenta equals $\frac{n}{2}\operatorname{Tr}\log(-\bar\Box/M^2) + \frac{1}{2}\log c\,\operatorname{Tr}\mathbf{1} + \frac{1}{2}\operatorname{Tr}\eta_\Pi(-\bar\Box)$ by Eq. (42). The trace of a constant vanishes in dimensional regularization, the last trace has a symbol that falls like $|z|^{-\sigma/2+\varepsilon}$, faster than $|z|^{-2}$, and has no pole, and the first has the pole of $\operatorname{Tr}\log(-\bar\Box)$ on vectors, a local functional of dimension four. This is the pole of the compensating factor of the local theory.

7.3. **The form of the pole.** In the local theory each graviton vertex carries the factor $c/(\kappa^2 M^{2n})$ of Eq. (74), times $\kappa^k$ for $k$ quantum legs, and each graviton propagator carries $\kappa^0 M^{2n}/c$; the quartic operators enter only through the ratios of their coefficients to $c$; the ghosts carry $c/M^{2n}$ at each vertex and the inverse on each propagator. Write the local action as $\lambda^{-1}\int\sqrt{g}\,\mathcal{L}$, with $\lambda = \kappa^2 M^{2n}/(2c)$ and $\mathcal{L}$ free of dimensionful constants. The one-loop effective action is of order $\lambda^0$, so its ultraviolet pole, as a local functional of the full metric $\bar g + \kappa Q$ and of the other fields, contains no dimensionful constant at all. By background gauge invariance [30, 37] its part

depending on $\bar{g}$ alone is a diffeomorphism-invariant local functional, and a dimensionless local invariant built from the metric has exactly four derivatives. In four dimensions these invariants are $\bar{R}^2$, $\bar{R}_{\mu\nu}\bar{R}^{\mu\nu}$, $\bar{R}_{\mu\nu\rho\sigma}\bar{R}^{\mu\nu\rho\sigma}$ and $\bar{\Box}\bar{R}$; the last is a total derivative, and the Riemann square may be traded for the Gauss–Bonnet density $\bar{E} = \bar{R}_{\mu\nu\rho\sigma}\bar{R}^{\mu\nu\rho\sigma} - 4\bar{R}_{\mu\nu}\bar{R}^{\mu\nu} + \bar{R}^2$. Hence

$$\Gamma_1^{\mathrm{div}} = \frac{1}{16\pi^2\epsilon}\int \mathrm{d}^4x\sqrt{\bar{g}}\,\big(\beta_1\bar{R}^2 + \beta_2\bar{R}_{\mu\nu}\bar{R}^{\mu\nu} + \beta_E\,\bar{E}\big), \tag{75}$$

with $\beta_1, \beta_2, \beta_E$ pure numbers that depend on $n$ and on the coefficients of $S_{\mathrm{q}}$ in units of $c$, and on nothing else.

In particular there is no pole in $\int\sqrt{\bar{g}}$ or $\int\sqrt{\bar{g}}\,\bar{R}$: neither the cosmological constant nor Newton's constant is renormalized at one loop. Those two terms need the dimensionful factors $M^4$ and $M^2$, and in the ultraviolet the only place they could come from is the correction to the monomial asymptotics of $a$. For a realization that correction is of relative order $|z|^{-\sigma/2+\varepsilon}$ with $\sigma/2-\varepsilon > 2$, and Sec. 7.2 shows that it contributes only convergent integrals. A single-brane response that saturated only as a power of $1/|sz|$ would give, through Eq. (21), corrections of relative order $x^{-1}$ or $x^{-2}$ and in general a running of both constants, and so would a gas whose fluctuations died out too slowly. Conditions (S3) and (G) prevent it. The explicit one-loop computation of Modesto, Rachwał and Shapiro [38] agrees. For form factors in front of $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$ and $R^2$ that grow like polynomials of degree $N$, they find a beta function of Newton's constant that is a linear combination of the ratios $\omega_{N-1}/\omega_N$ of the two leading coefficients of those polynomials. In Eq. (57) both form factors are multiples of $F$, and by Eq. (45) $F(z)$ equals $c\,z^{n-1}/M^{2n} - 1/z$ up to terms of order $|z|^{n-1-\sigma/2+\varepsilon}$. When $\sigma > 2n$ these fall faster than $1/z$, $F$ has no term in $z^{n-2}$, both ratios vanish, and so does the beta function. When $4 < \sigma \le 2n$ the correction is not a polynomial and lies outside the class of Ref. [38], but it is smaller than a term in $z^{n-2}$ by more than one power of $z$, and Sec. 7.2 shows directly that it produces no pole.

The numbers $\beta_i$ do not depend on the gauge condition or on its weight within the class used here. At one loop the change of the effective action under a change of gauge is proportional to the classical equations of motion [39], $\delta\Gamma_1 = \int(\delta S/\delta\bar{g}_{\mu\nu})\,Y_{\mu\nu}$ with $Y$ a functional of $\bar{g}$. Apply this to the local theory, whose poles these are. The pole of $\delta\Gamma_1$ is $\int(\delta S_{\mathrm{loc}}/\delta\bar{g}_{\mu\nu})\,Y^{\mathrm{div}}_{\mu\nu}$, with $Y^{\mathrm{div}}$ local. Every term of $\delta S_{\mathrm{loc}}/\delta\bar{g}_{\mu\nu}$ has $2n+2 \ge 10$ derivatives of the metric, and a local $Y^{\mathrm{div}}$ cannot remove derivatives, so every term of the product has at least that many and none can appear in Eq. (75), which has four. Hence $\delta\Gamma_1$ has no pole.

The values of $\beta_1$, $\beta_2$, $\beta_E$ without the quartic operators, which we call $\beta_i^{(0)}$, are fixed by $n$. Computing them is a well-defined exercise in the generalized Schwinger–DeWitt technique for a nonminimal operator of order $2n+2$ [37]. We do not need their values, only the fact that they are finite numbers.

7.4. **Three quartic operators.** Following Modesto and Rachwał [12], we add operators that are quartic in curvature and can shift the coefficients in Eq. (75). We need three, one for each coefficient:

$$S_{\rm q} = \frac{2}{\kappa^2 M^{2n}} \int \mathrm{d}^4x\sqrt{g}\,\Big[ s_1\, R^2(-\Box)^{n-3}R^2 + s_2\,(R_{\mu\nu}R^{\mu\nu})(-\Box)^{n-3}(R_{\rho\sigma}R^{\rho\sigma}) + s_3\,(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma})(-\Box)^{n-3}(R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta})\Big], \quad (76)$$

with real dimensionless $s_1, s_2, s_3$. Each term has $2n+2$ derivatives, as many as the leading term of $S_{\rm loc}$, so the vertex bound (71) still holds. Being quartic in curvature, the operators change neither the propagator nor the three-graviton vertex around flat space, so the spectrum of Sec. 5.3 and the Newtonian limit of Sec. 10 are untouched. Their contribution to Eq. (75) can be computed exactly.

*Which terms contribute.* The operators are linear in $s_i$. Every term of their second variation around $\bar g$ contains at least two factors of background curvature, so a contribution to $\Gamma_1$ of second or higher order in the $s_i$ carries at least four background curvatures and eight derivatives, and cannot appear in Eq. (75). To first order the one-loop contribution is the expectation value of the quadratic part of $S_{\rm q}$ in the free local theory, $\frac{1}{2}\,\mathrm{Tr}(\mathcal{D}\,S_{\rm q}'') = \langle S_{\rm q}^{(2)}\rangle$. Of the terms in $S_{\rm q}^{(2)}$ only those with exactly two background curvatures and with $(-\Box)^{n-3}$ acting on a quantum field can produce a pole in Eq. (75). When derivatives fall on the background the operator has more than four derivatives; and a term in which $(-\Box)^{n-3}$ acts on the product of two background curvatures carries $2n-6 \geq 2$ derivatives on the background in addition to the four in the two curvatures, six or more in all. Corrections to the linearized curvatures, to $\Box$ and to the index contractions that involve the background curvature add more background curvatures. Writing $R^2 = \bar R^2 + 2\bar R\,\delta R + \cdots$ and similarly for the other invariants, the relevant terms are

$$\frac{8}{\kappa^2 M^{2n}}\Big[ s_1\bar R^2\,\delta R\,(-\Box)^{n-3}\delta R + s_2\bar R^{\mu\nu}\,\bar R^{\rho\sigma}\,\delta R_{\mu\nu}(-\Box)^{n-3}\delta R_{\rho\sigma} + s_3\bar R^{\mu\nu\rho\sigma}\,\bar R^{\alpha\beta\gamma\delta}\,\delta R_{\mu\nu\rho\sigma}(-\Box)^{n-3}\delta R_{\alpha\beta\gamma\delta}\Big], \quad (77)$$

where $\delta R$, $\delta R_{\mu\nu}$, $\delta R_{\mu\nu\rho\sigma}$ are the flat-space linearized curvatures of $\delta g = \kappa h$ and the background curvatures are treated as constants.

*Correlators.* The linearized curvatures are gauge invariant, so only the gauge-invariant part $M^{2n}(P^{(2)} - \frac{1}{2}P_s^{(0)})/(ck^{2n+2})$ of $\langle hh\rangle$ contributes, and the result does not depend on the gauge-fixing term. From Eq. (58), $\delta R_{\mu\nu} = \frac{\kappa k^2}{2}[(P^{(2)}h)_{\mu\nu} + (\theta\cdot h)T_{\mu\nu}]$. Since $P^{(2)}P_s^{(0)} = 0$ and $P^{(2)}\theta = 0$ the mixed terms vanish; $\langle (P^{(2)}h)(P^{(2)}h)\rangle$ gives $P^{(2)}$ times the scalar factor, and $\langle(\theta\cdot h)^2\rangle$ gives $\theta\cdot(-\frac{1}{2}P_s^{(0)})\cdot\theta = -\frac{3}{2}$ times the same factor, because $\theta P_s^{(0)}\theta = D-1 = 3$.

Thus, in $D = 4$,

$$\langle \delta R_{\mu\nu}(k)\, \delta R_{\rho\sigma}(-k)\rangle = \frac{\kappa^2 M^{2n}}{4c\, k^{2n-2}} \Big[P^{(2)} - \frac{3}{2} T\otimes T\Big]_{\mu\nu\rho\sigma}, \tag{78}$$

$$\langle \delta R(k)\, \delta R(-k)\rangle = -\frac{3}{2}\, \frac{\kappa^2 M^{2n}}{c\, k^{2n-2}}, \tag{79}$$

the second following from the first with $\delta \cdot P^{(2)} \cdot \delta = 0$ and $T \cdot \delta = 2$.[21] For the Riemann tensor, the linearized curvature is $\delta R_{\mu\nu\rho\sigma} = -\frac{\kappa}{2}(k_\nu k_\rho h_{\mu\sigma} + k_\mu k_\sigma h_{\nu\rho} - k_\mu k_\rho h_{\nu\sigma} - k_\nu k_\sigma h_{\mu\rho})$. Contracted with an algebraic curvature tensor $\bar R^{\mu\nu\rho\sigma}$, the four terms are equal by its symmetries, and

$$\bar R^{\mu\nu\rho\sigma}\delta R_{\mu\nu\rho\sigma} = 2\kappa k^2\, W^{\nu\sigma} h_{\nu\sigma}, \qquad W^{\nu\sigma} = \bar R^{\mu\nu\rho\sigma}\hat k_\mu \hat k_\rho, \tag{80}$$

with $\hat k = k/|k|$. The tensor $W$ is symmetric and transverse, $\hat k_\nu W^{\nu\sigma} = 0$, so $W \cdot P^{(2)} \cdot W = \operatorname{tr} W^2 - \frac{1}{3}(\operatorname{tr} W)^2$ and $W \cdot P_s^{(0)} \cdot W = \frac{1}{3}(\operatorname{tr} W)^2$, and

$$\big\langle (\bar R \cdot \delta \mathrm{Riem})^2 \big\rangle = \frac{4\kappa^2 M^{2n}}{c\, k^{2n-2}} \Big[\operatorname{tr} W^2 - \tfrac{1}{2}(\operatorname{tr} W)^2\Big]. \tag{81}$$

*Poles.* In the loop integral $(-\Box)^{n-3} \to (k^2)^{n-3}$, and the correlators give integrands proportional to $k^{2n-6}k^{-2n+2} = k^{-4}$, whose ultraviolet pole is $1/(16\pi^2\epsilon)$. The tensors must be averaged over the direction of $k$, with $\langle \hat k_\mu \hat k_\nu\rangle = \delta_{\mu\nu}/4$ and $\langle \hat k_\mu \hat k_\nu \hat k_\rho \hat k_\sigma\rangle = (\delta_{\mu\nu}\delta_{\rho\sigma} + \delta_{\mu\rho}\delta_{\nu\sigma} + \delta_{\mu\sigma}\delta_{\nu\rho})/24$. Appendix B shows that

$$\bar R^{\mu\nu}\bar R^{\rho\sigma}\Big\langle P^{(2)} - \frac{3}{2} T\otimes T\Big\rangle_{\mu\nu\rho\sigma} = \frac{1}{2}\big(\mathrm{Ric}^2 - R^2\big), \tag{82}$$

$$\Big\langle \operatorname{tr} W^2 - \tfrac{1}{2}(\operatorname{tr} W)^2\Big\rangle = \frac{1}{48}\big(3\mathrm{Riem}^2 - R^2\big), \tag{83}$$

where on the right $R$, $\mathrm{Ric}^2$ and $\mathrm{Riem}^2$ are the background invariants. Multiplying by the coefficients $8s_i/(\kappa^2 M^{2n})$ of Eq. (77), the factors $\kappa^2 M^{2n}$ cancel, and in units of $1/(16\pi^2\epsilon)$ the three poles are

$$\begin{aligned}
s_1:&\quad 8s_1 \cdot \Big(-\frac{3}{2c}\Big) R^2 = -\frac{12 s_1}{c}\, R^2,\\
s_2:&\quad 8s_2 \cdot \frac{1}{4c} \cdot \frac{1}{2}\big(\mathrm{Ric}^2 - R^2\big) = \frac{s_2}{c}\big(\mathrm{Ric}^2 - R^2\big),\\
s_3:&\quad 8s_3 \cdot \frac{4}{c} \cdot \frac{3\mathrm{Riem}^2 - R^2}{48} = \frac{s_3}{c}\Big(2\mathrm{Riem}^2 - \frac{2}{3}R^2\Big).
\end{aligned}$$

[21] The negative sign in Eq. (79) is the conformal-factor sign of the Euclidean Einstein–Hilbert action, visible in the $-2P_s^{(0)}$ of Eq. (63). In dimensional regularization the one-loop computation is Gaussian algebra on the propagator (64), and the sign is carried through consistently.

| operator | averaged correlator | pole |
|---|---|---|
| $s_1 R^2 \Box^{n-3} R^2$ | $-\frac{3}{2}\, R^2$ | $-12R^2$ |
| $s_2 \mathrm{Ric}^2 \Box^{n-3} \mathrm{Ric}^2$ | $\frac{1}{8}(\mathrm{Ric}^2 - R^2)$ | $\mathrm{Ric}^2 - R^2$ |
| $s_3 \mathrm{Riem}^2 \Box^{n-3} \mathrm{Riem}^2$ | $\frac{1}{12}(3\mathrm{Riem}^2 - R^2)$ | $2E + 8\mathrm{Ric}^2 - \frac{8}{3}R^2$ |

TABLE 6. One-loop poles of the quartic operators (76), in units of $\frac{1}{16\pi^2\epsilon}\frac{s_i}{c}$. The second column is the correlator after contraction with the background and averaging over the direction of $k$, in units of $\kappa^2 M^{2n}/(c\, k^{2n-2})$.

With $\mathrm{Riem}^2 = E + 4\mathrm{Ric}^2 - R^2$ the third line is $(s_3/c)(2E + 8\mathrm{Ric}^2 - \frac{8}{3}R^2)$. Collecting coefficients,

$$\begin{pmatrix} \Delta\beta_1 \\ \Delta\beta_2 \\ \Delta\beta_E \end{pmatrix} = \frac{1}{c} \begin{pmatrix} -12 & -1 & -\frac{8}{3} \\ 0 & 1 & 8 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} s_1 \\ s_2 \\ s_3 \end{pmatrix}. \tag{84}$$

The triangular matrix has determinant $-24$. The full linear map from the three strengths to the three pole shifts therefore has determinant $-24/c^3$, which never vanishes. The diagrams are drawn in Fig. 15(b), and Table 6 collects the pieces. In the dimensional-regularization computation the angular averages are taken in $D$ dimensions; the difference from the four-dimensional values is of order $\epsilon$ and affects finite parts only.

Solving $\beta_i^{(0)} + \Delta\beta_i = 0$ from the bottom row up gives the tuning in closed form:

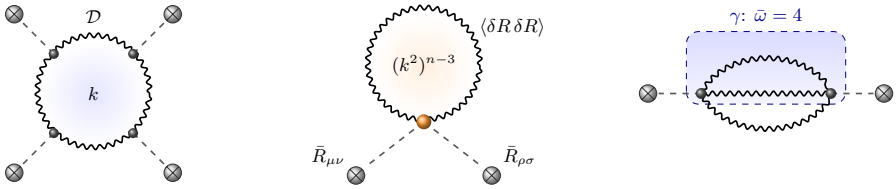
$$\begin{aligned} s_3 &= -\tfrac{c}{2}\,\beta_E^{(0)}, \qquad s_2 = -c\big(\beta_2^{(0)} - 4\beta_E^{(0)}\big), \\ s_1 &= \tfrac{c}{12}\big(\beta_1^{(0)} + \beta_2^{(0)} - \tfrac{8}{3}\beta_E^{(0)}\big). \end{aligned} \tag{85}$$

With this choice the one-loop background-field effective action has no ultraviolet pole of any kind, the Gauss–Bonnet one included.[22] By the gauge independence of the $\beta_i$, the tuning does not depend on the gauge.

## 8. Perturbative ultraviolet conclusions

Assume A1 and A2, let $n \geq 4$ be an integer, and take the realization-dependent $c = c_\Pi$ from Eq. (42). In the specified background-field calculation the one-loop invariant pole (75) is a sum of three curvature-squared integrals. Choosing $s_1, s_2, s_3$ from Eq. (85) cancels that pole. The determinant in Eq. (84) proves existence and uniqueness in this three-operator family. The actual numerical coefficients also need the three poles of the untuned local comparison theory.

[22] Cancelling the Gauss–Bonnet pole is not needed for the $S$ matrix in four dimensions, because $\int \sqrt{g}\, E$ is topological there. It is convenient all the same: in $D = 4 - 2\epsilon$ the density $E$ has vertices proportional to $D - 4$, and a $1/\epsilon$ counterterm in $E$ would leave finite evanescent contributions when inserted in higher-loop graphs. With $s_3$ tuned no such counterterm exists.




pole of full ring = pole of local ring
(a) one-loop ring, Sec. 7.2

$\propto \frac{s_i}{c} \times$ Table 6
(b) quartic-operator insertion, Sec. 7.4

$\bar{\omega}(2) = 6 - 2n < 0$ for $n \geq 4$
(c) two loops, Sec. 6.4

FIGURE 15. Graphs behind the argument. Wavy lines are quantum gravitons (or ghosts), dashed lines end on background insertions $\otimes$. (a) A one-loop ring contributing to $\Gamma_1[\bar{g}]$. Replacing any vertex or propagator by its nonlocal remainder lowers the degree by at least $2n$, so the ultraviolet pole is that of the local theory. (b) The insertion of a quartic operator with two background curvatures: the loop of $(-\Box)^{n-3}$ between two linearized curvatures has an integrand $\propto k^{-4}$ and gives the poles of Table 6. (c) A two-loop graph. The graph as a whole is convergent for $n \geq 4$; the shaded one-loop subgraph $\gamma$ has degree 4 and is the only possible source of a subdivergence.

Section 6.4 proves that a connected hard component with at least two loops has strictly negative degree. This leaves the one-loop subdivergences, including those in graphs with several loops. To infer all-order finiteness of renormalized physical amplitudes from these facts, the forest operation has to be valid for the momentum-dependent vertices in every mixed hard and soft region, with the relevant Taylor remainders improving the hard degree. The quantum-field and source counterterms have to satisfy the Slavnov–Taylor identity. Under those conditions the cohomological argument is as follows. The one-loop local invariant part is fixed by its value on background metrics. The background tuning sets that part to zero. Exact terms are absorbed in the BRST sector or by perturbative field redefinitions, and do not enter on-shell matrix elements under the usual equivalence and BRST hypotheses [40–42]. The negative degree then excludes additional overall higher-loop ultraviolet counterterms once the subdivergences are subtracted. The gauge-sector subtractions themselves are distinct from the three tuned physical couplings. The power-counting estimate alone does not supply all the mixed Taylor-remainder and Slavnov–Taylor identities for this nonlocal theory.

8.1. **Subtraction conventions.** The pole cancellation here uses dimensional regularization at fixed external kinematics and with the infrared separated from the ultraviolet. A momentum cutoff can produce power terms proportional to $\Lambda^4 \int \sqrt{g}$ and $\Lambda^2 \int \sqrt{g}\, R$. Their subtraction depends on the scheme [40]. A finite local term can also be added. The next calculation

specifies a common subtraction, after which an ordinary convergent integral remains. Soft graviton singularities at on-shell kinematics are governed by the low-momentum graviton pole and have to be treated separately from these ultraviolet statements [16].

8.2. **The zero-point energy of the graviton.** The absence of a pole in $\int\sqrt{\bar g}$ does not mean that the one-loop effective action has no such term. It has a finite one, and it can be computed exactly. On flat space $\Gamma_1 = (\rho_1 + \delta\rho_\Pi)\int \mathrm{d}^4x$, where $\rho_1 + \delta\rho_\Pi$ is the zero-point energy density of the graviton together with the ghosts and the compensating factor of Sec. 6.1; $\rho_1$ comes from the skeleton and $\delta\rho_\Pi$ from the fluctuation $R_\Pi$ of the realization. Around flat space the graviton operator is $p^2a(p^2)(\mathbf{1} - \frac{1}{2}\delta\otimes\delta)$ on the ten components of $h_{\mu\nu}$, the ghost operator is $p^2a(p^2)\,\delta_{\mu\nu}$ on four components, and the compensating factor is $\frac{1}{2}\operatorname{Tr}\log a$ on four components. A real bosonic component contributes $\frac{1}{2}\log$ of its eigenvalue and a ghost–antighost pair contributes $-\log$, so per momentum the integrand is

$$5\log(p^2a) - 4\log(p^2a) + 2\log a = \log p^2 + 3\log a, \tag{86}$$

up to a constant from $\det(\mathbf{1}-\frac{1}{2}\delta\otimes\delta)$ that, like $\int \mathrm{d}^Dk$ of any constant, vanishes in dimensional regularization. The same integrand follows without gauge fixing. The six transverse components of $h_{\mu\nu}$ each contribute $\frac{1}{2}\log(p^2a)$, and the Jacobian of the split of $h_{\mu\nu}$ into its transverse part and a gauge orbit contributes $-2\log p^2$, with no factor of $a$. The coefficient 3 of $\log a$ is therefore independent of the gauge.

The first term in Eq. (86) is that of general relativity. Its integral $\int \mathrm{d}^Dk\,\log k^2$ has no scale and vanishes in dimensional regularization. The second term is $3\log a = \frac{3n}{2}\,\mathrm{Ein}(k^4/M^4) + 3R_\Pi(k^2)$, by Eq. (18). We take the skeleton part first. With $w = k^4/M^4$ and $\Omega_D = 2\pi^{D/2}/\Gamma(D/2)$ the area of the unit sphere,

$$\int \mathrm{d}^Dk\;\mathrm{Ein}\Big(\frac{k^4}{M^4}\Big) = \frac{\Omega_D M^D}{4}\int_0^\infty w^{D/4-1}\;\mathrm{Ein}(w)\,\mathrm{d}w.$$

The last integral is a Mellin transform, and for Ein it is elementary. For $-1 < \operatorname{Re} s < 0$ one has $\int_0^\infty w^{s-1}(1-e^{-wt})\,\mathrm{d}w = -\Gamma(s)\,t^{-s}$. Integrating over $t\in[0,1]$ with the weight $\mathrm{d}t/t$, as in the definition of Ein, gives

$$\int_0^\infty w^{s-1}\;\mathrm{Ein}(w)\,\mathrm{d}w = \frac{\Gamma(s)}{s}, \qquad -1 < \operatorname{Re} s < 0. \tag{87}$$

The strip is where the integral converges, since $\mathrm{Ein}(w)\simeq w$ at small $w$ and $\mathrm{Ein}(w)\simeq\log w$ at large $w$. Dimensional regularization defines the radial integral by continuation in $D$ [40], that is in $s = D/4$, and the right side of Eq. (87) is meromorphic with poles only at $s = 0, -1, -2, \ldots$. At $D = 4$, that is $s = 1$, it is regular and equals $\Gamma(1) = 1$. Hence $\int \mathrm{d}^4k\,(2\pi)^{-4}\;\mathrm{Ein}(k^4/M^4) =$

$2\pi^2 M^4/(4\cdot 16\pi^4) = M^4/(32\pi^2)$, and

$$\rho_1 = \frac{3n}{2}\,\frac{M^4}{32\pi^2} = \frac{3n\,M^4}{64\pi^2}. \tag{88}$$

There is no pole, as Sec. 7.3 requires, and the value is positive. In the normalization of Eq. (57), where a cosmological term enters as $-\frac{2}{\kappa^2}\int\sqrt{g}\,(-2\Lambda)$, Eq. (88) is a finite cosmological constant $\Lambda_1 = \frac{\kappa^2}{4}\rho_1 = 3nGM^4/(8\pi)$. For a general admissible response the second term of Eq. (86) is $3n\Phi(k^2/M^2)$, and Eq. (34) gives

$$\rho_1 = \frac{3nM^4}{32\pi^2}\,\mu_\psi = \frac{3nM^4}{32\pi^2}\int_0^\infty t\,\big(1-\psi(t)\big)\,\mathrm{d}t, \tag{89}$$

positive and finite for every response that saturates; Table 1 lists three values.

The fluctuation term is the realization's own contribution. In dimensional regularization the integral of the constant $R_\infty$ vanishes, and what is left, $R_\Pi - R_\infty$, falls along the Euclidean axis like $k^{-\sigma+2\varepsilon}$ by Eq. (41), faster than $k^{-4}$, so its integral converges absolutely and is the same in every regularization. Inserting Eq. (38) and integrating brane by brane with Eq. (35),[23]

$$\begin{aligned}\delta\rho_\Pi &= 3\int\frac{\mathrm{d}^4k}{(2\pi)^4}\big[R_\Pi(k^2)-R_\infty\big]\\ &= -\frac{3M^4\mu_\psi}{16\pi^2}\int\frac{\vartheta(s)}{(sM^2)^2}\,(\Pi-\nu)(\mathrm{d}s).\end{aligned} \tag{90}$$

This is a compensated sum over the branes, with mean zero and, by Eq. (15), variance

$$\Big(\frac{3M^4\mu_\psi}{16\pi^2}\Big)^2 n\int_0^1\vartheta(\tau)\,\tau^{-4}\,\frac{\mathrm{d}\tau}{\tau}.$$

The integral converges when $V(t) = O(t^\sigma)$ with $\sigma > 4$, which (G) guarantees. For the power law (13) the variance is $(3M^4\mu_\psi/16\pi^2)^2\,n\vartheta_0/(\zeta-4)$, and relative to the mean (89) the standard deviation is

$$\frac{(\mathbb{E}\,\delta\rho_\Pi^2)^{1/2}}{\rho_1} = 2\Big[\frac{\vartheta_0}{n(\zeta-4)}\Big]^{1/2}, \tag{91}$$

independent of the response. Table 2 gives 0.37 for $n=4$ and $\zeta=9$, and 0.83 for $\zeta=5$. The zero-point energy of a realization is therefore $\rho_1+\delta\rho_\Pi$, a finite number that is the same for every cutoff under the common subtraction, whose mean over realizations is Eq. (89). Each brane lowers it, by an amount proportional to $\vartheta(s)/s^2$; the positive mean is the dimensionally continued sum over the infinitely many small branes, $\int_0^1\tau^{-2}\,\mathrm{d}\tau/\tau = -\frac{1}{2}$ by

[23] The exchange of the momentum integral and the compensated sum is justified by truncation. For the branes larger than $\eta/M^2$ the sum is finite and the exchange is trivial. As $\eta\to 0$ the right side of Eq. (90) converges in mean square by Eq. (15), and so does the left side, by Minkowski's inequality: the root-mean-square difference at momentum $k$ is at most $[nV(\eta)]^{1/2}$ and, by Eq. (40), at most $C|k|^{-\sigma}$ for large $k$, which is integrable.

continuation, which is how a sum of negative terms acquires the positive value (88).

The number in Eq. (88) is not a property of dimensional regularization. Cut the radial integral off at $w = W$, that is at $k = \Lambda$ with $W = \Lambda^4/M^4$. Integrating by parts with $\mathrm{Ein}'(w) = (1 - e^{-w})/w$ and using Eq. (25),

$$\begin{aligned}\int_0^W \mathrm{Ein}(w)\,\mathrm{d}w &= W\,\mathrm{Ein}(W) - W + 1 - e^{-W} \\ &= W\log W + (\gamma_{\mathrm{E}} - 1)W + 1 + O(e^{-W}).\end{aligned}$$

With a smooth cutoff the same integral is

$$\begin{aligned}\int_0^\infty \mathrm{Ein}(w)\,e^{-w/W}\,\mathrm{d}w &= W\log(1+W) \\ &= W\log W + 1 + O(W^{-1}),\end{aligned}$$

because $\int_0^\infty e^{-w/W}(1-e^{-wt})\,\mathrm{d}w = W^2t/(1+Wt)$ and $\int_0^1 W^2\,\mathrm{d}t/(1+Wt) = W\log(1+W)$. The terms that grow with $W$, of order $\Lambda^4\log(\Lambda/M)$ and $\Lambda^4$, differ between the two cutoffs. The constant is 1 in both, the value of $\Gamma(s)/s$ at $s = 1$, and this is true for any cutoff function.[24] The power-law terms are the scheme artifacts of Sec. 8.1; the finite part is universal. Figure 16 shows both halves of the argument: the Mellin transform with its strip of convergence and its value at $s = 1$, and the remainder after the growing terms are removed, for a one-parameter family of cutoffs that ends in the sharp one.

*Why the exponent in (G) is* 4. The fluctuation (90) used (G) as a sufficient condition. It is also necessary, up to factors that vary more slowly than any power, and the reason is dimensional. A brane of size $s$ changes the zero-point energy by

$$g(\tau) = -K\,\frac{\vartheta(\tau)}{\tau^2}, \qquad K = \frac{3M^4\mu_\psi}{16\pi^2}, \qquad \tau = sM^2,$$

an energy density set by its own scale, and the variance of the sum over the gas adds the squares of these changes, $\int g^2\,\mathrm{d}\nu = K^2 n\int \vartheta\,\tau^{-4}\,\mathrm{d}\tau/\tau$. The fourth power is the dimension of an energy density. Every regulator removes the branes too small to be resolved. Removing those smaller than $\eta/M^2$ leaves the compensated sum $X_\eta = \int_{\tau>\eta} g\,\mathrm{d}(\Pi - \nu)$, and the zero-point energy of the realization is its limit as $\eta \to 0$. The limit exists, almost surely, if and

[24] Let $r$ be smooth on $[0,\infty)$ and rapidly decreasing, with $r(0) = 1$, and let $\tilde r(s) = \int_0^\infty u^{s-1}r(u)\,\mathrm{d}u$. By the Mellin–Parseval formula, $\int_0^\infty \mathrm{Ein}(w)\,r(w/W)\,\mathrm{d}w = \frac{1}{2\pi i}\int_{\mathrm{Re}\,s=s_0}\frac{\Gamma(s)}{s}\,\tilde r(1-s)\,W^{1-s}\,\mathrm{d}s$ with $-1 < s_0 < 0$. Move the contour to the right. The double pole of $\Gamma(s)/s$ at $s = 0$ gives the terms $W\log W$ and $W$, whose coefficients involve $\tilde r(1)$ and $\tilde r'(1)$ and so depend on the cutoff. The simple pole of $\tilde r(1-s)$ at $s = 1$, where $\tilde r(1-s) \simeq r(0)/(1-s)$, gives the constant $\Gamma(1)\,r(0) = 1$ for every $r$. What remains decays as a negative power of $W$.

only if

$$n\int_0^1 \min\Big(\frac{\vartheta(\tau)}{\tau^4},\,\frac{1}{\tau^2}\Big)\frac{\mathrm{d}\tau}{\tau} < \infty, \tag{92}$$

and when the integral diverges $X_\eta$ has no limit even in distribution.[25] For a gas in which smaller branes are never stronger, $\vartheta$ nondecreasing, Eq. (92) is equivalent to

$$\int_0^1 V(\tau)\,\tau^{-5}\,\mathrm{d}\tau < \infty. \tag{93}$$

This requires $V(t) = o(t^4)$, and (G) implies it.[26] The two differ only by profiles that approach $t^4$ more slowly than any power, such as $V(t) = t^4/\log^2(1/t)$, which satisfies Eq. (93) but not (G). For the power law (13) both hold exactly when $\zeta > 4$.

A momentum cutoff shows the same threshold, and shows what goes wrong below it. With $|k| < \Lambda$ a brane contributes $g(\tau)\,\mu(\tau\Lambda^2/M^2)/\mu_\psi$ instead of $g(\tau)$, where $\mu(U) = \int_0^U u\,[1-\psi(u)]\,\mathrm{d}u$ rises from 0 to $\mu_\psi$. For the power law and $\psi_*$ the realization's part of the regulated zero-point energy, $\delta\rho_\Lambda$, has mean zero and, for $0 < \zeta < 4$ and $\Lambda \gg M$, the variance

$$\begin{aligned}\mathbb{E}\,\delta\rho_\Lambda^2 &\simeq K^2 n\vartheta_0\,C_\zeta\Big(\frac{\Lambda}{M}\Big)^{2(4-\zeta)},\\ C_\zeta &= \tfrac{1}{2}\,\Gamma\big(\tfrac{\zeta}{2}-2\big)\big(2^{\,2-\zeta/2}-2\big),\end{aligned} \tag{94}$$

[25]The increments of $X_\eta$ over disjoint ranges of size are independent, and Eq. (14) with $f = e^{iug} - 1$ gives $|\mathbb{E}\,e^{iuX_\eta}| = \exp\int_{\tau>\eta}[\cos(ug)-1]\,\mathrm{d}\nu$. The average of $1-\cos(ug)$ over $u \in [-h,h]$ is $1-\sin(hg)/(hg)$, which is at least $\frac{1}{10}\min(1,h^2g^2)$. If $\int \min(1,g^2/K^2)\,\mathrm{d}\nu = \infty$, the average of $-\log|\mathbb{E}\,e^{iuX_\eta}|$ over $[-h,h]$ therefore diverges as $\eta \to 0$, for every $h > 0$. A limit in distribution is excluded, because its characteristic function would equal 1 at $u = 0$, be continuous there, and be the uniform limit of those of $X_\eta$ on $[-h,h]$ [25]. If $\int \min(1,g^2/K^2)\,\mathrm{d}\nu < \infty$, split the branes at $|g| = K$. Those with $|g| \le K$ give a compensated sum with $\int g^2\,\mathrm{d}\nu < \infty$, which converges as in Sec. 4.4. Those with $|g| > K$ have a finite expected number, so almost surely there are finitely many and their sum converges; their compensator $\int_{\tau>\eta,\,|g|>K} g\,\mathrm{d}\nu$ converges if and only if $\int_{|g|>K}|g|\,\mathrm{d}\nu < \infty$, because $g$ has one sign, and otherwise $X_\eta \to +\infty$. The two conditions together are Eq. (92), since $g^2\,\mathrm{d}\nu = K^2 n\vartheta\tau^{-4}\,\mathrm{d}\tau/\tau$ and $|g|\,\mathrm{d}\nu = Kn\tau^{-2}\,\mathrm{d}\tau/\tau$.

[26]If $\vartheta$ is nondecreasing, the part of the integral (92) over $[t/2,t]$ is at least $n\log 2\,\min\big(\vartheta(t/2)\,t^{-4},t^{-2}\big)$ and tends to zero with $t$, so $\vartheta(\tau) = o(\tau^4)$ and the minimum is $\vartheta\tau^{-4}$ near $\tau = 0$. Since $\mathrm{d}V = \vartheta\,\mathrm{d}\tau/\tau$, Eq. (92) then reads $\int_0^1 \tau^{-4}\,\mathrm{d}V < \infty$. As $V(t)\,t^{-4} \le \int_0^t \tau^{-4}\,\mathrm{d}V$, which tends to zero, integration by parts gives $\int_0^1 \tau^{-4}\,\mathrm{d}V = V(1) + 4\int_0^1 V\tau^{-5}\,\mathrm{d}\tau$. Conversely $\int_t^{2t} V\tau^{-5}\,\mathrm{d}\tau \ge \frac{15}{64}V(t)\,t^{-4}$, so Eq. (93) forces $V(t)\,t^{-4} \to 0$ and the integration by parts runs backwards; since the minimum in Eq. (92) is at most $\vartheta\tau^{-4}$, this direction needs no monotonicity. Under (G), $\int_0^1 V\tau^{-5}\,\mathrm{d}\tau \le C/(\sigma-4)$.

which at $\zeta = 4$ grows like $2K^2 n\vartheta_0 \log(\Lambda/M)$ and for $\zeta > 4$ tends to the variance $K^2 n\vartheta_0/(\zeta - 4)$ of Eq. (90).[27] Under Eq. (92) every cutoff, on momenta or on sizes and of any shape, converges to the same $\delta\rho_\Pi$.[28] Without it the regulated value of a realization contains a random term whose spread grows like $(\Lambda/M)^{4-\zeta}$ and which has no limit, by the argument of the footnote to Eq. (92). A term that grows with the cutoff and has a fixed coefficient is a scheme artifact of the kind discussed in Sec. 8.1, and it changes in a known way when the regulator changes. This one differs from realization to realization, two cutoffs of different shape disagree about it by an amount whose spread grows in the same way, and the bare cosmological constant could absorb it only by being readjusted for each realization and each cutoff, without a limit. The zero-point energy then has no value that is the same for every cutoff under the common subtraction. Condition (G) is what keeps the finite answer of this section, and the exponent in it is the dimension of the quantity computed. Figure 17 shows the threshold.

The finite remainder in this section uses a fixed subtraction. Dominated convergence gives the same remainder for uniformly bounded cutoffs tending pointwise to one. A finite cosmological counterterm changes its interpretation as a contribution to the absolute vacuum energy.[29] In general relativity the zero-point integrand is $\log k^2$ alone. Its integral is zero in dimensional regularization and quartic with a cutoff, and it has no finite part that is independent of the regulator. In the theory of the gas the form factor turns it into a definite number set by $M$, $n$ and the realization. Newton's constant receives a finite shift of the same kind, proportional to $M^2$, which we have not computed. The cosmological constant is then the sum of a bare term, which is not renormalized, and $\Lambda_1$. Making the total small is still a choice of the bare term, but it is now a choice between two finite numbers rather than the subtraction of a divergence, and the theory does not explain why

[27] By Eq. (15) the variance is $\int g^2\,[\mu(\tau\Lambda^2/M^2)/\mu_\psi]^2\,\mathrm{d}\nu$. For $\psi_*$, $\mu(U)/\mu_\psi = 1 - e^{-U^2}$, and the substitution $\tau = vM^2/\Lambda^2$ turns the variance into $K^2 n\vartheta_0 (\Lambda/M)^{2(4-\zeta)} \int_0^{\Lambda^2/M^2} v^{\zeta-5}(1 - e^{-v^2})^2\,\mathrm{d}v$. For $0 < \zeta < 4$ the integral converges as $\Lambda \to \infty$. With $u = v^2$ and $\int_0^\infty u^{s-1}(1 - e^{-u})^2\,\mathrm{d}u = \Gamma(s)\,(2^{-s} - 2)$, which follows from $\int_0^\infty u^{s-1}(e^{-bu} - 1)\,\mathrm{d}u = b^{-s}\Gamma(s)$ for $-1 < \mathrm{Re}\,s < 0$ and extends to $-2 < \mathrm{Re}\,s < 0$ by continuation, it equals $C_\zeta$. At $\zeta = 2$ the formula is read as its limit, $C_2 = \log 2$, and $C_\zeta \simeq 1/(4 - \zeta)$ as $\zeta \to 4$. For $\zeta > 4$ the integral grows like $(\Lambda/M)^{2(\zeta-4)}/(\zeta - 4)$, and the product tends to $K^2 n\vartheta_0/(\zeta - 4)$.

[28] A cutoff of shape $r$ on momenta, with $0 \le r \le 1$ and $r(0) = 1$, replaces $g$ by $g\,m_r(\tau\Lambda^2/M^2)$, where $m_r(U) = \mu_\psi^{-1} \int_0^\infty u\,[1 - \psi(u)]\,r(u/U)\,\mathrm{d}u$ lies in $[0, 1]$ and tends to 1 as $U \to \infty$. A cutoff on sizes replaces it by $g$ times the indicator of $\tau > \eta$. In both cases the modified contribution is bounded by $|g|$ and tends to $g$ brane by brane. Split the branes at $|g| = K$. For those with $|g| \le K$ the compensated sums converge in mean square, by Eq. (15) and dominated convergence; those with $|g| > K$ are finitely many, and their compensator converges by dominated convergence.

[29] For the skeleton, $\mathrm{Ein}(u) - \log u - \gamma_\mathrm{E} = E_1(u)$, and $\int_0^\infty E_1(u)\,\mathrm{d}u = 1$ by Tonelli. This is an ordinary integrable remainder at both endpoints.

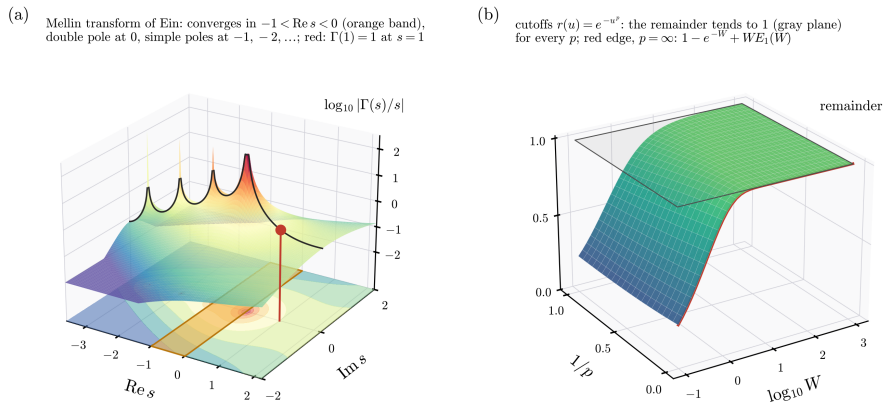



FIGURE 16. The finite zero-point energy. (a) $\log_{10} |\Gamma(s)/s|$ over the complex $s$ plane, the Mellin transform (87) of Ein. The integral converges in the strip $-1 < \mathrm{Re}\, s < 0$ (orange band on the floor, which also carries the contours of the surface); its continuation has a double pole at $s = 0$ and simple poles at $s = -1, -2, \ldots$ (black curve along the real axis), and is regular at $s = 1$, where it equals 1 (red). That value is the whole answer (88). (b) The integral $\int_0^\infty \mathrm{Ein}(w)\, r(w/W)\, \mathrm{d}w$ with the cutoffs $r(u) = e^{-u^p}$, after the terms that grow with $W$ are subtracted, $\tilde{r}(1)\, W \log W + [\tilde{r}'(1) + \gamma_{\mathrm{E}} \tilde{r}(1)]\, W$, with $\tilde{r}$ the Mellin transform of $r$. The surface is drawn over $\log_{10} W$ and $1/p \in [0, 1]$. The coefficients of the subtracted terms depend on $p$, the remainder does not: it tends to 1 (gray plane) for every $p$, and at $W = 10^3$ it lies within $10^{-3}$ of 1 across the whole family. The red edge $1/p = 0$ is the sharp cutoff, where the remainder is $1 - e^{-W} + WE_1(W)$.

the total is small. Matter fields, which this paper does not include, would add zero-point energies of their own.

## 9. Cutting rules and physical states

The spectrum of Sec. 5.3 is a statement about the tree-level propagator. Loops raise a separate question, because $1/a$ is unbounded near the imaginary $z$ axis, as Sec. 4.6 shows for the skeleton, and the loop energies cannot be Wick-rotated in the usual way. A theory with such a form factor has to be defined by a prescription. We use the one that goes back to Efimov [5] and was put in its modern form by Pius and Sen [14]. Loop integrals are defined with Euclidean loop momenta. Lorentzian amplitudes are the analytic continuations of the Euclidean ones in the external energies, and during the continuation the contours of the loop energies are deformed to avoid the poles of the propagators while their end points stay at $\pm i\infty$. Figure 18(a) shows

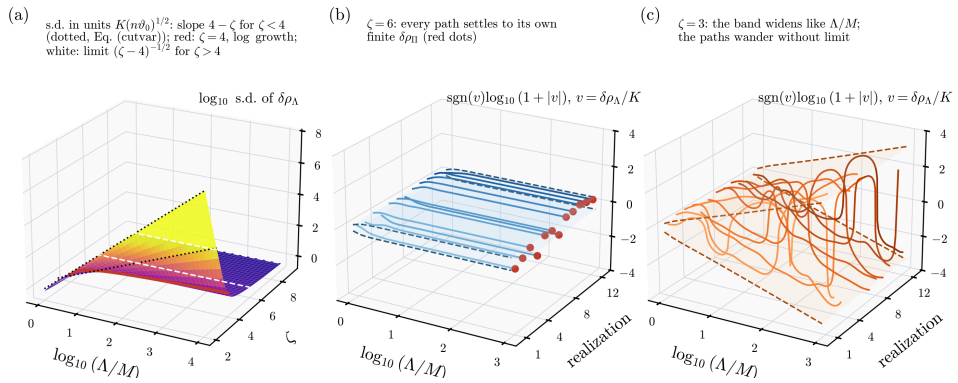



FIGURE 17. Why condition (G) needs $\zeta > 4$, for $n = 4$, $\vartheta_0 = \log 2$, $\psi_*$ and a sharp momentum cutoff $\Lambda$. (a) The standard deviation of the realization's part $\delta\rho_\Lambda$ of the zero-point energy, in units of $K(n\vartheta_0)^{1/2}$, as a surface over $\log_{10}(\Lambda/M)$ and $\zeta$, in $\log_{10}$, computed from the exact integral in the footnote to Eq. (94). For $\zeta > 4$ it levels off at $(\zeta-4)^{-1/2}$, the spread of the finite $\delta\rho_\Pi$ of Eq. (90); for $\zeta < 4$ it grows with slope $4-\zeta$ (dotted lines, Eq. (94)); the red curve is $\zeta = 4$, where the growth is logarithmic. (b) Twelve realizations of $v = \delta\rho_\Lambda/K$ as functions of the cutoff for $\zeta = 6$, on the compressed scale $\mathrm{sgn}(v)\log_{10}(1+|v|)$, with the band $\pm(\mathbb{E}\,\delta\rho_\Lambda^2)^{1/2}/K$ (translucent sheet with dashed edges). Every path settles to its own finite value, marked by a dot. (c) The same for $\zeta = 3$. The band widens like $\Lambda/M$ and the paths wander without limit, so the zero-point energy of a realization depends on where the regulator is put. In (b) and (c) the branes smaller than $0.3\,M^{-2}$ are represented, shell by shell in $\log\tau$, by their Gaussian limit; the larger ones are drawn individually.

why the end points matter: the propagator is small along the Euclidean contour and along the real axis away from its pole, and in the wedges between them it swings between very large and very small values.

In the Euclidean prescription the loop-energy contours begin on the imaginary axes and are deformed when the external energies approach the physical boundary. For an infrared-regulated amplitude with convergent subtracted integrals on the deformation domain, analyticity determines the continuation along the chosen path. The Euclidean prescription must be specified because large arcs need not vanish. Explicit comparisons of prescriptions for scalar bubbles are given in Ref. [49]. Section 9.3 evaluates a one-loop bubble in the brane-gas model with the same distinction between contour deformation and direct real-energy integration.

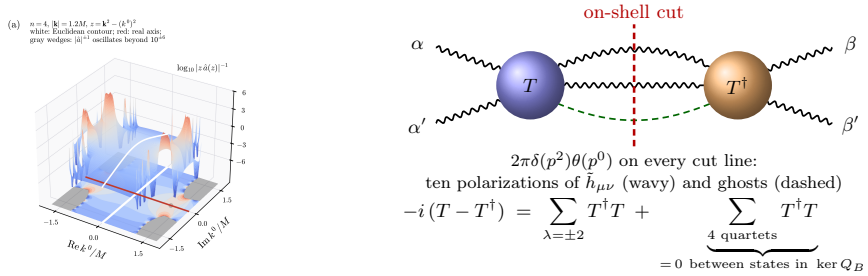



FIGURE 18. Unitarity in the Efimov–Pius–Sen prescription. (a) The size of a propagator, $\log_{10}|z\,\hat{a}(z)|^{-1}$ for the skeleton, over the complex plane of the loop energy $k^0$ at fixed spatial momentum $|\mathbf{k}| = 1.2M$, for $n = 4$, with $z = \mathbf{k}^2 - (k^0)^2$. Along the Euclidean contour (white, $\mathrm{Re}\,k^0 = 0$, drawn on the floor and on the surface) and along the real axis away from the poles at $k^0 = \pm|\mathbf{k}|$ (red, black dots) the propagator is small. In the wedges around $\arg k^0 = \pm\pi/4, \pm3\pi/4$, where $z$ is close to the imaginary axis, $\log|\hat{a}|$ has an oscillating part of size $\frac{n}{2}|E_1(z^2/M^4)|$, which grows like $e^{|\mathrm{Re}\,z^2|/M^4}$, so the propagator swings between extremely large and extremely small values; this is why the loop energies cannot be rotated to the real axis. The surface is clipped at $\pm6$ and left out where that oscillating part exceeds 6 decades (gray on the floor). (b) The cutting rule of Sec. 9.1: the anti-Hermitian part of an amplitude is the sum over cuts with all eighteen on-shell modes of the redefined fields. Between physical states, those in $\ker Q_B$, the contributions of the four BRST quartets cancel (Sec. 9.2), and only the two helicities of the graviton remain.

The following argument would give physical perturbative unitarity in this prescription, provided the cutting-rule theorem of Refs. [13, 14, 95] holds for every theory with the properties (P1)–(P3) below. The proof has the two steps familiar from local gauge theories. First, the amplitudes obey the Cutkosky cutting rules [79] with all ten components of the graviton and the ghosts on the cut lines, which is unitarity in the space of all asymptotic states, whose metric is indefinite (Sec. 9.1). Second, BRST invariance removes the states of negative and zero norm by the quartet mechanism of Kugo and Ojima (Sec. 9.2). Table 7 lists every input and where it is established.

9.1. **Cutting rules.** Briscese and Modesto proved the Cutkosky rules for nonlocal theories whose form factor $e^H$ has the following four properties [13]:

(U1) $H$ is an entire function;

(U2) $e^H$ has no zeros and no poles at finite momentum;

(U3) $e^H$ grows without bound at the ends of the loop-energy contours and as the spatial loop momentum grows;

(U4) $H$ is asymptotically logarithmic in a cone around the real axis of the squared momentum.

Their proof is written out for a scalar field with a local kinetic term, obtained by a field redefinition, and vertices that carry a factor $e^{-H/2}$ for every leg. It follows the proof of Pius and Sen for string field theory [14], and Chin and Tomboulis reached the same cutting rule for general entire vertices by way of the Landau equations [43]. In all three the Feynman rules enter through three properties only:[30]

(P1) the propagators are those of a local theory;

(P2) the vertices are entire functions of the momenta on their legs, with real Taylor coefficients in Euclidean variables;

(P3) on the deformed contours the integrand of every graph, together with its counterterms, is absolutely integrable at the ends of the contours, uniformly for external momenta in compact sets.

We verify them for the gravitational theory of the gas.

*(P1): a field redefinition.* In the Feynman–de Donder gauge of Sec. 6.1 the quadratic operators of $h_{\mu\nu}$ and of the ghosts around flat space are $a(p^2)$ times those of general relativity, Eq. (66). Put

$$\begin{aligned} h_{\mu\nu} &= a(-\Box)^{-1/2}\,\tilde{h}_{\mu\nu}, \\ c^\mu = a(-\Box)^{-1/2}\,\tilde{c}^\mu, &\qquad \bar{c}_\mu = a(-\Box)^{-1/2}\,\tilde{\bar{c}}_\mu, \end{aligned} \tag{95}$$

with $\Box$ the flat d'Alembertian. Since $a = e^H$ with $H$ entire, $a^{-1/2} = e^{-H/2}$ is entire as well, and the quadratic action of $\tilde{h}$ and the ghosts is that of general relativity in Feynman–de Donder gauge. That is (P1); the redefined propagators have a single pole, at $p^2 = 0$, with the residue of general relativity. The Jacobian of Eq. (95) does not depend on the fields and drops out. The redefinition does not change the $S$ matrix. It is linear, and on shell it is the identity, since $a(0) = 1$, so the residues that enter the reduction formula are unchanged.[31]

*(P2): entire vertices.* After the redefinition every vertex is the vertex of Sec. 6.3 multiplied by $a^{-1/2}(\ell_j^2)$ for each quantum leg. By Sec. 6.2 the vertex of Sec. 6.3 is a finite sum of polynomials in the momenta times divided differences of the entire functions $\mathcal{G}$ and $a - cx^n$, and a divided difference

[30] The Gaussian or exponential form of the vertices in Refs. [13, 14, 95] is used to show that the integrals converge at the ends of the deformed contours, which is (P3). The rest of the argument, the continuation of the Euclidean amplitude into the physical region from the upper half plane of the external energies, the pinching of the contours at the poles of the propagators, and the identification of the anti-Hermitian part with the sum over cuts, uses only the location of the poles and the analyticity of the vertices.

[31] This is the equivalence theorem for field redefinitions in its simplest form: a linear redefinition $h = Z(-\Box)\tilde{h}$ multiplies every amputated Green function by $Z(p_j^2)$ on each external leg, and at $p_j^2 = 0$ these factors are $a(0)^{-1/2} = 1$.

of an entire function is an entire function of all its arguments, by Eq. (68). Products of entire functions are entire, and all Taylor coefficients are real because $\psi$ is real on the real axis and the sizes and strengths of the branes are real. The counterterms of Sec. 8 are local, with real coefficients. This is (P2).

*(P3): the ends of the contours.* Take a line that carries the loop momentum $k$ plus a combination $q$ of external momenta. Near the end of a loop-energy contour the Euclidean loop energy is real and large, $k_4 = \kappa \to \pm\infty$, while the continued external energies are complex but bounded. The squared Euclidean momentum of the line is then $z = (\kappa + q_4)^2 + |\mathbf{k}+\mathbf{q}|^2 = (\kappa^2 + \mathbf{k}^2)\,[1 + O(|q|/|\kappa|)]$, so $\arg z \to 0$, and the same holds when $|\mathbf{k}| \to \infty$ at fixed $\kappa$. In the language of Appendix C, all squared momenta of lines and of running momenta in vertex chains lie in a region $\Omega = \operatorname{conv} K + S_\delta$, a compact set plus a cone around the positive axis. On $\Omega$ the cone asymptotics (42) give $|a(z)| \geq \frac{c}{2}|z/M^2|^n$ once $|z|$ is large, so the propagator obeys Eq. (67) with $p^2$ replaced by $|z|$, and Appendix C shows that the divided-difference bound (70), and with it the vertex bound (71), hold on $\Omega$ with $|\ell|$ replaced by $|\operatorname{Re}\ell|$ plus a constant. Every factor of the integrand therefore obeys on the contour the bound it obeys on the Euclidean section, and the power counting of Sec. 6.4, applied to the subtracted integrands,[32] gives (P3). Properties (U3) and (U4) are the special case of this statement for a single propagator.

The scalar and string-field proofs of the cutting rule [13, 14, 80, 95] motivate conditions (P1)–(P3). For the full gravitational amplitude the application also requires compatible contour estimates for the subtracted tensor and ghost graphs and the BRST identities after continuation. The bubble below is a direct check at one loop.

9.2. **Physical states.** The space $\mathcal{F}$ contains states of negative and zero norm, and the cuts of Sec. 9.1 run over all of them. They are removed by BRST symmetry, as in Einstein gravity. Kugo and Ojima proved [15, 44] that the physical $S$ matrix is unitary with a positive inner product if

(Q1) the gauge-fixed theory has a nilpotent BRST charge $Q_B$ that commutes with the $S$ matrix;

(Q2) the asymptotic fields are free fields on which $Q_B$ acts by the linearized BRST transformation, so that the unphysical modes form quartets;[33]

[32] The integrals that diverge are the one-loop subgraphs, with $\bar\omega = 4$. Subtract from each its Taylor expansion to fourth order in its external momenta at a Euclidean point, as in the BPHZ scheme [40]. Derivatives with respect to external momenta lower the degree of the vertex and propagator bounds by one each, because the functions involved are symbols in the sense of Eqs. (46) and (70), so the subtracted subgraphs have negative degree and the subtracted integrand of every graph is absolutely convergent by Weinberg's theorem. The subtraction differs from the counterterms of Sec. 8 by local terms with real coefficients, which are vertices of the kind allowed by (P2).

[33] A quartet is a pair of BRST doublets: a mode $\chi$ with $Q_B\chi = \gamma$ and an antighost mode $\bar\gamma$ with $Q_B\bar\gamma = \beta$, where $\gamma$ and $\bar\gamma$ have ghost number $\pm 1$ and the only nonzero inner products are $\langle\chi|\beta\rangle$ and $\langle\gamma|\bar\gamma\rangle$. Every state built with at least one quartet mode is either

(Q3) the $S$ matrix is unitary in the full space $\mathcal{F}$.

Under these conditions $S$ maps $\ker Q_B$ into itself, and it is unitary on the quotient $\ker Q_B / \operatorname{im} Q_B$, whose inner product is positive definite. For linearized gravity in de Donder gauge the eight unphysical polarizations of the graviton and the eight ghost modes form four quartets, one for each value of the vector index, and the quotient contains the two helicities of the graviton and nothing else [15].

Condition (Q3) requires the contour result discussed in Sec. 9.1. At the classical level of (Q1), the action (57) is invariant under diffeomorphisms, being built from curvatures and the covariant d'Alembertian, and the gauge-fixing and ghost terms of Sec. 6.1 are together the BRST variation of a gauge-fixing fermion, $\bar{c}^\mu a(-\bar{\Box})(\chi_\mu - \frac{1}{2} b_\mu)$ with a Nakanishi–Lautrup field $b_\mu$ that is integrated out. The weight depends on the background only and is inert under BRST, so the gauge-fixed action is BRST invariant and $Q_B^2 = 0$. The Slavnov–Taylor identities follow from that invariance and the invariance of the measure, as in local theories, and with them $[Q_B, S] = 0$ [15]; the counterterms of Sec. 8 are gauge invariant or BRST exact and preserve this. For (Q2), the redefinition (95) leaves the linear part of the BRST transformation unchanged. Indeed $s\, h_{\mu\nu} = \partial_\mu c_\nu + \partial_\nu c_\mu + O(\text{fields}^2)$ becomes

$$s\, \tilde{h}_{\mu\nu} = a^{1/2} \big( \partial_\mu c_\nu + \partial_\nu c_\mu \big) + \cdots = \partial_\mu \tilde{c}_\nu + \partial_\nu \tilde{c}_\mu + \cdots \,,$$

because $a^{1/2}$ and $a^{-1/2}$ act on the same momentum. The free action is that of general relativity in Feynman–de Donder gauge (P1). So the asymptotic fields and the asymptotic BRST charge are exactly those of linearized general relativity, and the quartets are the ones found by Kugo and Ojima. There are no further asymptotic states, because the redefined propagators have no other poles, which is the content of (U2).[34]

This cancellation gives physical perturbative unitarity if the contour cutting theorem extends to the full renormalized tensor and ghost amplitudes, the BRST identities hold for that prescription, and the free asymptotic states give the relevant cohomology. For massless gravitons this holds only after the soft divergences have been dealt with, by an infrared regulator or by passing to inclusive rates [16], exactly as in general relativity. The prescription has a price that local theories do not pay: it is acausal on scales of order $1/M$, where the time order of production and decay vertices can be reversed [45]. With $M$ near the Planck mass this is far from observable.

outside $\ker Q_B$ or in $\operatorname{im} Q_B$, or has zero norm, and the quotient $\ker Q_B / \operatorname{im} Q_B$ is spanned by states without quartet modes [44].

[34] This is a statement about perturbation theory. Shapiro has pointed out that the resummed propagator $1/(p^2 a + \Sigma)$ of a nonlocal theory may acquire complex-conjugate poles from the self-energy $\Sigma$ [4]. At any finite order of perturbation theory the self-energy enters as insertions on lines with the bare propagator, and no such pole appears. Section 9.5 estimates the one-loop self-energy and finds that the resummed propagator has no such pole in the region where the loop expansion is controlled; outside it the question is nonperturbative.

| input | content | established in |
|---|---|---|
| (P1) | local propagators after $h = a^{-1/2}\tilde{h}$ | Eq. (95), $a$ zero-free |
| (P2) | vertices entire, real coefficients | Secs. 6.2, 6.3 |
| (P3) | integrable at the ends of the contours | cone bounds plus estimates for the subtracted amplitudes |
| (Q1) | BRST invariance, $[Q_B, S] = 0$ | requires the renormalized Slavnov–Taylor identity |
| (Q2) | free asymptotic fields, four quartets | $a(0) = 1$, Eq. (95) |
| (Q3) | $S^\dagger S = \mathbf{1}$ on $\mathcal{F}$ | requires the cutting theorem in this gauge theory |
| — | asymptotic states free; soft gravitons treated inclusively | assumed, as for general relativity [15, 16] |

TABLE 7. Inputs of the perturbative unitarity criterion. The analytic and BRST requirements for the renormalized amplitudes are separated from the algebraic properties of the free propagator.

9.3. **One loop in closed form.** The general argument of Sec. 9.1 rests on Refs. [13, 14, 95]. The simplest loop can be continued by hand, and doing so tests it. Take two distinct massless scalars $\phi_1$ and $\phi_2$ coupled minimally to the metric, and the process $\phi_1\phi_1 \to \phi_2\phi_2$ at center-of-mass energy $E = \sqrt{s}$. At one loop it contains the graph in which each pair annihilates into two gravitons through the seagull vertex of $\sqrt{g}\, g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$, Fig. 19(c). That vertex depends only on the momenta of the scalars, so the graph is a tensor factor times the scalar bubble

$$A(p) = \int \mathrm{d}^4\ell\; F(k^2)\, F\big((p-k)^2\big), \qquad k = \frac{p}{2} + \ell, \tag{96}$$

where $F(z) = 1/[z\, a(z)]$ is the scalar part of the propagator of the gas. The tensor factor, the contraction of the two seagull vertices through two de Donder numerators, is $(t-u)^2/16$ per unit coupling, which vanishes only at $90°$.[35] The loop integral of this graph is therefore exactly $A$, with the propagator of the gas.

[35] Write the part of $\sqrt{g}\, g^{\mu\nu}$ quadratic in $h$ as $B^{\mu\nu}(h,h)$, with $B^{\mu\nu}(h,h) = h^{\mu}{}_{\rho}h^{\rho\nu} - \frac{1}{2}h\,h^{\mu\nu} + \big(\frac{1}{8}h^2 - \frac{1}{4}h_{\rho\sigma}h^{\rho\sigma}\big)\delta^{\mu\nu}$. The seagull vertex of two scalars with momenta $p_1, p_2$ is the bilinear form $\frac{1}{2}(p_{1\mu}p_{2\nu} + p_{2\mu}p_{1\nu})B^{\mu\nu}$ on the ten components of $h$, times the coupling, and the numerator of the propagator in Feynman–de Donder gauge is $P_{\mu\nu,\alpha\beta} = \frac{1}{2}(\delta_{\mu\alpha}\delta_{\nu\beta} +$

For Euclidean $p$ the integral converges absolutely, since $|F(z)| \le C|z|^{-n-1}$ on the real axis. Put $p = (p_4, \mathbf{0})$ and $\ell = (t, \boldsymbol{\ell})$ with $|\boldsymbol{\ell}| = r$, and continue $p_4$ from real values through the first quadrant to $p_4 = iE$, where $p^2 = -s$. In the plane of $t$ the poles of the two propagators sit at $t = \mp\frac{p_4}{2} \pm ir$. When $r < E/2$ two of them cross the real axis during the continuation, the one at $-\frac{p_4}{2} + ir$ downward and the one at $\frac{p_4}{2} - ir$ upward, and the deformed contour of Efimov, Pius and Sen is the real axis plus small circles around them, Fig. 19(d). At $p_4 = iE$ one has $k^2 = \xi$ and $(p-k)^2 = \bar{\xi}$, with

$$\xi = t^2 + r^2 - \frac{s}{4} + iEt,$$

and $F(\bar{\xi}) = \overline{F(\xi)}$ because $a$ has real Taylor coefficients. The two residues are equal and give $F(2rE - s)$ times $2\pi/r$. To obtain pieces that converge separately, subtract the same decomposition for the local propagator with a Pauli–Villars partner, $F_\mu(z) = 1/z - 1/(z+\mu^2)$ with $\mu^2 > s$, whose bubble $A_\mu$ is elementary. Then

$$A(s) = A_\mu(s) + \int_0^\infty 4\pi r^2 \mathrm{d}r \int_{-\infty}^{\infty} \mathrm{d}t \left[ \big|F(\xi)\big|^2 - \big|F_\mu(\xi)\big|^2 \right] + \frac{2\pi^2}{s} \int_{-s}^{0} (w+s) \big[F(w) - F_\mu(w)\big] \, \mathrm{d}w, \quad (97)$$

with

$$A_\mu(s) = \pi^2 \int_0^1 \mathrm{d}x \, \log \frac{D_{0\mu} D_{\mu 0}}{D_{00} D_{\mu\mu}},$$

where $D_{ab} = x m_a^2 + (1-x) m_b^2 - x(1-x)(s - i0)$, $m_0 = 0$ and $m_\mu = \mu$.[36] Every integral in Eq. (97) now converges absolutely, and every integrand is real, because $|F(\xi)|^2$ is real and $F$ is real on the real axis. Only $A_\mu$ has an imaginary part. It comes from the logarithm of $D_{00} = -x(1-x)(s - i0)$, whose argument is $\pi$, while for $\mu^2 > s$ the other three arguments, $D_{0\mu}$, $D_{\mu 0}$ and $D_{\mu\mu}$, are positive. Hence

$$\operatorname{Im} A(s) = \operatorname{Im} A_\mu(s) = -\pi^3 \qquad \text{for every } s > 0, \quad (98)$$

which is the imaginary part of the local massless bubble: the phase space of two on-shell gravitons, at which $a(0) = 1$. Nothing from the complex regions where $1/a$ is large reaches it. This is the cutting rule of Sec. 9.1 for

$\delta_{\mu\beta}\delta_{\nu\alpha} - \delta_{\mu\nu}\delta_{\alpha\beta}$). For on-shell massless scalars the trace $\mathrm{Tr}(B_{12} P B_{34} P)$ equals $(t-u)^2/16$; Appendix D lists a check at random kinematics.

[36] Each term of Eq. (97) without the subtraction has a logarithmic singularity at $t = 0$, $r = E/2$, where the crossing poles reach the real axis; the singularities cancel between the two terms, and the subtraction cancels them term by term. Near $\xi = 0$ both $F$ and $F_\mu$ equal $1/\xi$ plus a function that is bounded there, since $a(0) = 1$, so $|F|^2 - |F_\mu|^2 = O(1/|\xi|)$, which is integrable; at large $|\xi|$, $|F_\mu|^2 \simeq \mu^4/|\xi|^4$. On $[-s, 0]$, $F - F_\mu$ is bounded. The closed form of $A_\mu$ is the Feynman-parameter integral of the local bubble with masses 0 and $\mu$, with the $-i0$ inherited from $p_4 = iE + 0$. As a check, Eq. (97) applied to $F_\mu$ itself at complex $p_4$ reproduces that closed form (Appendix D).

this graph, obtained without the general argument, and it holds for every admissible $a$, the form factor of every realization included.

The real part tells a different story. The real section in Eq. (97) has the positive integrand $|F(\xi)|^2$, and this is where the wedges of Fig. 18(a) enter. As $(t,r)$ runs over $\mathbb{R}\times[0,\infty)$, the point $\xi$ covers the parabolic region

$$P_s=\Big\{\xi:\ \operatorname{Re}\xi\ge\frac{(\operatorname{Im}\xi)^2}{s}-\frac{s}{4}\Big\}$$

once, with $4\pi r^2\,\mathrm{d}t\,\mathrm{d}r=(2\pi/E)\,\rho(\xi)\,\mathrm{d}^2\xi$ and $\rho(\xi)=r=[\operatorname{Re}\xi+s/4-(\operatorname{Im}\xi)^2/s]^{1/2}$. The region contains the segment $\xi=iy$, $|y|<s/2$, of the imaginary axis, where $\xi^2=-y^2$, and for the skeleton

$$|F(iy)|^2=\frac{1}{y^2}\exp\Big[n\ \mathrm{Ein}^-\Big(\frac{y^2}{M^4}\Big)\Big],$$

with

$$\mathrm{Ein}^-(X)=\int_0^X\frac{e^u-1}{u}\,\mathrm{d}u\ge\frac{e^X-1-X}{X}.$$

Away from $\xi=0$ the function $\log|F|^2$ is harmonic, so its mean over a disk equals its value at the center, and by Jensen's inequality the mean of $|F|^2$ over the disk is at least its value at the center. A disk of radius $\delta s/8$ around $i(1-\delta)s/2$ lies in $P_s$, and on it $\rho^2\ge 7\delta s/64$. Bounding everything else in Eq. (97) by a power of $s$ gives, for $0<\delta<\frac12$ and $s\ge 6M^2$,

$$\operatorname{Re}A(s)\ge\frac{\pi^2\sqrt{7}\,\delta^{5/2}}{64\,(1-\delta)^2}\exp\Big[n\ \mathrm{Ein}^-\Big(\frac{(1-\delta)^2s^2}{4M^4}\Big)\Big]\\ -C_\delta\Big(1+\frac{s}{M^2}\Big)^2,\quad(99)$$

a double exponential in $s^2$.[37] The growth is not a property of $\psi_*$. For any admissible form factor, that of a realization included, $a=e^H$ with $H$ entire, and $H$ is not a polynomial, because $a$ grows like a power on the real axis, Eq. (43). So $1/a=e^{-H}$ is an entire function of infinite order [19]: for every $k$ its maximum on the circle $|\xi|=R$ exceeds $\exp(R^k)$ for arbitrarily large $R$. The region $P_s$ contains the disk $|\xi|<s/4$, and the same argument, with a disk of radius $s/16$ around the point of the circle $|\xi|=s/8$ where $|1/a|$ is largest, shows that $\operatorname{Re}A(s)$ is not bounded by $\exp(s^k)$ for any $k$.[38]

[37]The disk $D$ has area $\pi\delta^2s^2/64$, and on it $\rho^2\ge s/4-\delta s/8-s[(1-\delta)/2+\delta/8]^2=\delta s(16-9\delta)/64\ge 7\delta s/64$, so the real section over $D$ contributes at least $(2\pi/E)(7\delta s/64)^{1/2}(\pi\delta^2s^2/64)|F(i(1-\delta)s/2)|^2$, which is the first term of Eq. (99). Take $\mu^2=2s$. On $P_s$, $|\xi+\mu^2|\ge 7s/4$, so $\int_{P_s,\,|\xi|\ge M^2}\rho\,|F_\mu|^2\,\mathrm{d}^2\xi$ is bounded by a power of $s$; on $|\xi|<M^2$, which $D$ does not meet when $s\ge 6M^2$, $\big||F|^2-|F_\mu|^2\big|\le C/|\xi|$ and $\rho\le(1+s/4)^{1/2}$. The residue term is at most $\pi^2 s\sup_{[-s,0]}|F-F_\mu|$, and $|F(w)-1/w|\le C$ there because $a\ge 1$ on the real axis, while $|1/(w+\mu^2)|\le 1/s$. Finally $|\operatorname{Re}A_\mu|\le C\log(2+s/M^2)$. Dropping the positive part of the real section outside $D$, these bounds give the second term.

[38]On that disk $|\xi|\ge s/16$ and $\rho^2\ge -3s/16+s/4-(3s/16)^2/s=7s/256$, and the remainder of Eq. (97) is bounded by a power of $s$ exactly as in the previous footnote,

Table 8 and Fig. 19(b) give the numbers for the skeleton with $n = 4$, from Eq. (97) evaluated by quadrature. Below $1.3M$ the real part is of the same size as the imaginary part. It then rises steeply: $\mathrm{Re}\, A$ is about 475 at $E = 1.5M$, $4.5 \times 10^6$ at $1.7M$ and $4 \times 10^{88}$ at $1.9M$, while $\mathrm{Im}\, A$ stays at $-\pi^3$. The integrand peaks near the edge of $P_s$, in the wedge where $\mathrm{Re}\, \xi^2 < 0$, Fig. 19(a); as $E$ grows the peak moves toward the point $t^2 = 3s/4$, $r = 0$, where $\mathrm{Re}\, \xi^2$ reaches its most negative value, $-s^2/2$, and at $E = 1.9M$ it lies within 5% of that value.

Above $M$ this graph is not small. The tree amplitude of the same process is the exchange of one graviton in the $s$ channel, suppressed by $1/a(-s)$ as in Eq. (100), and up to a factor that depends on the scattering angle and on conventions the ratio of the graph to it is $Gs\, a(-s)\, \mathrm{Re}\, A(s)$. For every $M$ between $10^{-40} M_{\mathrm{P}}$ and $10^{-2} M_{\mathrm{P}}$ this ratio passes one between $E = 1.4M$ and $E = 1.9M$ (dotted lines in Fig. 19(b)), because $\mathrm{Re}\, A$ rises by more than eighty decades in that interval. The loop expansion of this process is therefore an expansion in a small number only below an energy between $1.4M$ and $1.9M$, which depends on $M/M_{\mathrm{P}}$ only through a double logarithm. We have not computed the other one-loop graphs of the process, nor the graviton loops of pure gravity. The graphs with two graviton lines in the $s$ channel contain the same factor $|F(\xi)|^2$ on the real section, multiplied by vertex factors, and we expect the same growth, but a cancellation among graphs cannot be excluded without computing them.

Koshelev and Tokareva found, for a nonlocal scalar theory defined by the same continuation, that the one-loop amplitude grows exponentially and the theory becomes strongly coupled around the nonlocality scale when the form factor grows along the timelike axis [48]. The form factor of the gas is bounded below by a power on both real half-axes, so that mechanism is absent here; the growth found above comes from the wedges between the half-axes, which the continued real section crosses. Buoninfante verified the cutting rule for one-loop bubbles of a string-inspired model in the same prescription [49], and Eq. (98) is that verification for the form factor of the gas, in closed form.

9.4. **The size of amplitudes.** Unitarity in this sense constrains the cuts of amplitudes, not their size. For actions quadratic in the Ricci tensor and the scalar curvature, with form factors inserted, the on-shell tree amplitudes coincide with those of general relativity for any number of gravitons [46], and $R + G_{\mu\nu} F(\Box) R^{\mu\nu}$ is of that type. The quartic operators are different. Those built from $R$ and $R_{\mu\nu}$ add nothing to on-shell graviton scattering, because the linearized Ricci tensor of an on-shell graviton vanishes. The Riemann operator does add something: a contact term of order $G|s_3| E^{2n+2}/M^{2n}$ at center-of-mass energy $E$, which grows faster than any amplitude of general

using only $a(0) = 1$ and the two-sided bound on the real axis. A zero-free entire function of finite order $\varrho$ is $e^H$ with $H$ a polynomial of degree at most $\varrho$, by Hadamard's factorization theorem [19]; applied to $1/a = e^{-H}$, finite order would make $H$ a polynomial.

| $E/M$ | $\operatorname{Re} A$ | $\operatorname{Im} A$ | bound | $s\,a(-s)\operatorname{Re} A$ |
|---|---|---|---|---|
| 1.00 | 11.7 | $-31.006$ | 0.37 | 57.3 |
| 1.25 | 24.7 | $-31.006$ | 0.54 | 770 |
| 1.40 | 82.1 | $-31.006$ | 0.80 | $7.60 \times 10^{3}$ |
| 1.50 | 475 | $-31.006$ | 1.14 | $8.70 \times 10^{4}$ |
| 1.60 | $1.27 \times 10^{4}$ | $-31.006$ | 3.78 | $4.45 \times 10^{6}$ |
| 1.70 | $4.54 \times 10^{6}$ | $-31.006$ | 156 | $2.90 \times 10^{9}$ |
| 1.80 | $2.42 \times 10^{26}$ | $-31.006$ | $1.94 \times 10^{5}$ | $2.74 \times 10^{29}$ |
| 1.90 | $3.81 \times 10^{88}$ | $-31.006$ | $2.00 \times 10^{11}$ | $7.42 \times 10^{91}$ |

TABLE 8. The one-loop bubble (96) for the skeleton with $n = 4$, in units $M = 1$, from Eq. (97). The real section was integrated by adaptive quadrature in polar coordinates about $\xi = 0$, the focus of the parabola that bounds $P_s$, and the result is the same to ten digits for the two Pauli–Villars masses $\mu = 2E + 2$ and $\mu = 3E + 3$; a uniform grid in $(t, r)$ agrees to 0.1%. The imaginary part equals $-\pi^3 = -31.006$ at every energy, Eq. (98); in Eq. (97) it comes entirely from the closed-form term $A_\mu$. The fourth column is the leading term of the bound (99), maximized over $\delta$. It bounds from below the contribution of the disk $D$ used in deriving Eq. (99) to the real section, while the inequality itself is stated for $s \geq 6M^2$, where the remainder is controlled; the table shows how far below the true value it lies. The last is the factor $s\,a(-s)\operatorname{Re} A$ whose product with $(M/M_{\mathrm{P}})^2$ is the ratio of the graph to the tree amplitude, up to angular factors.

relativity and reaches order one at $E_* = (M^{2n}/G|s_3|)^{1/(2n+2)}$. That scale can be avoided. Set $s_3 = 0$ and solve for $s_1$ and $s_2$ alone, with the upper-left $2 \times 2$ block of Eq. (84), whose determinant is $-12/c^2$. The Gauss–Bonnet pole then stays, but $\int \sqrt{g}\, E$ is topological in four dimensions and does not enter the $S$ matrix.

With $s_3 = 0$ every tree amplitude of gravitons is that of general relativity, and at fixed angle it grows like $GE^2$, reaching order one near the Planck energy $E_{\mathrm{P}} = G^{-1/2}$. The loops set a lower scale. The one-loop graph of Sec. 9.3, finite at every energy, overtakes the tree amplitude of its process between $1.4M$ and $1.9M$ for any $M$ between $10^{-40}M_{\mathrm{P}}$ and $10^{-2}M_{\mathrm{P}}$. The loop expansion, finite term by term, is therefore an expansion in a small number only up to energies of order $M$, and for $M \ll M_{\mathrm{P}}$ the theory becomes strongly coupled well below the Planck energy at which general relativity does. The form factor that removes the divergences also lowers the scale at which perturbation theory ends.

Matter that couples minimally to the metric behaves differently, because its gravitational scattering at tree level goes through the propagator (64) itself. Between conserved sources the propagator gives $T_1 \cdot T_2 - \frac{1}{2} T_1 T_2$ divided

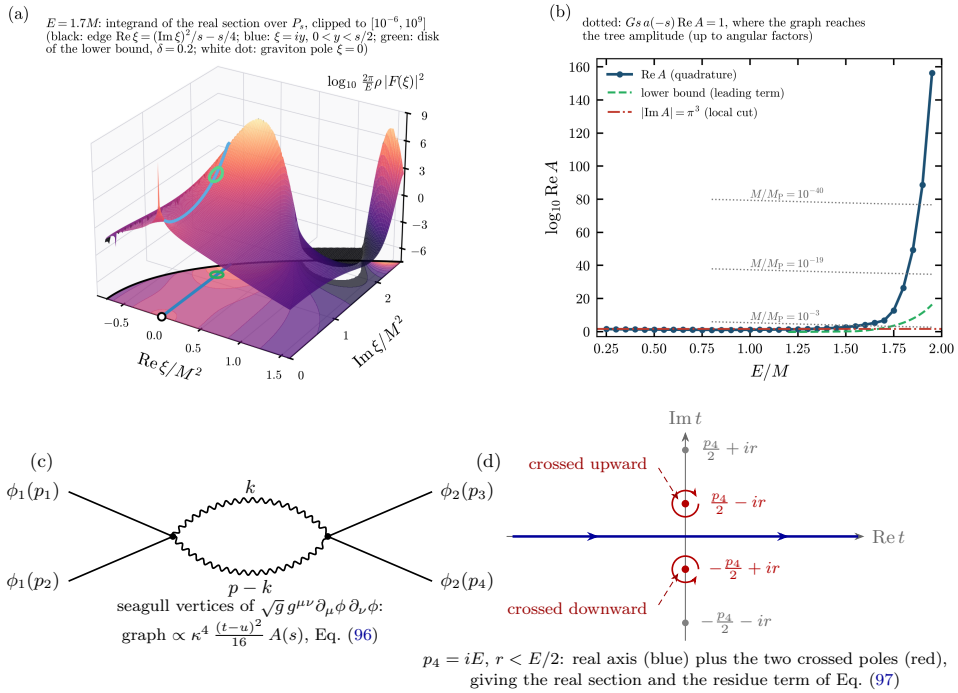



FIGURE 19. The bubble for the skeleton at $n = 4$. (a) Real-section integrand at $E = 1.7M$, with the disk used in Eq. (99) and the subtracted singularity at $\xi = 0$. (b) Quadrature, the lower bound and the constant imaginary part. The dotted curves indicate the tree scale for the three indicated values of $M/M_\mathrm{P}$. (c) Two scalar seagull vertices joined by graviton lines. (d) The fixed-end contour and the two crossed poles.

by $p^2 a(p^2)$, and for two distinct massless scalars $T_1{\cdot}T_2 - \frac{1}{2}T_1T_2 = -su$ in terms of the Mandelstam variables. The one-graviton exchange is therefore the amplitude of general relativity divided by the form factor at the momentum transfer,

$$|A(s,t)| = \frac{8\pi G\,|su|}{|t|\,a(-t)}, \tag{100}$$

with $a$ evaluated at the Euclidean point $p^2 = -t > 0$. At a fixed scattering angle $\theta$ in the center-of-mass frame, $-t = s\sin^2(\theta/2)$ and $-u = s\cos^2(\theta/2)$, and the lower half of the two-sided bound (43) gives

$$|A| \le 8\pi G M^2\, C_\Pi\, \frac{\cos^2(\theta/2)}{\sin^4(\theta/2)}\, \frac{x}{(1+x^2)^{n/2}}, \qquad x = \frac{-t}{M^2},$$

with $C_\Pi = e^{-nm_\psi + \|R_\Pi\|}$. The last factor is at most $(n-1)^{(n-1)/2}/n^{n/2}$, which is 0.325 for $n = 4$, and it falls like $x^{1-n}$. At every fixed angle the tree amplitude for the gravitational scattering of matter therefore rises like

$GE^2$ only up to $E \sim M$, then falls like $GM^2(M/E)^{2n-2}$, and never exceeds a number of order $GM^2 = (M/M_{\mathrm{P}})^2$. For $M$ below the Planck mass, matter does not scatter strongly through one graviton at any energy. This is a tree-level statement. At one loop the same kind of process has the graph of Sec. 9.3, which overtakes the tree amplitude above an energy between $1.4M$ and $1.9M$, so the tree-level bound describes the theory only below that energy. The finiteness proof of Sec. 8 concerns pure gravity, and loops of matter fields are not treated here. The forward peak at $|t| \to 0$ is the Newtonian one of general relativity, an infrared effect untouched by the form factor.

9.5. **The free Euclidean field.** The perturbative bounds do not control the large-order expansion or construct interacting correlation functions. The one-loop example of Sec. 9.3 is a diagnostic of the continuation at energies of order $M$. At the free level there is a Euclidean Gaussian construction.

The spin-two part of the quadratic action (63), $\frac{1}{2}p^2 a(p^2)\, hP^{(2)}h$, is positive on the Euclidean axis, so it defines a Gaussian probability measure, in infinite volume, on the gauge-invariant field $P^{(2)}h$, with covariance $P^{(2)}/(p^2 a)$; the covariance is integrable at small $p$ in four dimensions, so there is no infrared obstruction either. For each Fourier mode the linearized Weyl tensor obeys $C \cdot C = \frac{p^4}{2}\, hP^{(2)}h$, and $\operatorname{tr} P^{(2)} = 5$, so the Weyl tensor of $\kappa h$ has

$$\mathbb{E}\, C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}(x) = \frac{5\kappa^2}{2}\int \frac{\mathrm{d}^4 p}{(2\pi)^4}\,\frac{p^2}{a(p^2)} = \frac{5GM^6}{\pi}\int_0^\infty \frac{x^2\,\mathrm{d}x}{a(x)}. \quad (101)$$

By the two-sided bound (43) the integral converges exactly when $n \geq 4$, and at $n = 3$ it diverges logarithmically, the same threshold as in the power counting of Sec. 6.4. For the skeleton with $n = 4$ the integral lies between $\pi/(4\hat{c})$ and $\pi/4$ and equals 0.4237, so the root-mean-square Weyl curvature at a point is $0.82\, M^3/M_{\mathrm{P}}$, small compared with $M^2$ whenever $M \ll M_{\mathrm{P}}$. The same estimate applied to $C(x) - C(0)$, with $1 - \cos y \leq 2|y/2|^{2\alpha}$, gives $\mathbb{E}|C(x) - C(0)|^2 \leq C_\alpha |Mx|^{2\alpha}$ for every $\alpha < \min(1, n-3)$. The field is Gaussian, so Kolmogorov's continuity criterion applies, and the curvature of almost every free configuration is a Hölder-continuous function. In general relativity the integral in Eq. (101) grows like the sixth power of a cutoff, the free field is a distribution, and every product of fields at a point has to be renormalized. Here no product of curvatures at a point needs renormalization: the free graviton of the gas has no short-distance singularity at all. This Gaussian measure is a Euclidean random field, and it is not reflection positive, so the Osterwalder–Schrader reconstruction [50, 51, 81], or Nelson's construction from a Markov field [82], does not turn it into a quantum field with a positive Hilbert space. Figure 20 shows the failure on the time-slice covariance of the free field, which for a reflection-positive Gaussian field would be a completely monotone function of the Euclidean time and

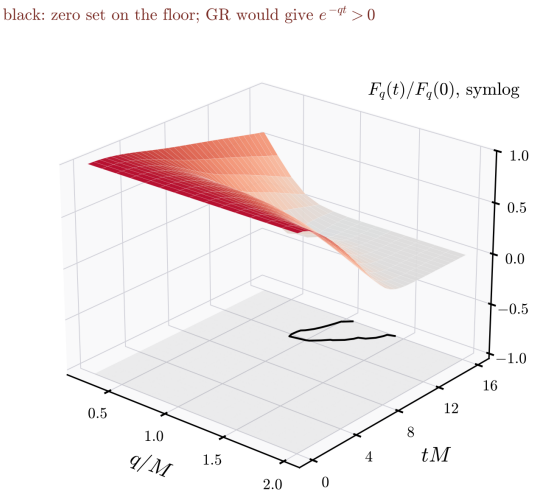



FIGURE 20. The time-slice covariance of the free Euclidean graviton of the skeleton with $n = 4$, $F_q(t) = \int \mathrm{d}p_0\, e^{ip_0 t}/[2\pi(p_0^2+q^2)a(p_0^2+q^2)]$, normalized by $F_q(0)$, over the spatial momentum $q/M$ and the Euclidean time $tM$, on a symmetric logarithmic scale. For a reflection-positive Gaussian field $F_q$ is the Laplace transform of a positive measure in the energy, hence positive and completely monotone in $t$; here it is negative inside the pocket bounded by the black curve, which is the zero set of $F_q$ drawn on the floor, and reaches it at $tM \simeq 12$ for $q = M$; the negative values are small on the scale of $F_q(0)$ but far above the quadrature error. For general relativity the same function is $e^{-qt}/(2q)$ and never changes sign. This is the failure of reflection positivity stated in the text, made visible on the covariance itself.

here changes sign. Reflection positivity of a free field requires its covariance to be a Stieltjes function, $\int \mathrm{d}\varrho(m^2)/(z+m^2)$ with a positive measure, and a Stieltjes function that is analytic across the negative real axis is $c/z$, since $\varrho$ is the jump across that axis; $1/(za)$ is analytic there except at $z = 0$ and is not $c/z$.[39] The Hilbert space of the theory is the Fock space of the asymptotic gravitons of Sec. 9.2, as in string field theory, and the statement about the curvature is a statement about the Euclidean field.

The full Euclidean quadratic action also has the familiar conformal-sign problem [47]. Continuation of the conformal mode addresses a different question from positivity of the spatial curvature observable. A Dyson resummation involves $1/(za+\Sigma)$, and zeros of its denominator cannot be excluded by

[39]The same holds for the covariance $P^{(2)}/(p^2 a)$ of the spin-two field, whose components are $1/a(p^2)$ times rational functions of the momentum; the obstruction is the factor $1/a$, which is entire and not constant.

| divergence | fate | reason |
|---|---|---|
| one loop, $\int\sqrt{g}$ and $\int\sqrt{g}\,R$ (running of $\Lambda$, $G$) | absent for $n \geq 4$ | one-loop poles are those of a local theory with no dimensionful coupling; saturation (S3) and fine graining (G), Sec. 7.3; the finite vacuum energy has mean $3nM^4/(64\pi^2)$, Sec. 8.2 |
| one loop, $R^2$, $R_{\mu\nu}R^{\mu\nu}$, $E$ | removed | three quartic operators, triangular matrix with determinant $-24/c^3$, Eqs. (84), (85) |
| connected hard component with two or more loops | negative degree for $n \geq 4$ | vertex bound and saturation of hard subspaces, Sec. 6.4 |
| one-loop background curvature poles | cancelled | three-operator tuning in dimensional regularization |
| one-loop quantum-field and source poles | forest and BRST inputs | the on-shell argument requires the identities discussed in Sec. 8, Sec. 8 |
| power-law terms ($\Lambda^4$, $\Lambda^2$ with a cutoff) | scheme dependent | dimensional regularization and momentum subtraction have different power terms, Sec. 8.1 |
| soft gravitons on shell (infrared) | physical | as in general relativity; cancel in inclusive rates [16] |
| static potential at $r \to 0$ (classical) | removed | $\Phi_N(0)$ finite; Gamma-function bounds for the skeleton, Sec. 10 |
| extra poles of the propagator | absent | almost every realization entire and zero-free, by (S5); resonant branes would give complex pairs, Eq. (7) |

TABLE 9. Separate ultraviolet, infrared and quadratic statements. The background tuning is (85); a claim about physical amplitudes also requires the forest and BRST inputs in Sec. 8.

the absence of zeros of $a$ alone [4]. The pole computation at the quadratic level and the following positivity test have no such resummation assumption.

9.6. **A compact-time curvature test for reflection positivity.** A sign test on a time-slice covariance can miss a negative Osterwalder–Schrader form even if that covariance is positive at every time. We give a test using two nearby positive times and a linearized curvature component. It applies to every admissible realization with $n > 1/2$. For $q = |\mathbf{q}| > 0$ put

$$C_q(u) = \frac{1}{(u^2+q^2)a_\Pi(u^2+q^2)},$$
$$K_q(t) = \int_{\mathbb{R}} \cos(ut) C_q(u) \frac{\mathrm{d}u}{2\pi},$$
$$M_j(q) = \int_{\mathbb{R}} u^j C_q(u) \frac{\mathrm{d}u}{2\pi} \qquad (j = 2, 4).$$

For $M_2(q) < \infty$, evenness and differentiation under the integral give $K_q'(0) = 0$ and $K_q''(0) = -M_2(q) < 0$. The reflected norm of the difference of the two time evaluations at $t$ and $2t$ is proportional to

$$K_q(2t) - 2K_q(3t) + K_q(4t) = -M_2(q)t^2 + o(t^2) < 0. \tag{102}$$

For $n > 3/2$ the fourth moment is finite. The elementary bound

$$\left|\cos y - 1 + y^2/2\right| \leq |y|^4/24$$

gives error at most $434M_4(q)t^4/24$. Thus $0 < t^2 < 12M_2(q)/(434M_4(q))$ suffices to give a norm at most $-M_2(q)t^2/2$.

To place the test in the curvature algebra, choose a small spatial Fourier ball about a nonzero $\mathbf{q}_0$ and a smooth orthonormal frame $v(\mathbf{q}), w(\mathbf{q}), \hat{\mathbf{q}}$ there,

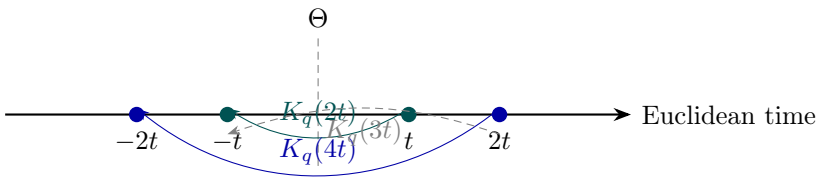



FIGURE 21. The two positive-time packets are paired with their reflected partners. The mixed pair occurs twice in the quadratic form with a minus sign. The test uses only the kernel at strictly positive separations.

with $v, w$ transverse. The linearized curvature $R^{(1)}_{v\hat{\mathbf{q}}w\hat{\mathbf{q}}}$ is $q^2 h_{vw}/2$ up to the overall curvature sign convention, since differentiation in either transverse direction vanishes on each Fourier mode. Its defining tensor has only spatial indices and is even under time reflection. The traceless polarization $(v \otimes w + w \otimes v)/\sqrt{2}$ is transverse to the full Euclidean momentum $(u, \mathbf{q})$, and the spin-two projector acts as the identity on it. The covariance of the curvature component is therefore $q^4 C_q(u)/8$ in the normalization of the spin-two propagator. The reflection form has precisely the sign in (102). Linearized diffeomorphisms add commuting third derivatives with opposite signs to the curvature, so this test survives the quotient by linearized gauge transformations.

Choose a smooth spatial packet in this ball and its conjugate at the opposite momentum, giving a real observable. The bounds for $M_2(q)$ and $M_4(q)$ are uniform on the support by the two-sided real-axis estimate and $q$ bounded away from zero. The integrand in (102) is continuous in $q$ and is negative throughout a sufficiently small ball at the chosen $t$. Replace each point evaluation by a smooth unit-mass bump of width less than $t/4$. Continuity of $K_q$ and dominated convergence show that sufficiently narrow bumps still give a strictly negative form. The spatial packet is Schwartz. Cutting it off on larger spatial balls preserves the sign in the limit, because the tempered curvature covariance is continuous in the Schwartz topology. Thus there are smooth compactly supported positive-time test tensors with strictly negative reflected norm. The test also applies to the averaged covariance from (56), since its ultraviolet power and positive real-axis sign are the same.

9.7. **A quantitative obstruction for positive spectral mixtures.** Suppose a scalar physical polarization has a positive Källén–Lehmann representation at spacelike $z > 0$. The monotonicity argument used for a different brane gas in Ref. [57] takes the form

$$C_{\mathrm{pos}}(z) = Z/z + \int_{(0,\infty)} \frac{\rho(\mathrm{d}m^2)}{z + m^2}, \qquad Z \geq 0, \quad \rho \geq 0,$$

with finite $C_{\mathrm{pos}}(z)$ there. Possible contact terms are removed before this comparison.[40] Then $zC_{\mathrm{pos}}(z) \geq Z$ and is nondecreasing, since each $z/(z + m^2)$ has this property. In particular it cannot tend to zero with a positive massless residue. There is a two-point version. Let $z_1 < z_2$ with $a_i = a_\Pi(z_i)$ and $a_2 > a_1 > 0$. If the relative error of $C_{\mathrm{pos}}$ against $1/(za_\Pi(z))$ at both points is at most $\epsilon$, then

$$\frac{1-\epsilon}{a_1} \leq z_1 C_{\mathrm{pos}}(z_1) \leq z_2 C_{\mathrm{pos}}(z_2) \leq \frac{1+\epsilon}{a_2}.$$

It follows that $\epsilon \geq (a_2 - a_1)/(a_2 + a_1)$. This bound is sharp for a free choice of $Z$, by taking $C_{\mathrm{pos}}(z) = 2/((a_1 + a_2)z)$. With $Z = 1$ the lower bound at any point with $a_\Pi(z) > 1$ is already $a_\Pi(z) - 1$. Neither estimate uses a series expansion or a limit at infinity.

Finally, a fixed smeared reflection form is a continuous linear functional of the two-point distribution. If a family of reflection-positive Euclidean interacting measures on the same observable algebra converged to the proposed covariance in the free-coupling limit, every such form would be nonnegative in that limit. Equation (102) contradicts this conclusion for the rescaled linearized curvature, provided the observable and its reflection have the specified free limit. This excludes that class of positive Euclidean completions of the given perturbative model. It leaves open a construction with a different physical observable map or a genuinely Lorentzian definition with its own positivity proof.

9.8. **A cutoff-free curvature interaction in a bounded window.** Fix a realization with the two-sided propagator bound and $n \geq 4$. Let $\mathcal{R}$ denote the full linearized curvature of the spin-two Gaussian field, as a real vector of components. On the same Gaussian probability space let $\mathcal{R}_\Lambda$ retain just the Euclidean Fourier modes of length at most $\Lambda$. Fix a nonnegative, nonzero $w \in C_c^\infty(\mathbb{R}^4)$ even in the time coordinate. For $g \geq 0$ define

$$V_\Lambda = \int_{\mathbb{R}^4} w(x) |\mathcal{R}_\Lambda(x)|^4 \, \mathrm{d}^4x,$$

$$V = \int_{\mathbb{R}^4} w(x) |\mathcal{R}(x)|^4 \, \mathrm{d}^4x,$$

$$\mathrm{d}\mu_{\Lambda,g} = \frac{e^{-gV_\Lambda}}{\mathbb{E} e^{-gV_\Lambda}} \, \mathrm{d}\mu_0.$$

The norm sums the real tensor components, so the interaction is even under time reflection. The construction fixes a spacetime window and uses linearized curvature.

[40] Distributions supported at coincident Euclidean times do not contribute to the test in Fig. 21. The spacelike comparison concerns the spectral part after such terms have been removed.

The covariance bound counts two derivatives per curvature and uses the bounded spin-two projector. Uniformly in $x$, for $\Lambda \geq M$,

$$\mathbb{E}|\mathcal{R}_\Lambda(x) - \mathcal{R}(x)|^2 \leq C \int_\Lambda^\infty r^{3+4-2n-2}\,\mathrm{d}r \leq C'\Lambda^{6-2n}.$$

The covariance of both Gaussian vectors is bounded independently of $\Lambda, x$, including the small-momentum contribution. By diagonalizing the covariance and applying the one-dimensional Gaussian moment formula, the $L^p$ norm of a centered Gaussian vector is bounded by a constant depending on $p$ and its dimension times its $L^2$ norm. Thus for $1 \leq p < \infty$,

$$\|\mathcal{R}_\Lambda(x) - \mathcal{R}(x)\|_{L^p(\mu_0)} \leq C_p\Lambda^{3-n}, \qquad \|\mathcal{R}_\Lambda(x)\|_{L^p(\mu_0)} + \|\mathcal{R}(x)\|_{L^p(\mu_0)} \leq C'_p.$$

For vectors $a, b$, $\big||a|^4 - |b|^4\big| \leq 4(|a|+|b|)^3|a-b|$, by differentiating the fourth power along the connecting segment. Hölder's inequality with exponents $4/3, 4$ and integration against $w$ give

$$\|V_\Lambda - V\|_{L^p(\mu_0)} \leq C_{p,w}\Lambda^{3-n}.$$

The integrals are finite almost surely by their finite expectation.

The exponential $e^{-gs}$ is $g$-Lipschitz for $s \geq 0$. Jensen's inequality gives $\mathbb{E}e^{-gV_\Lambda} \geq e^{-g\mathbb{E}V_\Lambda} > 0$, uniformly in the cutoff at fixed $g$. The normalized densities therefore converge in $L^p(\mu_0)$ at rate $O_{p,g,w}(\Lambda^{3-n})$ to $e^{-gV}/\mathbb{E}e^{-gV}$. In particular the measures converge in total variation on the underlying Gaussian probability space. Expectations of products of any finite number of pointwise or compactly smeared linearized curvatures converge to their expectations under the limiting law $\mu_g$. To see this, telescope the products using the Gaussian moment estimates to obtain convergence in $L^2(\mu_0)$, and then apply Cauchy–Schwarz to the product and the density. The same proof works for uniformly bounded real even Fourier multipliers equal to one below $\Lambda$, since the same tail integral controls their difference from one.

The limit is non-Gaussian for $g > 0$. Choose $x_0$ with $w(x_0) > 0$ and a normalized nondegenerate curvature component $X$ at $x_0$. In the free law write $\mathcal{R}(x) = Xf(x) + Y(x)$, where $f(x) = \mathbb{E}[\mathcal{R}(x)X]$ is continuous and nonzero at $x_0$. Every finite set of $Y$-evaluations is jointly Gaussian and independent of $X$ by zero covariance. In the interacting law the conditional density of $X$ given $Y$ is proportional to

$$\exp\left(-X^2/2 - g\int w(x)|Xf(x) + Y(x)|^4\,\mathrm{d}^4x\right).$$

The polynomial in $X$ in the integral has leading coefficient $\int w|f|^4 > 0$. The conditional law has a strictly positive density with a quartic logarithm. A Gaussian field would instead give Gaussian conditional laws for this linear decomposition, possibly point masses. This proves non-Gaussianity.

Let $F$ be the linear observable from Sec. 9.6 with strictly negative reflected norm. The free expectation of $|(\Theta F)F|$ is finite. Dominated convergence as

$g \downarrow 0$ gives

$$\lim_{g\downarrow 0} \int (\Theta F) F \, \mathrm{d}\mu_g = \int (\Theta F) F \, \mathrm{d}\mu_0 < 0.$$

Thus this explicit cutoff-free interacting curvature law still fails reflection positivity at sufficiently small positive coupling. Translation invariance at nonzero coupling would require removing the window, and replacing linearized curvature by a dynamical metric would require a nonlinear construction.

## 10. Newtonian limit

The quartic operators do not touch the propagator, so the static field of a point mass follows from Eq. (64) alone. For a source $T_{\mu\nu} = m\,\delta^3(\mathbf{x})\,u_\mu u_\nu$ at rest, the linearized field is the one of general relativity with $1/\mathbf{k}^2$ replaced by $1/(\mathbf{k}^2 a(\mathbf{k}^2))$, because the form factor multiplies both projectors by the same scalar. In general relativity the computation gives $\tilde{\Phi}_N(\mathbf{k}) = -4\pi G m/\mathbf{k}^2$, so here

$$\tilde{\Phi}_N(\mathbf{k}) = -\frac{4\pi G m}{\mathbf{k}^2\, a(\mathbf{k}^2)}, \tag{103}$$

and the Fourier transform of a radial function in three dimensions,

$$\int \frac{\mathrm{d}^3 k}{(2\pi)^3}\, e^{i\mathbf{k}\cdot\mathbf{x}} f(k) = \frac{1}{2\pi^2 r} \int_0^\infty k \sin(kr) f(k)\, \mathrm{d}k,$$

gives

$$\Phi_N(r) = -\frac{2Gm}{\pi r} \int_0^\infty \frac{\sin kr}{k\, a(k^2)}\, \mathrm{d}k. \tag{104}$$

With $a = 1$ the integral is $\pi/2$ and Eq. (104) is Newton's potential. With the form factor of the gas, the integrand is bounded by $1/a(k^2)$ uniformly in $r$, and $1/a$ falls like $k^{-2n}$. Below we find how deep the well is, how fast the potential approaches $-Gm/r$, and how large a mass the linearized description can carry; all three have closed answers, and Fig. 22 shows the potential of the skeleton for $n = 4, \dots, 9$.

10.1. **Depth of the well.** At the origin $\sin(kr)/(kr) \to 1$, and

$$\Phi_N(0) = -\frac{2Gm}{\pi}\, I_n, \qquad I_n = \int_0^\infty \frac{\mathrm{d}k}{a(k^2)}. \tag{105}$$

Since $|\sin(kr)/(kr)| \le 1$ and the integrand of $I_n$ is positive, $|\Phi_N(r)| \le |\Phi_N(0)|$ for every $r$: the potential is deepest at the center, and it is finite there. The value of $I_n$ is bracketed from both sides by elementary integrals, and the bracket is narrow. We give the numbers for the skeleton, $a = \hat{a}$. A realization has $a = \hat{a}\, e^{R_\Pi}$ with $R_\Pi$ bounded on the real axis, so its $I_n$ lies between $e^{-\sup R_\Pi}$ and $e^{-\inf R_\Pi}$ times the skeleton value, the supremum and infimum being taken over the real axis; the qualitative statements below, a finite well, $2n-2$ continuous derivatives and an approach to Newton's law faster than any power, hold for almost every realization as they stand, because they use only Eq. (43), the derivative bounds that follow Eq. (42),

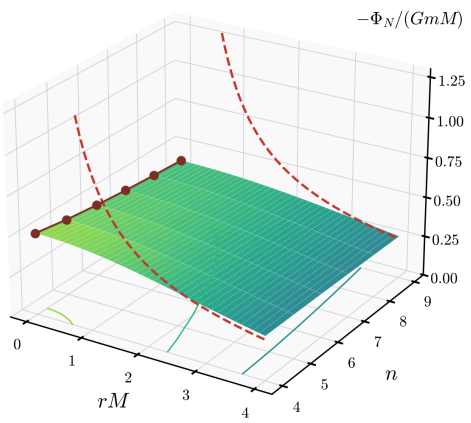



FIGURE 22. The potential $-\Phi_N(r)/(GmM)$ of a point mass, Eq. (104), for the skeleton with $n = 4, \dots, 9$, as a surface over the distance $rM$ and the strength $n$. The surface is finite everywhere; its height at $r = 0$ (dots) is the depth of the well, $2I_n/\pi$ of Eq. (105), which falls from 0.510 at $n = 4$ to 0.404 at $n = 9$, and it meets the Newtonian potential $1/(rM)$ (dashed, drawn along the edges $n = 4$ and $n = 9$) within a few units of $1/M$, the difference falling faster than any power of $1/(Mr)$ by Eq. (113). The contours on the floor are the level lines of the surface.

which take the place of Eq. (28), and $a - 1 = O(k^4)$; only the constants change from one realization to another.

For an upper bound we use the left inequality of Eq. (26), $a \geq (1 + k^4/M^4)^{n/2}$. With $t = k^4/M^4$,

$$I_n \leq \int_0^\infty \frac{\mathrm{d}k}{(1 + k^4/M^4)^{n/2}} = \frac{M}{4} \int_0^\infty \frac{t^{-3/4}\,\mathrm{d}t}{(1+t)^{n/2}} = J_n, \tag{106}$$

where

$$J_n = \frac{M}{4}\, B\Big(\frac{1}{4}, \frac{n}{2} - \frac{1}{4}\Big) = \frac{M}{4}\, \frac{\Gamma(\frac{1}{4})\Gamma(\frac{n}{2} - \frac{1}{4})}{\Gamma(\frac{n}{2})}. \tag{107}$$

For even $n = 2m$ the gamma functions collapse. From $\Gamma(m - \frac{1}{4}) = \Gamma(\frac{3}{4}) \prod_{j=1}^{m-1} (j - \frac{1}{4})$ and the reflection formula $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \pi\sqrt{2}$,

$$J_{2m} = \frac{\pi\sqrt{2}}{4}\, M \prod_{j=1}^{m-1} \Big(1 - \frac{1}{4j}\Big), \tag{108}$$

so $J_4 = \frac{3\pi\sqrt{2}}{16} M$, $J_6 = \frac{21\pi\sqrt{2}}{128} M$, $J_8 = \frac{77\pi\sqrt{2}}{512} M$, and so on.

For a lower bound the right inequality of Eq. (26) is too weak, since $\hat{c} = e^{n\gamma_{\mathrm{E}}/2}$ grows exponentially with $n$. A better one comes from the definition of Ein itself. The integrand $(1-e^{-t})/t$ is at most 1, so $\mathrm{Ein}(w) \le w$ for $w \ge 0$, and therefore $a(k^2) \le \exp(\frac{n}{2}k^4/M^4)$. Then

$$I_n \ge \int_0^\infty e^{-nk^4/(2M^4)}\,\mathrm{d}k = \frac{M}{4}\,\Gamma\Big(\frac{1}{4}\Big)\Big(\frac{2}{n}\Big)^{1/4} \equiv K_n. \tag{109}$$

The two bounds are close. Their ratio is $J_n/K_n = (n/2)^{1/4}\,\Gamma(\frac{n}{2}-\frac{1}{4})/\Gamma(\frac{n}{2})$, and Wendel's inequality for ratios of gamma functions [52],[41] $\Gamma(x+s) \ge x\,\Gamma(x)\,(x+s)^{s-1}$ with $x = \frac{n}{2}-\frac{1}{4}$, $s = \frac{1}{4}$, gives $J_n/K_n \le 2n/(2n-1)$. Hence

$$\Phi_N(0) = -\frac{\Gamma(\frac{1}{4})}{2\pi}\Big(\frac{2}{n}\Big)^{1/4} GmM\Big[1+\frac{\theta_n}{2n-1}\Big], \qquad 0 \le \theta_n \le 1. \tag{110}$$

For $n = 4$, Eqs. (106) and (109) give $0.4852 \le |\Phi_N(0)|/(GmM) \le 0.5303$, and the value of the integral itself, obtained by quadrature as a check, is 0.5100; for $n = 6$ the bracket is $[0.4385, 0.4640]$ around 0.4519. For comparison, the Gaussian form factor $e^{p^2/M^2}$ gives $\Phi_N = -Gm\,\mathrm{erf}(Mr/2)/r$ and $|\Phi_N(0)| = GmM/\sqrt{\pi} \approx 0.564\,GmM$ [10, 53]. The well of the gas is about ten percent shallower for $n = 4$, and it becomes slowly shallower as the gas becomes denser, as $n^{-1/4}$.

The smoothness of $\Phi_N$ is fixed by the decay of Eq. (103). Since $\tilde{\Phi}_N = O(k^{-2n-2})$, the integral $\int \mathrm{d}^3k\,|\mathbf{k}|^j|\tilde{\Phi}_N|$ converges for $j < 2n-1$, and $\Phi_N$ has $2n-2$ continuous derivatives everywhere, the origin included. Equivalently, $\nabla^2\Phi_N = 4\pi G\rho_{\mathrm{eff}}$ with an effective source

$$\rho_{\mathrm{eff}}(\mathbf{x}) = m\int \frac{\mathrm{d}^3k}{(2\pi)^3}\,\frac{e^{i\mathbf{k}\cdot\mathbf{x}}}{a(\mathbf{k}^2)}, \qquad \int \rho_{\mathrm{eff}}\,\mathrm{d}^3x = m, \tag{111}$$

the total being $m$ because $a(0) = 1$. The point mass is spread over a region of size $1/M$. The same two estimates of $a$ give

$$\frac{\Gamma(\frac{3}{4})}{8\pi^2}\Big(\frac{2}{n}\Big)^{3/4} \le \frac{\rho_{\mathrm{eff}}(0)}{mM^3} \le \frac{1}{8\pi^2}\,B\Big(\frac{3}{4}, \frac{n}{2}-\frac{3}{4}\Big),$$

and the linearized tidal tensor at the center, $\partial_i\partial_j\Phi_N(0) = \frac{4\pi G}{3}\rho_{\mathrm{eff}}(0)\,\delta_{ij}$, is finite. The curvature singularity of the linearized Schwarzschild field is gone.

[41] For $x > 0$ and $0 < s < 1$, Hölder's inequality with exponents $1/s$ and $1/(1-s)$ applied to $\Gamma(x+s) = \int_0^\infty t^{s\cdot x}\,t^{(1-s)(x-1)}e^{-t}\,\mathrm{d}t$ gives $\Gamma(x+s) \le \Gamma(x+1)^s\Gamma(x)^{1-s} = x^s\Gamma(x)$. The same inequality with $x \to x+s$ and $s \to 1-s$ gives $\Gamma(x+1) \le (x+s)^{1-s}\Gamma(x+s)$, that is $\Gamma(x+s) \ge x\,\Gamma(x)/(x+s)^{1-s}$. Together, $(x/(x+s))^{1-s} \le \Gamma(x+s)/(x^s\Gamma(x)) \le 1$. We use the lower half with $x = \frac{n}{2}-\frac{1}{4}$ and $s = \frac{1}{4}$.

10.2. **Approach to Newton's law.** Using $\int_0^\infty \sin(kr)\,\mathrm{d}k/k = \pi/2$, write

$$\Phi_N(r) = -\frac{Gm}{r}\big[1-\delta(r)\big], \qquad \delta(r) = \frac{2}{\pi}\int_0^\infty \sin(kr)\,\chi(k)\,\mathrm{d}k, \tag{112}$$

with $\chi(k) = [1 - 1/a(k^2)]/k$. Because $a(k^2)$ is an even entire function of $k$ and $1 - 1/a = O(k^4)$ (for the skeleton $\frac{n}{2}k^4/M^4 + O(k^8)$ by Eq. (27)), $\chi$ is odd, smooth, and $O(k^3)$ at the origin. At large $k$, Eq. (28) and the chain rule give $\partial_k^j[1/a(k^2)] = O(k^{-2n-j})$, hence $\chi^{(j)}(k) = O(k^{-1-j})$. Now integrate by parts twice:

$$\int_0^\infty \sin(kr)\,\chi(k)\,\mathrm{d}k = -\frac{1}{r^2}\int_0^\infty \sin(kr)\,\chi''(k)\,\mathrm{d}k.$$

The boundary terms at infinity vanish because $\chi$ and $\chi'$ do, and those at $k = 0$ vanish because $\chi(0) = 0$ and $\sin 0 = 0$. The function $\chi''$ is again odd and smooth, so the step can be repeated, and after $N$ repetitions

$$|\delta(r)| \le \frac{2}{\pi}\,\frac{\|\chi^{(2N)}\|_{L^1}}{r^{2N}} = \frac{C_N}{(Mr)^{2N}}, \tag{113}$$

where $C_N$ depends on $n$ and $N$ only, since $\chi(k) = M^{-1}\hat{\chi}(k/M)$ with $\hat{\chi}$ independent of $M$; for a realization $C_N$ depends also on the sizes $sM^2$ of its branes. As $N$ is arbitrary, the correction to Newton's law falls faster than any power of $1/(Mr)$. Figure 23 shows the potential for $n = 4$ and the deviation for $n = 4, \dots, 8$; the oscillations of $\delta(r)$ come from the oscillating Fourier kernel and do not affect the bound.

This is the classical counterpart of the infrared statement of Sec. 8. Every term of the low-momentum expansion of $1/a$ is a polynomial in $k^4$, and its Fourier transform is a contact term supported at $r = 0$, so no inverse power of $r$ can appear at any order. The long-range quantum corrections of the effective field theory of gravity, which come from loops of massless particles and are non-analytic in $k$ [54, 83, 84], are not affected, because the infrared of the theory is that of general relativity. The inverse-square law has been tested down to $52\,\mu$m [55], and by Eq. (113) these tests require only $M \cdot 52\,\mu\mathrm{m} \gg 1$, that is $M$ well above $4 \times 10^{-3}$ eV.

10.3. **Mass scale of the linear regime.** The linearized description is self-consistent at every point when $2|\Phi_N| < 1$ everywhere, which by Eq. (105) and $|\Phi_N(r)| \le |\Phi_N(0)|$ happens exactly when

$$m < m_c = \frac{\pi}{4G\,I_n}, \qquad \frac{\pi}{4GJ_n} \le m_c \le \frac{\pi}{4GK_n}. \tag{114}$$

In units with $\hbar = c = 1$ and $G = M_{\mathrm{P}}^{-2}$, $m_c = \mu_n\, M_{\mathrm{P}}^2/M$ with $\mu_n$ of order one; Table 10 gives it for several $n$. Below $m_c$ the weak-field expansion holds down to $r = 0$ and the linearized field has no horizon. Above it, the expansion fails near the center and the question becomes nonlinear, which

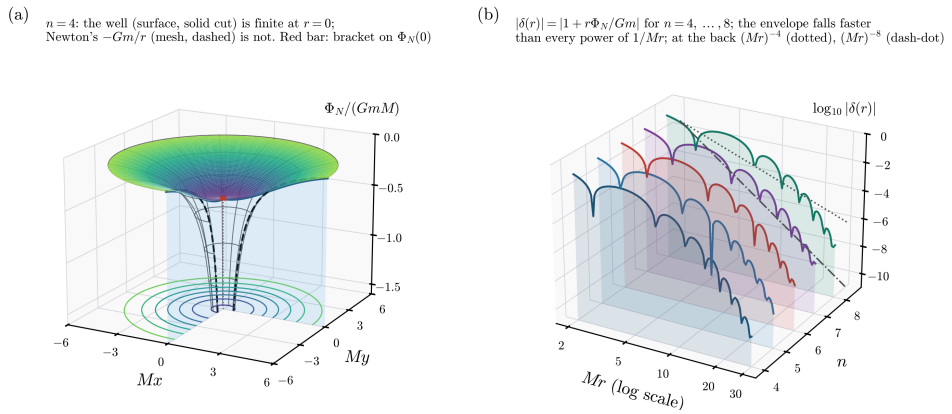



FIGURE 23. Newtonian limit. (a) The potential (104) of a point mass for the skeleton with $n = 4$, in units of $GmM$, drawn over a plane through the source with one quarter cut away. The surface and the solid profiles on the cut faces are $\Phi_N$; the mesh and the dashed profiles are Newton's $-Gm/r$. The red bar on the axis is the bracket $0.4852 \le |\Phi_N(0)|/GmM \le 0.5303$ of Eqs. (106) and (109); the floor carries the contours of $\Phi_N$. (b) The relative deviation $|\delta(r)|$ of Eq. (112) for $n = 4, \ldots, 8$, on logarithmic axes. The dips are zeros of $\delta$, which oscillates because of the Fourier kernel. The envelope falls faster than the reference lines $(Mr)^{-4}$ (dotted) and $(Mr)^{-8}$ (dash-dot) drawn at the back, as Eq. (113) requires. The curves are numerical evaluations of the closed-form integrals and serve as illustration only.

is beyond this paper. For $M$ near the Planck mass, $m_c$ is itself of order the Planck mass, so this is a statement about the smallest objects, not about astrophysical black holes.

| $n$ | $\hat{c}$ | $K_n$ | $I_n$ | $J_n$ | $\frac{\lvert\Phi_N(0)\rvert}{GmM}$ | $m_c$ |
|---|---|---|---|---|---|---|
| 4 | 3.172 | 0.7622 | 0.8011 | 0.8330 | 0.5100 | $[0.943, 1.030]$ |
| 5 | 4.234 | 0.7208 | 0.7485 | 0.7725 | 0.4765 | $[1.017, 1.090]$ |
| 6 | 5.650 | 0.6887 | 0.7099 | 0.7289 | 0.4519 | $[1.077, 1.140]$ |
| 7 | 7.540 | 0.6627 | 0.6797 | 0.6953 | 0.4327 | $[1.130, 1.185]$ |
| 8 | 10.06 | 0.6409 | 0.6550 | 0.6682 | 0.4170 | $[1.175, 1.225]$ |
| 10 | 17.92 | 0.6061 | 0.6165 | 0.6264 | 0.3925 | $[1.254, 1.296]$ |

TABLE 10. Constants of the Newtonian limit, in units $M = 1$ (for $J_n$, $K_n$, $I_n$) and $M_{\mathrm{P}}^2/M$ (for $m_c$). $K_n$ and $J_n$ are the closed-form bounds (109) and (107); the column $I_n$ is a quadrature of Eq. (105), given as a check. The bracket for $m_c$ is Eq. (114), and $\hat{c} = e^{n\gamma_{\mathrm{E}}/2}$. All entries are for the skeleton $\hat{a}$.

## 11. Discussion

The intersection nerve and the gas serve different roles. The nerve controls the incidence complex and admits a curvature-coupled effective action. Positive Gaussian cluster exchange cannot raise the inverse covariance of a light mode whose bare block is fixed. The gas gives ultraviolet suppression by prescribing a multiplicative response. A dynamical derivation of that response from brane degrees of freedom would have to account for the Schur-complement sign in Sec. 2.3.

The one-loop cancellation uses the same strategy as earlier finite higher-derivative gravity [12]. The gas changes the origin and random coefficient of the form factor. For the power-law size profile its annealed inverse moments can be calculated without selecting a realization, although correlation between different propagator factors prevents replacing them by independent averages. The Euclidean power counting applies to all linear hard subspaces. The one-loop calculation cancels the background ultraviolet pole. The nonlocal forest estimates and the extension of cutting and BRST identities discussed above are distinct parts of an all-orders physical amplitude theorem.

The compactly supported curvature test rules out an interacting reflection-positive Euclidean limit that retains the specified linearized curvature covariance in the free limit. This result does not decide whether a different Lorentzian construction or a different physical observable map can realize the pole and amplitude data. Neither the Gaussian construction nor the perturbative tuning constructs such a theory.

## Author contributions

The contributions to the intersection framework follow the posted v2 [85]. Sanjeevan Singha Roy developed the homological description and helped formulate the cluster algebra. Riddhima Sadhu developed the entropy and

information analysis. Deep Bhattacharjee conceived the brane-cluster framework and led the integration of its geometric, algebraic and gravitational parts. Priyanka Samal developed models of cluster topology and their comparison with Regge geometry. Pallab Nandi contributed the algebraic discussion of renormalization and the analysis of quantum error correction. Soumendra Nath Thakur contributed to geometric modeling and the topological field-theory interpretation. Ushashi Bhattacharya coauthored the Poisson brane-gas analysis, including the propagator pole structure and the perturbative ultraviolet cancellation [57, 86].

## Data availability

The argument uses analytic calculations. The plots and sample-value checks for the Poisson model accompany the public verification archive [87]. Appendix D describes the scope of those checks. The figure inputs and standalone LaTeX source are in the `creelie/brane-cluster-gravity` repository. The PDF and source archive for this revision are available in release `v3.0.1`.

## Use of AI tools

AI tools assisted with manuscript revision, algebraic checks, and preparation of diagrams and source files.

## Competing interests

The posted v2 reported no competing interests.

## Appendix A. Conventions and notation

Spacetime has dimension $D$, set to 4 except in dimensional regularization, where $D = 4 - 2\epsilon$. Momentum-space computations are done in Euclidean signature, with $\Box \to -p^2$. The curvature conventions are $R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}$ and $R_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}$, and $\kappa^2 = 32\pi G$. The Gauss–Bonnet density is $E = \mathrm{Riem}^2 - 4\mathrm{Ric}^2 + R^2$. The Euclidean one-loop effective action is $\Gamma_1 = \frac{1}{2}\,\mathrm{STr}\log S''$, where the supertrace includes the ghosts and the compensating factor of Sec. 6.1. The ultraviolet pole of $\int \mathrm{d}^D k\,(2\pi)^{-D} k^{-4}$ over $|k| \geq M$ is $1/(16\pi^2\epsilon)$. In estimates, $C$ and $K$ denote constants that may change from line to line and depend on the indicated indices and on $n$, but not on momenta. Table 11 collects the symbols used most often.

| symbol | meaning |
|---|---|
| $M$ | inverse of the largest brane size |
| $n$ | strength per $e$-fold of size, Eq. (11) |
| $s$, $\tau$ | brane size (squared length); $\tau = sM^2 \in (0,1]$ |
| $\vartheta$, $\vartheta_0$, $\zeta$ | strength of a brane; power law $\vartheta_0\tau^\zeta$ |
| $V(t)$, $\sigma$ | fine graining, Eq. (12) and condition (G) |
| $\Pi$, $\nu$ | a realization of the gas (Poisson process) and its intensity |
| $\psi$, $\psi_*$ | exponent of one brane; $\psi_*(w) = 1 - e^{-w^2}$ |
| $\theta$ | opening of the sectors in (S4) |
| $z$, $x$ | $z = p^2$ (Euclidean), $x = z/M^2$ |
| $a(z)$ | form factor of a realization, Eqs. (1), (17) |
| $\hat{a}$, $\hat{c}$ | skeleton $e^{n\Phi}$ and its coefficient $e^{nC_+}$ |
| $R_\Pi$, $R_\infty$ | fluctuation of $\log a$ and its limit, Eqs. (18), (36) |
| Ein, $E_1$ | entire and ordinary exponential integrals |
| $c$ | $\hat{c}\, e^{R_\infty}$, leading coefficient of $a$ |
| $\mathcal{G}$ | nonlocal remainder of $F$, Eq. (45) |
| $\mathcal{D}(p)$ | graviton propagator, Eq. (64) |
| $P^{(2)}$, $P_s^{(0)}$ | spin-two and spin-zero projectors |
| $\theta_{\mu\nu}$, $T_{\mu\nu}$ | $\delta - \hat{k}\hat{k}$ and $\theta/3 + \hat{k}\hat{k}$ |
| $f[z_0, \dots, z_m]$ | divided difference, Sec. 6.2 |
| $\bar{\omega}(L)$ | superficial degree from the vertex bound |
| $\beta_1, \beta_2, \beta_E$ | one-loop poles of $R^2$, $\mathrm{Ric}^2$, $E$ |
| $s_1, s_2, s_3$ | couplings of the quartic operators |
| $J_n$, $K_n$, $I_n$ | bounds and value of $\int_0^\infty \mathrm{d}k/\hat{a}(k^2)$ |

TABLE 11. Notation.

## APPENDIX B. ANGULAR AVERAGES

*Ricci.* Let $\bar{R}_{\mu\nu}$ be a symmetric tensor in four dimensions with trace $\bar{R}$, and write $\mathrm{Ric}^2 = \bar{R}_{\mu\nu}\bar{R}^{\mu\nu}$. With $\theta = \delta - \hat{k}\hat{k}$ and $T = \theta/3 + \hat{k}\hat{k}$,

$$\bar{R}\cdot P^{(2)}\cdot\bar{R} = \mathrm{tr}(\theta\bar{R}\theta\bar{R}) - \tfrac{1}{3}\big(\mathrm{tr}\,\theta\bar{R}\big)^2,$$
$$\mathrm{tr}(\theta\bar{R}\theta\bar{R}) = \mathrm{Ric}^2 - 2\,\hat{k}\bar{R}^2\hat{k} + (\hat{k}\bar{R}\hat{k})^2,$$
$$\mathrm{tr}\,\theta\bar{R} = \bar{R} - \hat{k}\bar{R}\hat{k}, \qquad T\cdot\bar{R} = \tfrac{1}{3}\bar{R} + \tfrac{2}{3}\hat{k}\bar{R}\hat{k}.$$

The averages over the unit sphere in four dimensions follow from the moments quoted in Sec. 7.3: $\langle\hat{k}\bar{R}\hat{k}\rangle = \bar{R}/4$, $\langle\hat{k}\bar{R}^2\hat{k}\rangle = \mathrm{Ric}^2/4$ and $\langle(\hat{k}\bar{R}\hat{k})^2\rangle = (\bar{R}^2 +$

$2\mathrm{Ric}^2)/24$. Hence

$$\begin{aligned}\langle \mathrm{tr}(\theta\bar R\theta\bar R)\rangle &= \tfrac{7}{12}\mathrm{Ric}^2 + \tfrac{1}{24}\bar R^2,\\ \langle (\mathrm{tr}\,\theta\bar R)^2\rangle &= \tfrac{13}{24}\bar R^2 + \tfrac{1}{12}\mathrm{Ric}^2,\\ \langle \bar R\cdot P^{(2)}\cdot\bar R\rangle &= \tfrac{5}{9}\mathrm{Ric}^2 - \tfrac{5}{36}\bar R^2,\\ \langle (T\cdot\bar R)^2\rangle &= \tfrac{13}{54}\bar R^2 + \tfrac{1}{27}\mathrm{Ric}^2,\end{aligned}$$

and

$$\langle \bar R\cdot P^{(2)}\cdot\bar R\rangle - \tfrac{3}{2}\langle (T\cdot\bar R)^2\rangle = \tfrac{1}{2}\big(\mathrm{Ric}^2 - \bar R^2\big),$$

which is Eq. (82). For $\bar R_{\mu\nu} = \lambda\delta_{\mu\nu}$ the first average vanishes, as it must since $P^{(2)}$ annihilates $\delta_{\mu\nu}$, and the second equals $4\lambda^2$.

*Riemann.* Let $\bar R_{\mu\nu\rho\sigma}$ be an algebraic curvature tensor, with the symmetries of the Riemann tensor and the cyclic identity, and $W^{\nu\sigma} = \bar R^{\mu\nu\rho\sigma}\hat k_\mu\hat k_\rho$. Contracting the second and fourth indices gives the Ricci tensor, as contracting the first and third does, by pair symmetry; so $\mathrm{tr}\,W = \hat k\bar R\hat k$ in the notation above, and $\langle(\mathrm{tr}\,W)^2\rangle = (\bar R^2 + 2\mathrm{Ric}^2)/24$. For the other term,

$$\langle \mathrm{tr}\,W^2\rangle = \bar R^{\mu\nu\rho\sigma}\bar R^{\alpha}{}_{\nu}{}^{\beta}{}_{\sigma}\,\langle \hat k_\mu\hat k_\rho\hat k_\alpha\hat k_\beta\rangle.$$

The three pairings of the fourth moment give, in turn, $\bar R^{\mu\nu}{}_{\mu}{}^{\sigma}\bar R^{\alpha}{}_{\nu\alpha\sigma} = \mathrm{Ric}^2$, $\bar R^{\mu\nu\rho\sigma}\bar R_{\mu\nu\rho\sigma} = \mathrm{Riem}^2$, and $\bar R^{\mu\nu\rho\sigma}\bar R_{\rho\nu\mu\sigma}$. For the last one, pair symmetry gives $\bar R_{\rho\nu\mu\sigma} = \bar R_{\mu\sigma\rho\nu} = -\bar R_{\mu\sigma\nu\rho}$, and the cyclic identity $\bar R_{\mu\sigma\nu\rho} = -\bar R_{\mu\nu\rho\sigma} - \bar R_{\mu\rho\sigma\nu}$ gives

$$\begin{aligned}\bar R^{\mu\nu\rho\sigma}\bar R_{\rho\nu\mu\sigma} &= \mathrm{Riem}^2 + \bar R^{\mu\nu\rho\sigma}\bar R_{\mu\rho\sigma\nu}\\ &= \mathrm{Riem}^2 - \bar R^{\mu\nu\rho\sigma}\bar R_{\mu\rho\nu\sigma}.\end{aligned}$$

The last contraction is the classical $\bar R^{\mu\nu\rho\sigma}\bar R_{\mu\rho\nu\sigma} = \frac{1}{2}\mathrm{Riem}^2$,[42] so the third pairing equals $\frac{1}{2}\mathrm{Riem}^2$. Altogether $\langle \mathrm{tr}\,W^2\rangle = (\mathrm{Ric}^2 + \frac{3}{2}\mathrm{Riem}^2)/24$, and

$$\begin{aligned}\Big\langle \mathrm{tr}\,W^2 - \tfrac{1}{2}(\mathrm{tr}\,W)^2\Big\rangle &= \frac{2\mathrm{Ric}^2 + 3\mathrm{Riem}^2 - \bar R^2 - 2\mathrm{Ric}^2}{48}\\ &= \frac{3\mathrm{Riem}^2 - \bar R^2}{48},\end{aligned}$$

which is Eq. (83). The Ricci terms cancel, which is why the third column of the matrix (84) has no entry in $\mathrm{Ric}^2$ before $\mathrm{Riem}^2$ is traded for $E$. In dimensional regularization the averages are taken in $D$ dimensions; the change is of order $\epsilon$ and affects finite parts only.

## Appendix C. Divided differences

*Proof of Eq.* (69). For $f(w) = w^k$, expand $(A+B)^k$ and keep the words with $m$ letters $B$. They are $A^{j_m}BA^{j_{m-1}}\cdots BA^{j_0}$ with $j_0+\cdots+j_m = k-m$.

[42] Write $X = \bar R^{\mu\nu\rho\sigma}\bar R_{\mu\rho\nu\sigma}$. Contracting the cyclic identity $\bar R_{\mu\nu\rho\sigma} + \bar R_{\mu\rho\sigma\nu} + \bar R_{\mu\sigma\nu\rho} = 0$ with $\bar R^{\mu\nu\rho\sigma}$ gives $\mathrm{Riem}^2 + \bar R^{\mu\nu\rho\sigma}\bar R_{\mu\rho\sigma\nu} + \bar R^{\mu\nu\rho\sigma}\bar R_{\mu\sigma\nu\rho} = 0$. By antisymmetry in the last pair the second term is $-X$, and relabeling $\rho\leftrightarrow\sigma$ in the third term and using antisymmetry in the last pair of the first factor shows that it is also $-X$. So $\mathrm{Riem}^2 = 2X$.

| statement | Py | C | Jl | Lean | where |
|---|---|---|---|---|---|
| cells of a brane: Poisson sum over the cells, characteristic function of the Gaussian shift, Monte Carlo of the coherent phase, sector $\theta = \pi/4$ | 4 | 1 | 1 | — | Eqs. (9), (10) |
| $\gamma_{\mathrm{E}}$ as a difference of two integrals; $\mathrm{Ein}(y) = \log y + \gamma_{\mathrm{E}} + E_1(y)$, real and complex $y$ | 2 | 10 | 8 | — | Eqs. (24), (25) |
| $\int_0^1 \psi_*(x\tau)\,\mathrm{d}\tau/\tau = \frac{1}{2}\,\mathrm{Ein}(x^2)$; power series of the exponent at complex $z$ | 8 | — | — | — | Eqs. (19), (20) |
| probability generating functional, Monte Carlo over the gas | 1 | — | — | — | Eq. (14) |
| $0 \le \mathrm{Ein}(y) - \log(1+y) \le \gamma_{\mathrm{E}}$, monotone; two-sided bound (26), $n = 1, \dots, 12$ | 3 | 14 | 4 | — | Eq. (26) |
| low-momentum coefficients $n/2$ and $n(n-1)/8$ | 4 | 3 | 2 | — | Eq. (27) |
| $\lvert E_1(w)\rvert \le e^{-\mathrm{Re}\,w}/\lvert w\rvert$; cone asymptotics with explicit error; decay on the imaginary axis | 2 | 2 | 12 | — | Eq. (31) |
| resonances of $2 - e^{-s^2 z^2}$ outside the cones; expected count $nR^2/(\pi M^4)$ (ratio 1.0003 at $R = 60M^2$) | 16 | 1 | — | — | Eq. (7) |
| spin projectors, linearized curvature, gauge invariance, $D = 4$ and $D = 5$ | 10 | — | 1 | — | Eqs. (58)–(63) |
| angular averages: Monte Carlo, isotropic tensors, and the exact 600-cell cubature of degree 8 | 8 | 2 | 6 | — | App. B |
| correlators of $\delta R$, $\delta R_{\mu\nu}$, $\delta R_{\mu\nu\rho\sigma}$ with random algebraic curvature tensors | 5 | — | — | — | Eqs. (78)–(81) |
| quartic-operator poles, matrix (84), determinant $-24$, tuning (85) (exact rationals in C) | 9 | 3 | 4 | 3 | Sec. 7.4 |
| Hermite–Genocchi; $w^k[z_0, \dots, z_m] = h_{k-m}$; scaling of the bound (70) in mixed configurations, for $\mathcal{G}$ and for a symbol of order $\mu = \frac{3}{2}$ | 11 | — | — | — | App. C |
| $\bar{\omega}(L) < 0$ for all $L \ge 2$ exactly when $n \ge 4$, for every $m_\sigma \le n - 2$; $n = 3$ marginal; both blocks of Table 5; merging inequality | 15 | 6 | 5 | 13 | Eq. (73) |
| insertion lemma, random four-momenta over seven decades; the two-insertion chain of Fig. 11 | 2 | 1 | 1 | 1 | Eq. (72) |
| growth exponent equals the strength per $e$-fold | 3 | — | — | — | Eq. (11) |

TABLE 12. Independent checks. "Py" is `verify.py` (195 checks), "C" is `formfactor_check.c` (84 checks) and "Jl" is `verify.jl` (114 checks); all pass. "Lean" is the number of theorems of `PowerCounting.lean` that prove the statement for all values of its integer arguments; for the insertion lemma only the arithmetic of the constant $C_r$ is formalized. The Julia angular averages use the 24-cell, a spherical design of degree 5, in exact rational arithmetic. The last column gives the equation or section checked.

Acting on the plane waves, $A^{j_i}$ becomes $z_i^{j_i}$, and the sum over the $j_i$ is the complete homogeneous symmetric polynomial $h_{k-m}(z_0, \dots, z_m)$, which is the divided difference of $w^k$ on $z_0, \dots, z_m$ [34].[43] Both sides of Eq. (69) are linear in $f$. For entire $f = \sum_k f_k w^k$ the series $\sum_k \lvert f_k\rvert \binom{k}{m} \max\lvert z_i\rvert^{k-m}$ converges, since $\lvert h_{k-m}\rvert \le \binom{k}{m} \max\lvert z_i\rvert^{k-m}$, so the identity passes to the limit. The functions to which we apply the formula, $\mathcal{G}$ and $a - cx^n$, are entire, so

[43] The identity $w^k[z_0, \dots, z_m] = h_{k-m}(z_0, \dots, z_m)$ follows by induction on $m$ from the recursion and $h_r(z_1, \dots, z_m) - h_r(z_0, \dots, z_{m-1}) = (z_m - z_0)\, h_{r-1}(z_0, \dots, z_m)$, which holds because both sides count the monomials of degree $r$ that contain $z_m$ but not $z_0$, minus those that contain $z_0$ but not $z_m$.

| statement | Py | C | Jl | Lean | where |
|---|---|---|---|---|---|
| $\mathrm{Ein}(w) \le w$; $J_n$ formula; $K_n \le I_n \le J_n \le \frac{2n}{2n-1}K_n$; $J_{2m}$ product; $\rho_{\mathrm{eff}}(0)$ bounds; Wendel's inequality; $n = 4, \dots, 14$ | 38 | 28 | 16 | — | Eqs. (107)–(111) |
| decay of $\delta(r)$ | 1 | — | — | — | Eq. (113) |
| zero-point count $\log p^2 + 3\log a$; Mellin transform $\Gamma(s)/s$; sharp and smooth cutoffs, constant term 1; $\rho_1 = 3nM^4/(64\pi^2)$ | 6 | 4 | — | — | Eqs. (86)–(88) |
| admissible class: $C_+$, moments $\int t(1-\psi)\,\mathrm{d}t$, boundedness of $\Phi - \frac{1}{2}\log(1+x^2)$, Mellin continuation | 12 | 3 | 1 | — | Eqs. (32)–(34) |
| incomplete-gamma variances (48), (49) against quadrature; $\mathbb{E}R_\infty^2 = n\vartheta_0/\zeta$; Eq. (35) | 3 | 2 | 14 | — | Sec. 4.8 |
| Monte Carlo over $4 \times 10^4$ gases: Campbell mean, isometry (15), variance of Eq. (38), $\mathbb{E}\,a_\Pi \ne \hat{a}$ by Eq. (16), variance of $\delta\rho_\Pi$ | 4 | — | — | — | Secs. 4.4, 8.2 |
| $R_\infty$ sampled from its Lévy measure: variance, Eq. (50), lower tail (51), $n = 4, 6, 8$ | 1 | 1 | 5 | — | Sec. 4.8 |
| identical branes: $\Psi_*$ in closed form and its asymptotics; variance and characteristic function by Monte Carlo; decay of $\lvert\mathbb{E}\,e^{iuR_\Pi}\rvert$ | 7 | 1 | 4 | — | Eq. (47) |
| mean-square decay (40) along rays; one realization: slope $-\zeta/2$ over dyadic annuli, and $a - cx^n$ bounded for $\zeta = 9$, growing for $\zeta = 5$ | 2 | — | — | — | Eqs. (41), (46) |
| argument principle on $\lvert z\rvert = 2M^2$: winding 0 for a realization, equal to the count of Eq. (7) for a resonant product | 1 | — | — | — | Sec. 4.1 |
| Table 2, both $\zeta = 5$ and $\zeta = 2n+1$; relative fluctuation (91); mean of $\delta\rho_\Pi$ by continuation | 3 | 1 | 8 | — | Eqs. (90), (91) |
| remainder $\to 1$ for the cutoffs $e^{-u^p}$; sharp-cutoff remainder $1 - e^{-W} + WE_1(W)$ | 2 | — | — | — | Sec. 8.2 |
| cutoff variance: $C_\zeta$ in closed form against quadrature, $C_2 = \log 2$, $(4-\zeta)C_\zeta \to 1$, and the limit $1/(\zeta-4)$ for $\zeta > 4$ | 3 | 1 | 1 | — | Eq. (94) |
| one-loop bubble: $\mathrm{Im}\,A_\mu = -\pi^3$; Eq. (97) for a local propagator against the closed form of $A_\mu$; the skeleton independent of $\mu$; the bound and ratio columns of Table 8 | 3 | — | 2 | — | Eqs. (97), (98) |
| the disk of the bound: $\rho^2 \ge 7\delta s/64$ on it, and its contribution above the leading term; seagull factor $(t-u)^2/16$, at random angles and in exact rational arithmetic | 2 | — | 1 | — | Eqs. (96), (99) |
| matter scattering: $T_1 \cdot T_2 - \frac{1}{2}T_1T_2 = -su$ for massless scalars; maximum of $x(1+x^2)^{-n/2}$, $n = 4, \dots, 8$; the bound below Eq. (100) for the skeleton | — | — | 8 | — | Sec. 9.4 |
| free graviton: $C \cdot C = \frac{p^4}{2}hP^{(2)}h$ for random modes, $\mathrm{tr}\,P^{(2)} = 5$; $\int_0^\infty x^2(1+x^2)^{-n/2}\mathrm{d}x$ in closed form; the skeleton integral 0.4237 and its bounds; logarithmic growth at $n = 3$ | — | — | 10 | — | Eq. (101) |
| $I_4$, $I_5$ and $\lvert\Phi_N(0)\rvert$ for $n = 4, 5$; $J_4 = 3\pi\sqrt{2}/16$; the constants $c$ for $n = 4, \dots, 8$ | 4 | — | — | — | Table 10 |

TABLE 13. Independent checks, continued from Table 12.

this argument covers them; for functions that are merely smooth the same formula follows from the functional calculus of self-adjoint operators [33].

*Proof of Eq.* (70). The proof is by induction on $m$; the case $m = 0$ is the hypothesis $|f(z)| \le C_0(1+z)^\mu$. By symmetry of the divided difference let $z_0 \le z_1 \le \cdots \le z_m$, so that $z_{\min} = z_0$ and $z_{\max} = z_m$, and write $W = (1+z_0)^{\mu-}(1+z_m)^{\mu+}$. For every $\xi \in [z_0, z_m]$ we have $(1+\xi)^\mu \le W$, and $W$ does not increase when $z_0$ is replaced by a larger argument or $z_m$ by a smaller one. There are two cases.

If $1 + z_m \le 2(1+z_0)$, all arguments are comparable. In Eq. (68) the point $\xi = \sum t_i z_i$ lies in $[z_0, z_m]$, so $|f^{(m)}(\xi)| \le C_m(1+\xi)^\mu(1+\xi)^{-m} \le$

$C_m W(1+z_0)^{-m}$ there, and the simplex has volume $1/m!$. Hence

$$|f[z_0,\dots,z_m]| \le \frac{2^m C_m}{m!}\, W \prod_{i\ge 1}(1+z_i)^{-1},$$

using $(1+z_0)^{-1} \le 2(1+z_i)^{-1}$ for each $i \ge 1$.

If $1+z_m > 2(1+z_0)$, then $z_m - z_0 > \frac{1}{2}(1+z_m)$ and we use the recursion. By the induction hypothesis applied to $z_1,\dots,z_m$, whose smallest element is $z_1$ and whose largest is $z_m$, and by the monotonicity of $W$,

$$|f[z_1,\dots,z_m]| \le K_{m-1} W \prod_{i=2}^{m}(1+z_i)^{-1}.$$

Dividing by $z_m - z_0 > \frac{1}{2}(1+z_m) \ge \frac{1}{2}(1+z_1)$ gives at most $2K_{m-1}W\prod_{i=1}^{m}(1+z_i)^{-1}$. The other term, $f[z_0,\dots,z_{m-1}]$, whose largest argument is $z_{m-1} \le z_m$, is bounded by $K_{m-1}W\prod_{i=1}^{m-1}(1+z_i)^{-1}$, and dividing by $z_m - z_0 > \frac{1}{2}(1+z_m)$ gives the same bound. So $K_m = \max(2^m C_m/m!,\, 4K_{m-1})$ works, which completes the induction.

*The two symbols.* For $f = \mathcal{G}$ the hypothesis holds with $\mu = m_\sigma - 1$ by Eq. (46), and Eq. (70) gives $|\mathcal{G}[z_0,\dots,z_m]| \le K_m(1+z_{\max})^{m_\sigma}\prod_{i=0}^{m}(1+z_i)^{-1}$, the form used in Sec. 6.3. For $f = a - cx^n$ the derivative bounds follow from Eq. (28) and the large-momentum behavior on the real axis: $a - cx^n = O(x^{n-\sigma/2+\varepsilon})$ for $x \ge 1$, and each derivative is one power of $x$ smaller (Sec. 4.8), so the hypothesis holds with $\mu = m_\sigma$.

*Proof of Eq.* (72). Write $\Lambda = (1+|q|)(1+|q'|)\prod_a(1+|\ell_a|)$ and label the legs so that $|\ell_1| \ge |\ell_2| \ge \cdots$, with $\ell_2 = 0$ if $r = 1$. Each of the products $|q||q'|$, $|q||\ell_1|$, $|\ell_1||\ell_2|$ and $|\ell_1|(|q|+|q'|)$ is at most $\Lambda$, because every factor on the left appears on the right with 1 added to it. From $q = q' - \sum_a \ell_a$ we have $|q| \le |q'| + s$, and $s \le r|\ell_1|$, so

$$|q|^2 \le |q||q'| + r|q||\ell_1| \le (1+r)\Lambda, \qquad 2|q|s \le 2r\Lambda.$$

Momentum conservation solved for the largest leg, $\ell_1 = q' - q - \sum_{a\ge 2}\ell_a$, gives $|\ell_1| \le |q| + |q'| + (r-1)|\ell_2|$, so that $s \le |\ell_1| + (r-1)|\ell_2| \le |q| + |q'| + 2(r-1)|\ell_2|$, and therefore

$$\begin{aligned} s^2 \le r|\ell_1|\, s &\le r|\ell_1|\big(|q|+|q'|\big) + 2r(r-1)|\ell_1||\ell_2| \\ &\le (2r^2 - r)\Lambda. \end{aligned}$$

Adding the three pieces gives $(|q|+s)^2 \le (1 + 2r + 2r^2)\Lambda$, which is Eq. (72) with $C_r = 1 + 2r + 2r^2$. The powers cannot be improved: for $r = 2$, $q = q' = 0$ and $\ell_1 = -\ell_2 = \lambda e$ with $|e| = 1$, both sides are of order $\lambda^2$.[44]

[44] The estimate is the point at which a vertex with many hard legs differs from one with all momenta comparable. Without it one would bound an insertion by $(1+q_{i-1}^2)\prod_a(1+\ell_a^2)$, which charges two powers to each leg and none to the outgoing running momentum. The divided difference still cancels the running momenta, but each hard leg then costs $\lambda^2$ instead of $\lambda$, and the counting of Sec. 6.4 closes only for $n \ge 6$.

*Complex arguments.* Section 9 needs Eq. (70) for complex arguments. Let $K$ be a compact set that contains 0, let $S_\delta = \{|\arg z| \le \pi/4 - \delta\}$, and put $\Omega = \operatorname{conv} K + S_\delta$. Because $S_\delta$ is a convex cone, $\Omega$ is convex and contains every convex combination of its points, in particular the point $\sum_i t_i z_i$ of Eq. (68). Writing each $z_i = k_i + v_i$ with $k_i \in \operatorname{conv} K$ and $v_i \in S_\delta$, and using $\operatorname{Re} v \ge |v|/\sqrt{2}$ on $S_\delta$, one finds $|\sum_i t_i z_i| \ge (\min_i |z_i| - C_K)/\sqrt{2} - C_K$, so that $1 + |\sum_i t_i z_i| \ge \kappa\,(1 + \min_i |z_i|)$ with $\kappa > 0$ depending on $K$ and $\delta$ only. Suppose $f$ is analytic near $\Omega$ with $|f^{(j)}(z)| \le C_j(1 + |z|)^{\mu - j}$ there. Then the proof above goes through with $z_i$ replaced by $|z_i|$ in the case distinction: the recursion case uses $|z_m - z_0| \ge |z_m| - |z_0|$, and the comparable case uses the lower bound just derived and, for $\mu > 0$, the upper bound $|\sum_i t_i z_i| \le \max_i |z_i|$. So Eq. (70) holds on $\Omega$ with $1 + z_i$ replaced by $1 + |z_i|$. For $f = \mathcal{G}$ and $f = a - cx^n$ the derivative bounds on $\Omega$ follow from the cone asymptotics (42): far from the origin $\Omega$ lies in the wider cone $S_{\delta/2}$, where $\mathcal{G} = -1/z$ up to a term of order $|z|^{n-1-\sigma/2+\varepsilon}$, at most $|z|^{m_\sigma - 1}$, and Cauchy's estimate on a disk of radius $\frac{1}{2}|z|\sin(\delta/2)$ around $z$, which stays in $S_{\delta/2}$, bounds the $j$th derivative of that term by $C_j|z|^{n-1-\sigma/2+\varepsilon-j}$.

## Appendix D. Numerical checks

The closed-form identities, inequalities and coefficients derived in the text were re-evaluated at sample points by three independent programs, in Python, C and Julia, which share no code. Tables 12 and 13 list the groups of checks. Where an inequality is checked, it is checked on a grid or at random points, and the tolerance is far below the margin by which it holds; where an identity is checked, the two sides are computed by different routes (closed form against quadrature, series against continued fraction, exact rational arithmetic against floating point). The Julia program works in 256-bit arithmetic with its own power series, continued fractions, arithmetic-geometric mean and double-exponential quadrature. These checks confirm the algebra and guard against transcription errors; they are not part of any proof.

The integer statements on which the power counting and the one-loop tuning rest are, in addition, proved in Lean 4 without external libraries: the merging inequality and its iterated form, the bound $\bar{\omega}(L)$ on the superficial degree of any graph, its sign for all $n \ge 4$ and $L \ge 2$ and for $n \le 3$ at two loops, its growth with $m_\sigma$ and its sign for every fine graining that (G) allows, the entries of both blocks of Table 5, the determinants $-24$ and $-12$ of the pole matrix, and the cancellation of the three poles by the tuning (85). The public archive reports that its kernel checks use the standard ones of Lean's core logic, propositional extensionality and quotient soundness. These proofs repeat arguments given in the text in full; the analytic estimates, on special functions, integrals and the gas, are not formalized.

## Appendix E. Geometric observables of the cluster model

E.1. **Regge geometry and its independent data.** The intersection complex records incidences. A Regge metric requires edge lengths forming nondegenerate Euclidean simplices, with equal lengths on shared faces. In four dimensions the curvature is concentrated at triangles $f$, with area $A_f$ and deficit $\delta_f = 2\pi - \sum_{\sigma \supset f} \theta_{\sigma,f}$ at an interior hinge. Transport in the normal plane around the hinge rotates by $\delta_f$. The scalar-curvature integral has the normalization $2\sum_f A_f \delta_f$, since scalar curvature on the normal plane is twice its Gaussian curvature. The Einstein–Hilbert normalization thus gives $(8\pi G)^{-1} \sum_f A_f \delta_f$ for the curvature part of the Euclidean Regge functional [90]. Rescaling all lengths by $\lambda$ leaves the nerve, every dihedral angle, and every homology class unchanged, while multiplying this functional by $\lambda^2$. Incidence therefore supplies no canonical cluster–Regge action or ultraviolet scale.

E.2. **Stationary entropy for a local cluster coupling.** Consider the local Lorentzian density $\sqrt{-g}\, f(R, \Phi)$ with

$$f(R, \Phi) = \frac{R - 2\Lambda}{16\pi G} - \frac{\xi}{2} R\Phi^2 + \lambda_2 R^2.$$

On a regular bifurcation surface $\mathcal{H}$ of a stationary solution, the binormal has $\epsilon_{\mu\nu}\epsilon^{\mu\nu} = -2$. Using $\partial R / \partial R_{\mu\nu\rho\sigma} = g^{\mu[\rho} g^{\sigma]\nu}$, the contraction with two binormals is $-2$. The Noether-charge formula [89] then gives

$$S_{\text{Wald}} = 4\pi \int_{\mathcal{H}} f_R(R, \Phi)\, \mathrm{d}A = \frac{A(\mathcal{H})}{4G} - 2\pi\xi \int_{\mathcal{H}} \Phi^2\, \mathrm{d}A + 8\pi\lambda_2 \int_{\mathcal{H}} R\, \mathrm{d}A.$$

Kinetic and potential terms with no explicit curvature give no direct contribution to this derivative, although they change the stationary solution. For constant $\Phi$ and $R$ the formula is a constant times area. It is a local-action calculation. The entropy of the nonlocal action would require a separate variational and boundary analysis.

E.3. **Entanglement and the code condition.** If a finite collection of cluster sectors admits a tensor product factorization and a density matrix $\rho$, then tracing out a complement defines $S(A) = -\operatorname{Tr} \rho_A \log \rho_A$. For finite-dimensional sectors strong subadditivity gives

$$S(AB) + S(BC) - S(B) - S(ABC) \geq 0$$

by the Lieb–Ruskai theorem [94]. Entropies add for product states because the eigenvalues of a tensor product are the products of the component eigenvalues. For a correlated cluster state neither additivity nor an area law follows from the nerve. A holographic area law uses further geometric and dynamical hypotheses [91]. Likewise, encoding into a tensor product by an isometry $V$ corrects erasure of a factor $E$ only if

$$V^*(X_E \otimes I)V = c(X_E) I$$

for every operator on $E$. Necessity follows because the erased marginal must be independent of the encoded state. Polarization of diagonal expectations gives the displayed operator equality. The geometric discussion of quantum error correction in holography [92] uses a specified code and reconstruction map. No such code follows from intersection numbers alone. Similarly Page's random-state entropy calculation [93] depends on its ensemble. Applying it to an evaporating geometry would require a law for the state and for its time evolution.

E.4. **Fixed points of a proposed finite-dimensional flow.** For a differentiable beta function $\mathrm{d}g/\mathrm{d}t = \beta(g)$, $t = \log(k/k_0)$, a fixed point satisfies $\beta(g_*) = 0$. Linearization yields $\mathrm{d}\delta g/\mathrm{d}t = D\beta(g_*)\delta g$. Its linear solution is $\delta g(t) = e^{tD\beta(g_*)}\delta g(0)$. If the matrix is diagonalizable, modes with negative real exponent decay as $k \to \infty$, and modes with positive real exponent require tuning in the linearized flow. Jordan blocks add polynomial factors in $t$. A proposed flow with one cluster mode per simplex can change this matrix, but the dimension of its attractive subspace cannot be read from the number of modes. Deriving the beta function and controlling truncation are additional requirements for an interacting fixed-point argument.

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